

■ Dedicated link such as dark fiber configured as a routed link

■ L3 over MPLS L2VPN (E-line, E-LAN)

Although all these connectivity types will deliver a routed DCI, each one has its strengths and weaknesses with regard to a routed DCI solution. Table 8-7 compares the primary design aspects of each of the connectivity types listed here.

Connectivity	Strengths	Weaknesses/Concerns
MPLS L3VPN	Scalable. Seamless end-to-end QoS support Simplifies routing over the WAN with reduced operational complexity.	Achieving path visualization (isolation) over the DCI can be a challenging task because it is controlled by the WAN provider. Using dual providers for high availability (HA) can increase operational complexity. Quality of service (QoS) differentiation or high bandwidth can add additional cost to the DC operators.
Tunneling over the Internet	Offers a low-cost connectivity. Simple, quick to set up. Fully controlled by the DC operator. Can be secured to some extent with IPsec.	Low end-to-end service reliability. Higher latency. Security and compliance concerns. Bandwidth limitation for any DCI that needs gigabits of bandwidth.

Dedicated link (for example, dark fiber)	High bandwidth. Full control of routing, virtualization, QoS and security setup Highly reliable (ideally with protected/redundant links).	High capex. Limited geographic coverage. Higher opex to operate both physical and control plane.
L3 over MPLS L2VPN	High bandwidth with lower cost compared to other options. Full control of routing, virtualization, and security setup. Supports end-to-end seamless QoS. Global Carrier Ethernet can support global DCI between countries.	Weaknesses depend on the deployed L2VPN technologies (Virtual Private LAN Services [VPLS], Hierarchical VPLS [H-VPLS], or Ethernet VPN [EVPN]).* Apart from EVPN, there is active-active multihoming limitation with VPLS. VPLS with link state such Open Shortest Path First (OSPF) can introduce operational complexities.

*The limitations from the L2VPN itself, such as MAC scalability, are not an issue with this design model because the edge nodes are deployed in Layer 3 mode (routed connectivity).

Table 8-7 Routed DCI Connectivity Options Comparison

Layer 2 DCI

As discussed earlier, data center applications and services must always drive the design decisions of the DCI solution in terms of distance (latency), capacity, and network communication required (L2 versus L3). In today's data centers, there are several systems, especially around the clustering of servers (geoclustering) and system virtualization, that mandate stretching Layer 2 LANs between the interconnected data centers. This stretching enables the resiliency and clustering mechanisms offered by the different applications at the web, application, and database layers to achieve specific design goals, such as application resiliency and workload allocation flexibility via workload mobility and distributed workload across multiple data centers. In today's data centers, many system virtualization vendors require nonroutable (Layer 2) adjacency between the distributed workloads as well to achieve VM mobility, as shown in Figure 8-21.

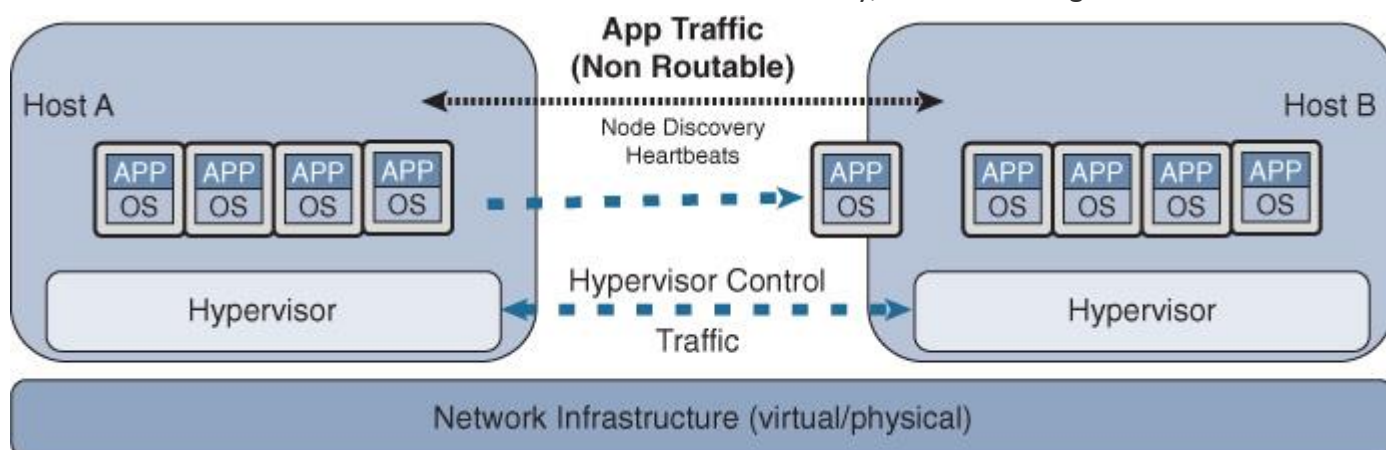


Figure 8-21 LAN extension for distrusted workload and VM mobility

Therefore, LAN extension between data centers became one of the most popular DCI solutions even though it is one of the most complex designs for several reasons, which are covered in this section (as are with the recommended solutions and optimization techniques later in this chapter). Table 8-8 summarizes the main strengths and design concerns of the Layer 2-based DCI.

Strengths	Design Concerns
Supports distributed workload over the WAN/Internet.	Design complexity, because special design considerations are required to overcome asymmetrical routing and what is also known as <i>traffic trombone</i> or <i>ping-pong traffic</i> .
Supports live VM migration (VM nobility) that disassociates IP/VM from a physical location.	<i>Dual-active</i> situations for the distributed workloads and network security services such as firewall after can lead to a serious outage of interconnected data centers.
Any IP (physical or virtual) entity can retain its IP address when moved between DCs.	Operational complexity, due the increased scope of the failure domain (large Layer 2 domain), and therefore special consideration is required with regard to Layer 2 flooding.
Supports active-standby and active-active DR models.	Active-active is a challenging design to achieve because it has to align well with other services and components at higher layers such as local FHRP, global site load balancers.
Supports stretched network services and security appliances such as load balancers and firewalls between interconnected data centers.	
Facilitates data center migration projects to maintain minimal to zero downtime of resource availability.	

Table 8-8 Layer 2-Based DCI Strengths and Design Concerns

Layer 2 DCI Connectivity Options

Today's DCI solutions, which are capable of providing LAN extension to meet modern data center requirements, can be classified into three primary connectivity models:

■ **Dark-fiber based:** Typically, this DCI model is applicable to DC operators that own the physical layer (interconnect), which offers them the freedom to enable any of the desired solutions appropriate for their requirements.

■ **SP controlled:** This type of DCI connectivity is based on the fact the DCI (WAN/MAN) provider deploys and controls the service for the DC customers (LAN extension), such as E-Line and E-LAN services.

■ **DC operator controlled (overlay):** This type of DCI is usually based on an overlaid LAN extension solution, where the DC operator deploys and controls the emulated Layer 2 service over an IP transport such as VxLAN, OTV, and L2VPN, over GRE.

Note

Technically, any of the listed Layer 2 technologies and protocols can be enabled over a dark fiber; all are valid and proven solutions. However, this section focuses on enabling classic 802.1Q over a dark fiber. The other technologies, like VxLAN and OTV, are discussed as a transport agnostic where any IP transport is applicable, including dark fiber configured as routed link. An L2VPN deployed by the DC operator, however, follows the same concept with only one fundamental difference: The underlay transport must be MPLS enabled. You can either use GRE for this purpose or you can enable MPLS at the link level in case of dark fiber.

Layer 2 DCI solutions (SP controlled versus DC operator controlled) can be divided into two primary layers:

- The service layer is the actual extended Layer 2 services; it is almost always between the customer edges (CEs) facing the PEs.
- The network layer represents the service emulation across the IP/MPLS network (overlay), as shown in Figure 8-22. This layer varies based on who is deploying and controlling the service (SP controlled versus DC operator controlled)

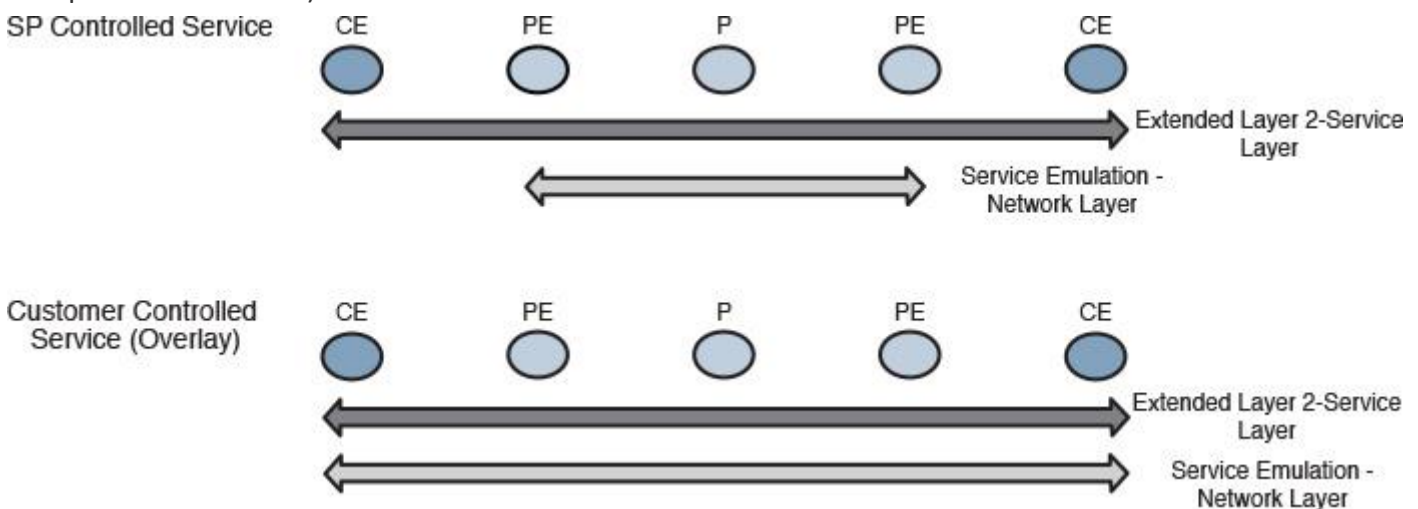


Figure 8-22 Layer 2 DCI Service Provisioning Models

Note

The services offered by SPs are almost always L2VPN based, whereas the ones that are deployed and controlled by the DC operator (CE side) can be any DCI overlaid solution.

For the purpose of simplicity, this section covers the characteristics of these Layer 2 DCI solutions on a per-technology basis, as listed here:

- Dark fiber based
- Layer 2 over Metro Ethernet (ME) transport such as E-line and E-LAN, including the following:
 - SP provisioned and controlled ME
 - DC operator provisioned and controlled ME over any IP transport
- TRILL/FabricPath-based DCI
- EoMPLSoGRE (OTV) over any IP transport
- VxLAN over any IP transport

Dark Fiber-Based DCI

This DCI solution is based on a direct physical dark fiber link controlled by the data center operator. With this solution, the DC operator has the freedom to run any of the protocols listed earlier to establish the DCI connectivity. However, this section focuses on the usually extended LAN over 802.1Q link (dark fiber) along with mLAG, as shown in Figure 8-23. Although the following design enables data center operators to use all the mLAG member links to carry the traffic passing between the two data centers, several design optimizations must be considered, such as FHRP localization and Layer 2 flooding control (DCI path

optimization, covered in greater detail later in this chapter). In addition, as covered earlier in this book, every traffic flow normally uses a single LAG/mLAG member link; this means that each traffic flow will be limited to a single member link capacity (unless the *flowlet* concept is used).⁴

4. “Dynamic Load Balancing Without Packet Reordering,” IETF draft, chen-nvo3-load-banlancing, <http://www.ietf.org>

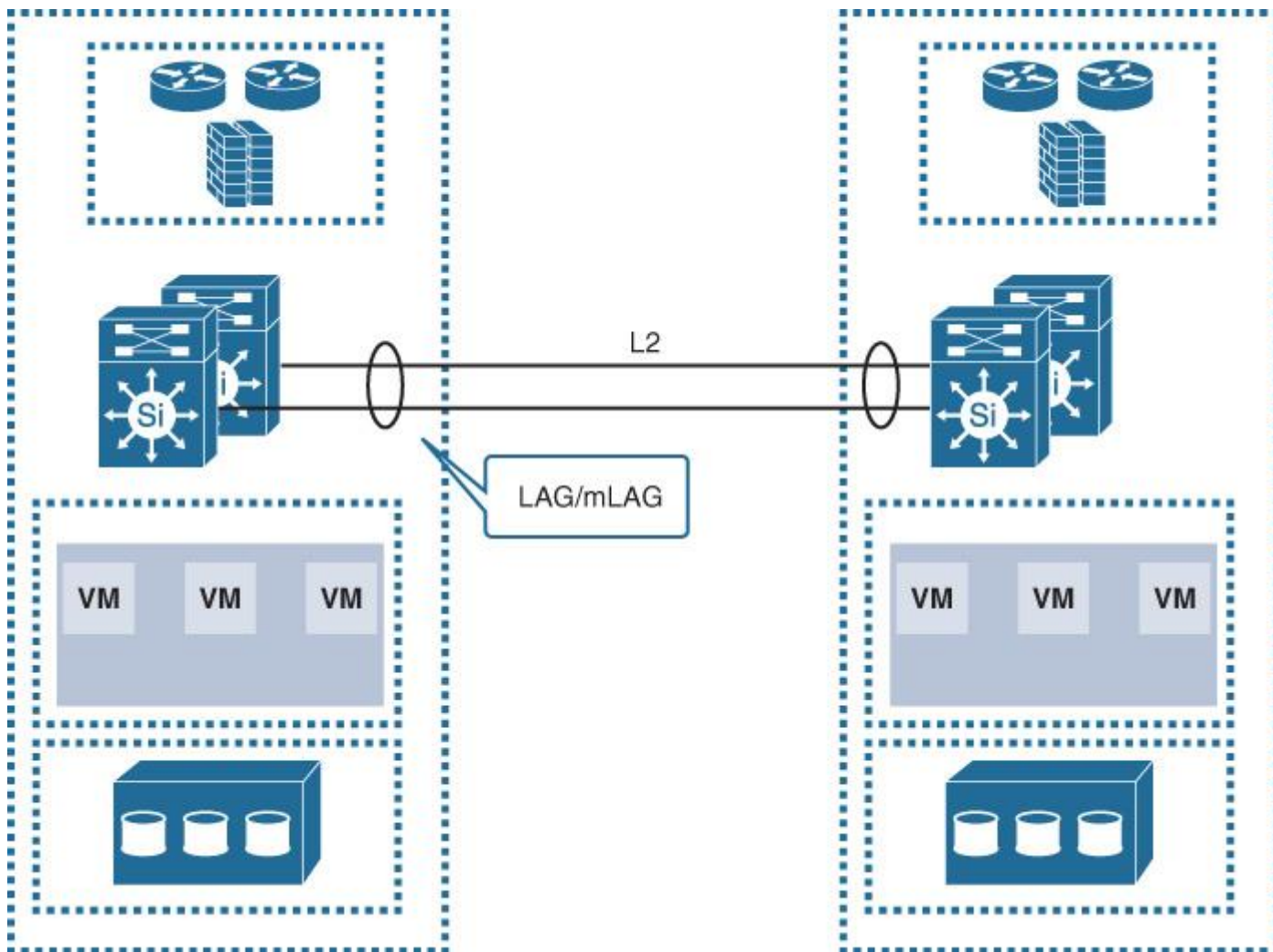


Figure 8-23 802.1Q Layer 2 over Dark Fiber DCI

Note

The dynamic load-balancing capability available today in Cisco Application Centric Infrastructure (ACI) “adjusts the traffic allocations according to congestion levels. It measures the congestion across the available paths and places the flows on the least congested paths, which results in an optimal or near optimal placement of the data, DLB can be configured to place traffic on the available uplinks using the granularity of flows or of flowlets.” [84]

Besides the design challenges and the operational complexity that can be introduced to the DCN by the extended LAN across multiple locations, interconnecting more than two data centers over 802.1Q dark fiber links in a triangle or square topology can lead to serious design complexity and operational inefficiencies because STP is the Layer 2 control protocol in this particular design model. This means that there will be inefficient utilization of the links associated with suboptimal routing, which can be very sensitive in data center environments that require low-latency communications across the DCI. As shown in Figure 8-24, if the hosts belonging to VLAN X distributed across DC1 and DC2 need to communicate, traffic has to traverse DC3 because of STP blocking.

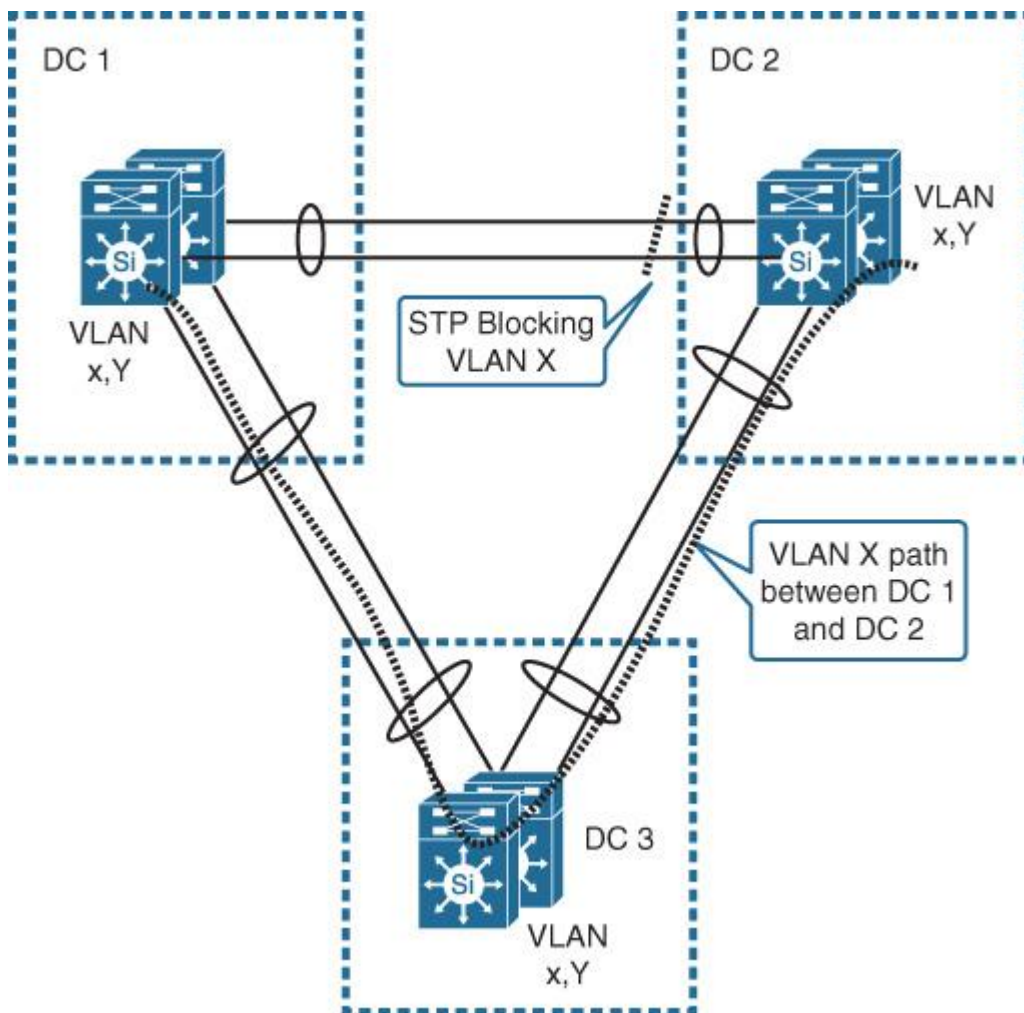


Figure 8-24 STP Blocking with Triangle DCI Topology

Although this can be fixed with STP traffic engineering (where network operators can control which link can be blocked for which VLAN from the STP point of view), this will introduce a high degree of operational complexity and limit optimal communications across to data centers only. This issue can be resolved by either considering a protocol that does not rely on STP such as TRILL, VPLS, or VxLAN (which is beyond the scope of this subsection) or by introducing the concept of *core DCI* to transform the design model into one that is more flexible and scalable to interconnect multiple data centers, as shown in Figure 8-25.

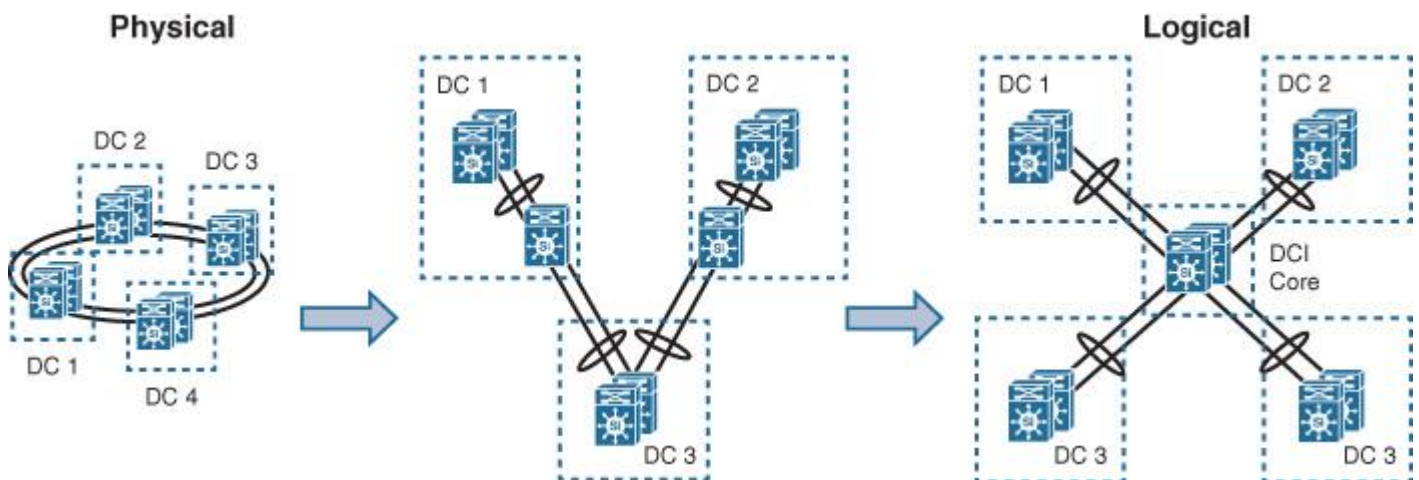


Figure 8-25 Dark Fiber DCI over xWDM: Logical Star Topology

Core DCI is based on introducing two core switches deployed in switch clustering mode, ideally distributed across two different data centers to avoid introducing any single point of failure to the design. The physical

connectivity of the underlying fiber, however, should be provisioned over an x wavelength-division multiplexing (xWDM) ring, where the logical point-to-point links from each DC to the core DC are created via xWDM transport lambda or wavelength over the optical ring. Ultimately, this will create a logical star topology that will eliminate STP limitations and offer a flexible and reliable DCI solution, as shown in Figure 8-25.

However, in addition to the fact that this design can be an expensive option in terms of capex and opex, not every location (geographic area) has xWDM available. Therefore, these design constraints have to be considered during the planning and design optimization stage to avoid providing impractical design recommendations. (An *impractical* design is one that is technically valid but practically cannot be deployed because of certain design constraints.)

Layer 2 DCI over ME Transport

This design option is primarily based on emulating a Layer 2 LAN over an IP MPLS-enabled infrastructure, as discussed earlier in this chapter. The most common connectivity models for the data center DCI solution are either the E-Line or E-LAN connectivity models.

This DCI connectivity design option can be provisioned either by the SP or by the DC operator, as described here:

■ **SP controlled:** In this scenario, the SP owns the MPLS infrastructure and emulates the Layer 2 transport transparently to the DC side for the DCI (E-line, E-LAN).

■ **DC operator controlled:** In this scenario, the DC operators (enterprise or SP DC) design and set up their own overlaid ME solution, such as E-Line or E-LAN, over any IP transport with the aid of a tunneling mechanism to enable MPLS for the pseudo wires (PWs) of the emulated LAN, such as VPLS-over-GRE-over-IP. Similarly, if the DC operator has a dark fiber-based DCI MPLS, it can be enabled across this DCI. Run either Ethernet over MPLS (EoMPLS) or VPLS based on the physical layout.

Furthermore, as discussed earlier in Chapter 6, “Service Provider MPLS VPN Services Design,” EVPN and Advanced VPLS (A-VPLS) can provide DCI capabilities similar to dark fiber with mLAG but on a larger scale over an emulated L2 LAN and with a cheaper price, as shown in Figure 8-26. In addition, EVPN with PBB can meet very large-scale public cloud-grade DCI requirements (scales to millions of MACs) with a simplified and optimized control plane (BGP based) combined with the PBB MAC hiding principle.

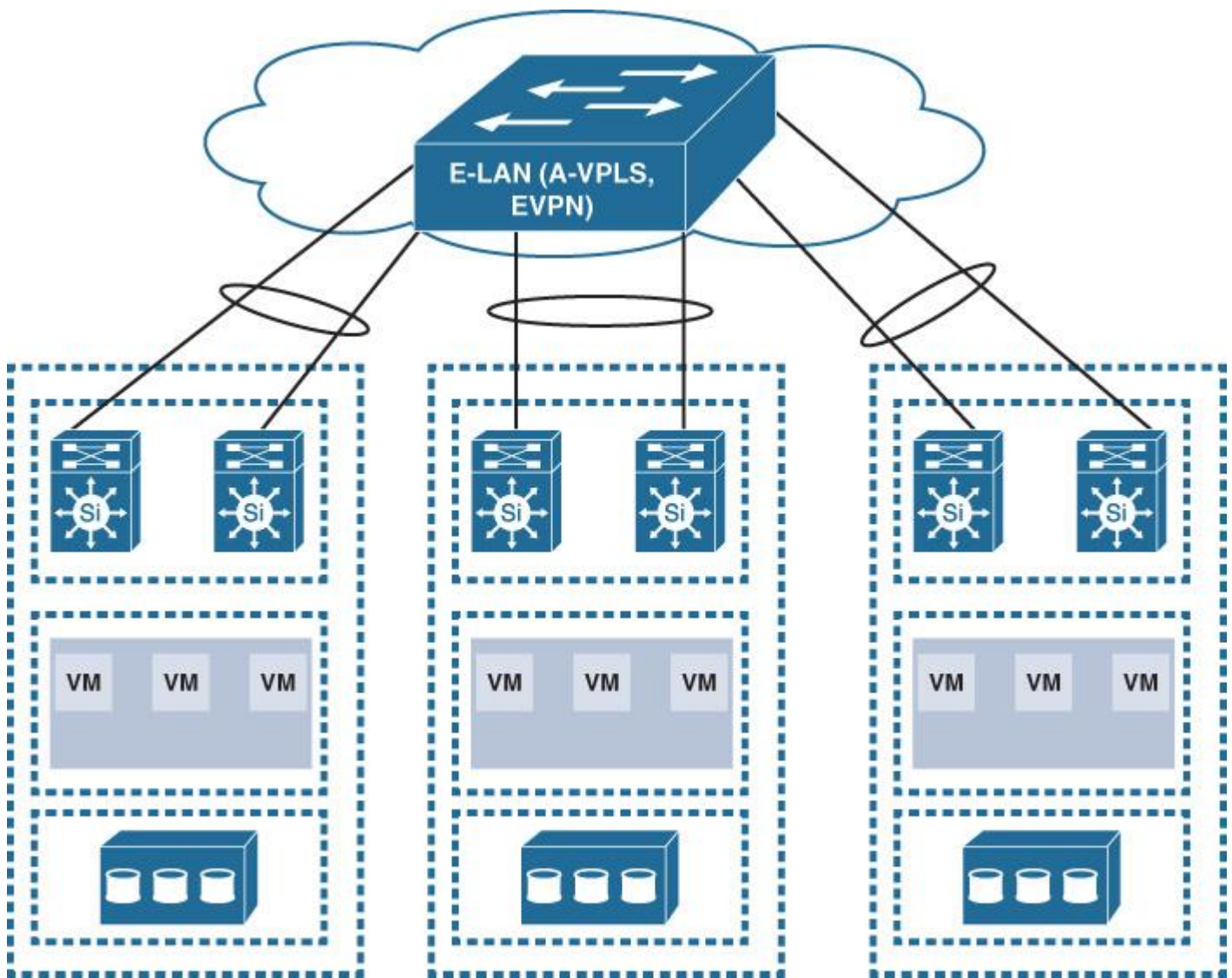


Figure 8-26 DCI over E-LAN

TRILL-FabricPath-Based DCI

As discussed earlier, the MDT in FabricPath (FP) controls the forwarding of multidestination traffic. Normally, a root for each MDT is assigned per FP topology in a multipotology environment. From the perspective of design performance and efficiency within a single LAN/DCN environment, this is not a concern because the traffic is still within the boundary of the same high-speed LAN/DCN.

In contrast, when FP is extended over the DCI, this behavior can impact the entire DCI solution simply because the multidestination traffic (BUM traffic) will traverse the DCI link all the way to the MDT root that might be residing at the remote DC. This will result in high utilization (most probably congestion) to the DCI link, especially if there are applications using multicast. In other words, the placement of the MDT root with FabricPath enabled over DCI is critical. The scenario in Figure 8-27 shows how multidestination traffic from host B's access switch (for example, multicast or unknown unicast) is sent back to DC1 to the MDT root then back to DC2, even though both host A and host B are located within the same physical data center.

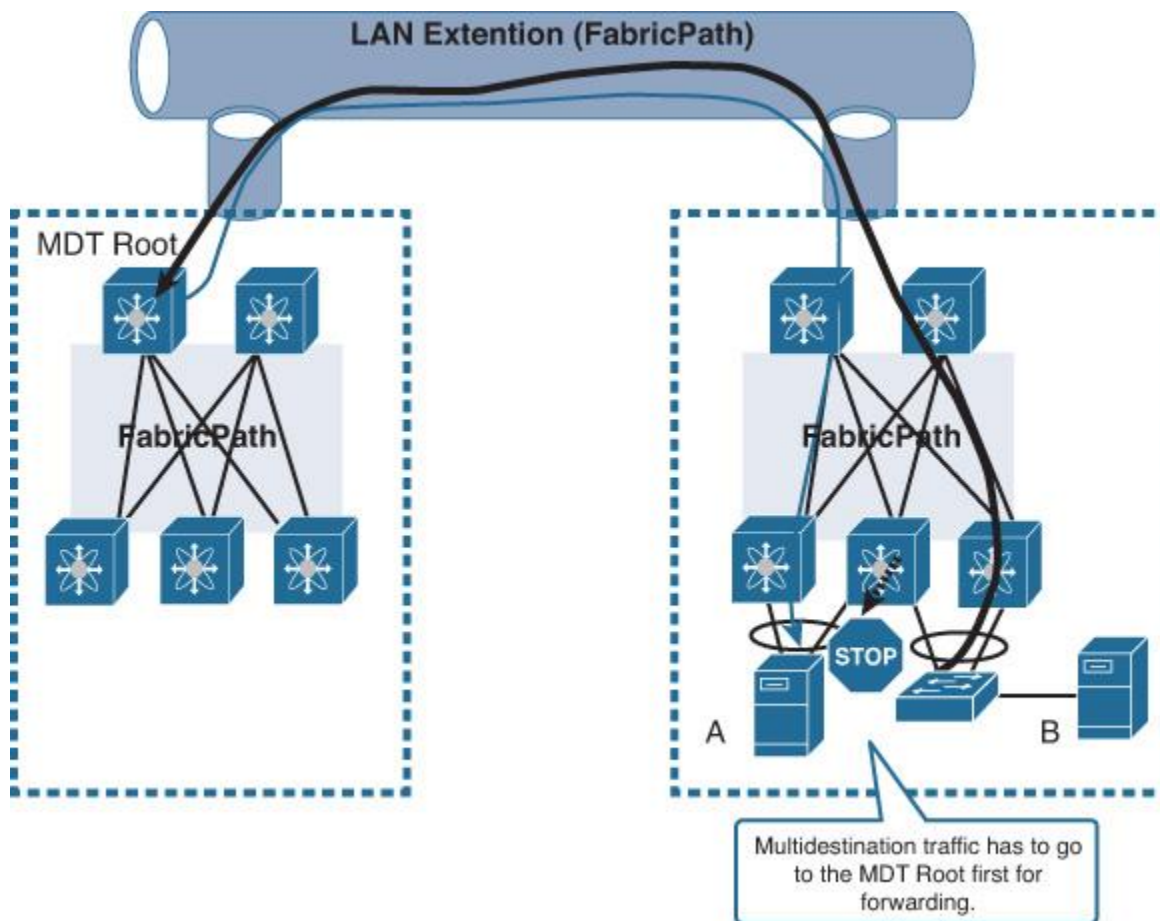


Figure 8-27 FP-Based DCI and MDT Issue

However, TRILL/FP can be considered a valid and feasible DCI solution in scenarios where the distance between the interconnected DCs is short, with connectivity over high-capacity links, such as 10/40/100-Gbps interconnections (ideally overprovisioned). In this case, traffic trombone may not be an issue, and TRILL/FabricPath can offer a scalable design of multiple interconnected data centers in terms of number of sites and MAC address table scaling, along with failure domain containment, as shown in Figure 8-28.

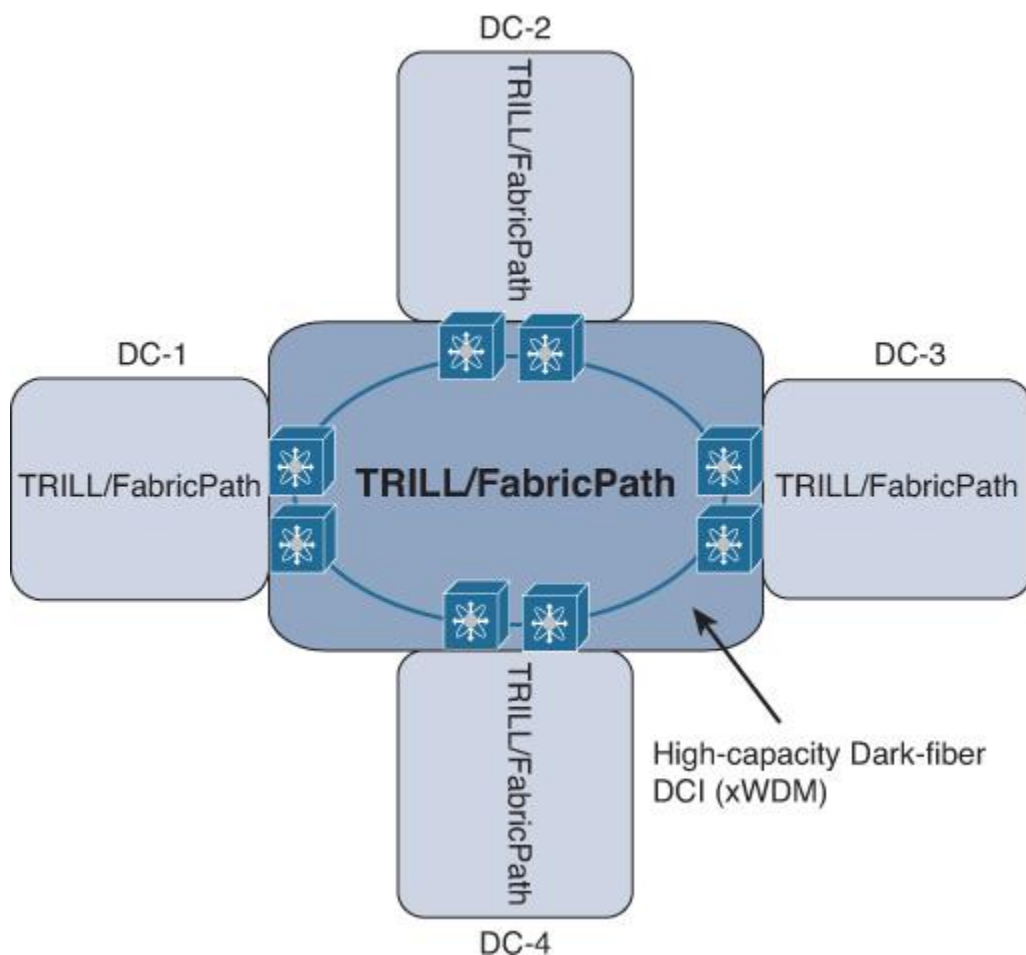


Figure 8-28 TRILL/FP Multiple Data Center DCI

Overlay Transport Virtualization

Overlay Transport Virtualization (OTV) is an IP-based functionality designed by Cisco Systems specifically to offer simplified and scalable Layer 2 extension capabilities over any transport infrastructure, such as Layer 2, Layer 3, or label switched. OTV introduced the concept of *MAC routing*, where IS-IS, as a control plane protocol, exchanges MAC reachability information between OTV edge nodes, providing seamless LAN extension functionality. Furthermore, OTV provides a native built-in multihoming capability, increasing the HA of the overall solution.

As a result, OTV provides the capability to extend Layer 2 segments over multiple DCI edges of multiple data centers without introducing any loop or unicast flooding that can impact the effectiveness of the DCI solution and the overall stability of the interconnected data centers. Concurrently, STP is kept contained within each data center domain, as shown in Figure 8-29. Therefore, using OTV (when applicable and the choice is available) over any transport, including dark fiber, to provide LAN extension can simplify and improve designing, implementing, and maintaining the DCI solution, especially when multiple data centers need to be interconnected.

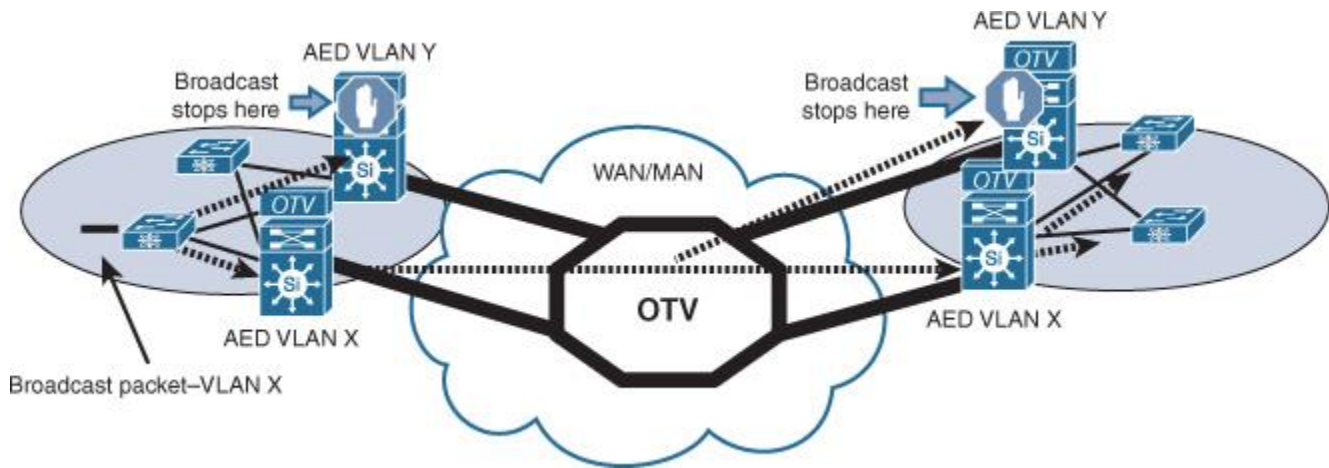


Figure 8-29 OTV-Based DCI

OTV selects an authoritative edge device (AED) for each group of the extended VLANs to control the MAC address advertisement for these VLANs and the ingress and egress traffic forwarding for these VLANs per site. As shown in Figure 8-29, only the AED of VLAN X will forward broadcast traffic sourced from the local site across the OTV DCI. Similarly, only the AED of VLAN X at the remote site will forward the broadcast traffic internal to VLAN. This capability offers an optimized behavior for an extended LAN over a DCI (mitigate the impact of broadcast storms across the DCI).

There are some scenarios involving large data centers with several PODs interconnected over Layer 3 links (such as Layer 3-based Clos fabric) per DC. In these types of scenarios, if the DC operator needs to extend Layer 2 VLANs across multiple PODs within the same DC and across multiple DCs in different regions, an overlay solution is required to achieve this, as shown in Figure 8-30. With the typical flat OTV design, this can be a challenging design, especially if many DC sites need to be interconnected. The OTV has limits in terms of maximum supported sites to be interconnected; this number varies based on the platform/software release.

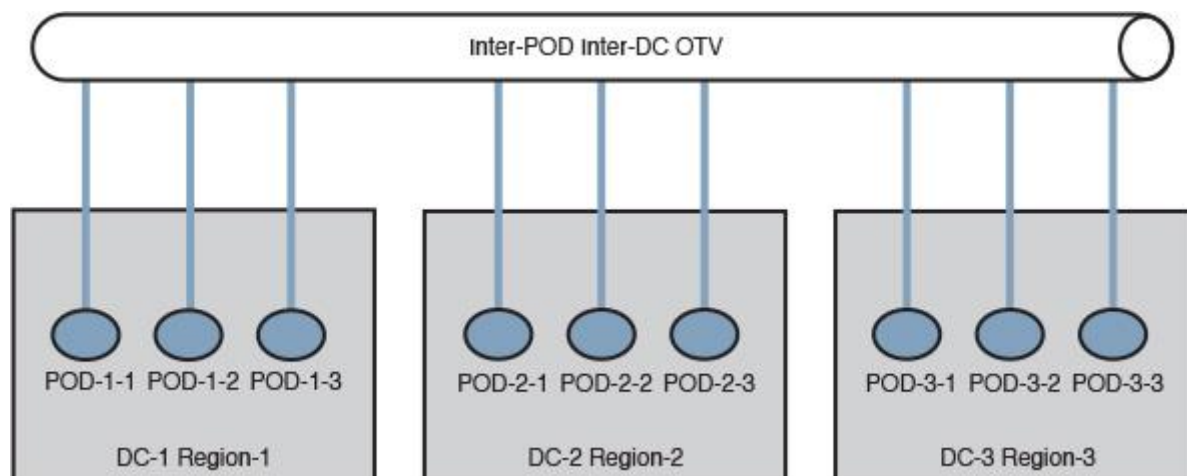


Figure 8-30 Single-Tier OTV Limitation

By introducing the multitier OTV design approach, network designers can accommodate the requirements of extending Layer 2 within each DC (across the different PODs) and at a global level (across different data centers in different regions) [60]. This design can overcome scaling limitations with regard to the supported number of interconnected sites with OTV and achieve a more structured Layer 2 extension, where VLANs can be extended selectively between the different tiers/regions, as shown in Figure 8-31. In spite of these advantages, this design approach requires careful path optimization considerations, which may lead to increased design and operational complexity. (The more data centers to be interconnected, the more complex to design and manage.)

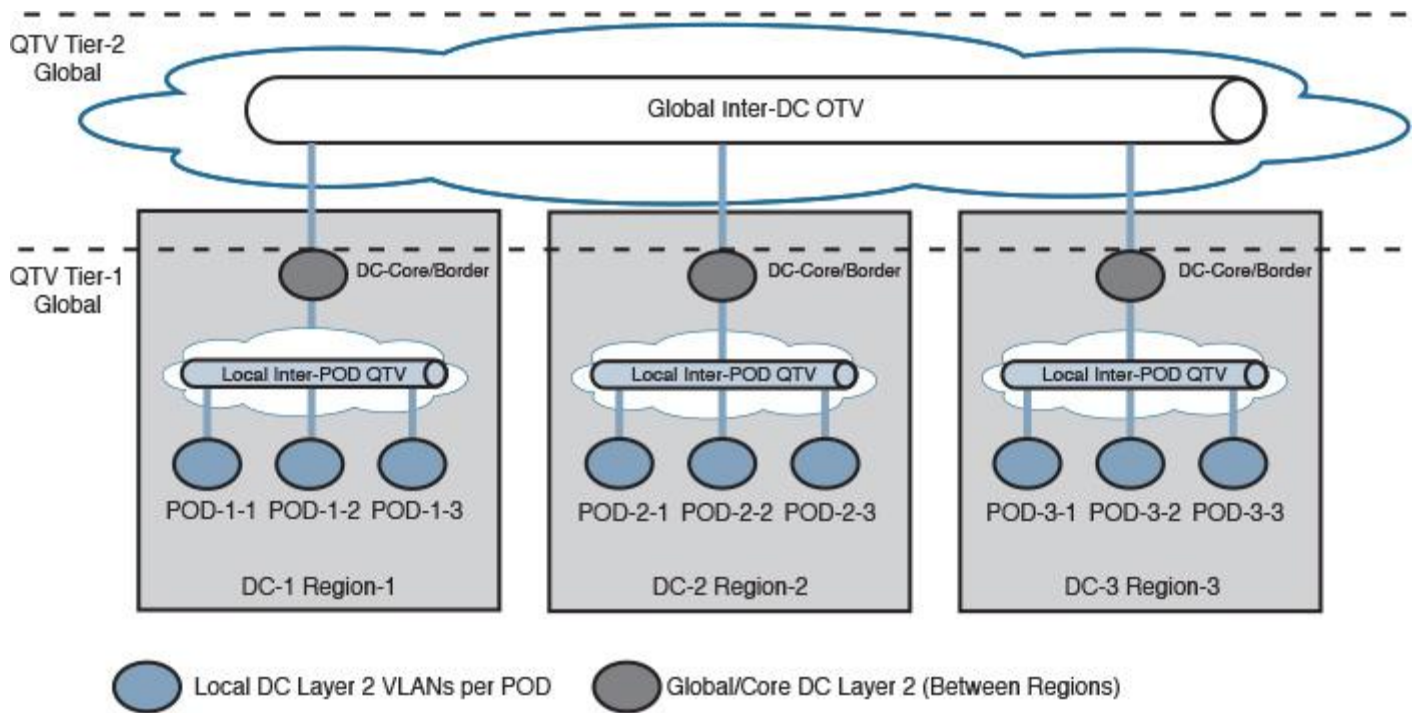


Figure 8-31 Two-Tier OTV

VxLAN-Based DCI

The IETF standard VxLAN was created and commonly used as a data center fabric overlay protocol to facilitate and optimize modern data center performance and agility in response to virtualized and cloud-based data center requirements. In VxLAN, VMs can be moved anywhere within the DC, because of abstraction between the location of the VM/application within the DC and the physical access/leaf switch and port VM mobility. VxLAN adds simplicity, transparency to applications, and flexibility to modern Clos fabric architectures; in addition, it enhances its ability to scale an extremely large number of VMs/MACs, especially when an MP-BGP EVPN control plane is used for VxLAN.

BGP as a control plane, together with VxLAN headend replication, can optimize VxLAN behavior with regard to multdestination traffic for MAC learning (broadcast and unknown unicast), especially as the behavior has become like Layer 3 routing (MAC learning is based on MAC address advertisement). Flooding will be eliminated unless there is a host that has not advertised or learned MAC, and only in this case will the typical Layer 2 flooding be used. At the time of this writing, VxLAN began to be considered a DCI solution, especially the unicast mode of VxLAN, because it can be more controllable at the DCI edge (can be policed more deterministically than multicast mode). This is applicable to any of the VxLAN design models (host based or network based), as shown in Figure 8-32.

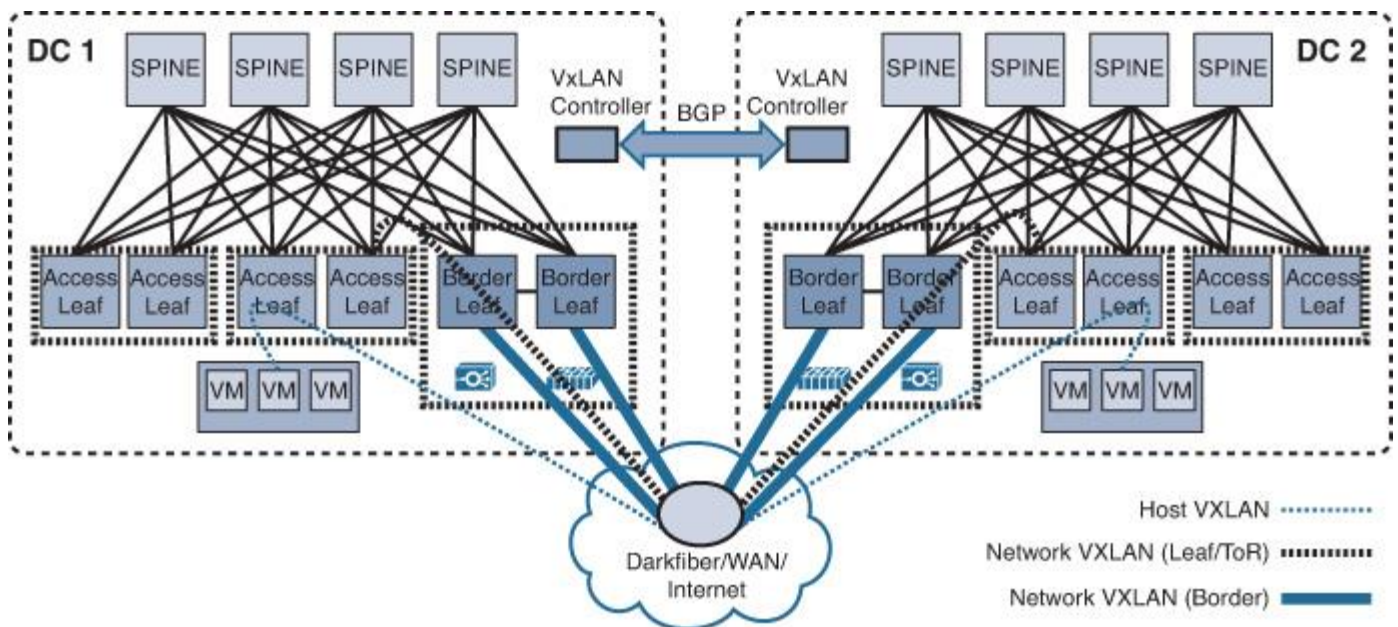


Figure 8-32 VxLAN-Based DCI

Note

The placement of the functionality of a VxLAN controller with BGP (EVPN) as a control plane (where all the hosts' MAC addresses are hosted and updated) can vary from vendor to vendor and from solution to solution. For example, it can be placed at the spine nodes of the Clos architecture, as well as cloud be placed at the virtualized (software-based) controller, such as the Cisco nexus 1000v virtual supervisor module VSM. It also can be used as a route reflector to exchange VTEP list between multiple VSMs.

Note

Technically, you can use all the VxLAN models discussed earlier (host and network based) at the same time. However, from a design point of view, this can add design complexity and increase operational complexity. Therefore, it is always recommended to keep it simple and start with one model that can achieve the desired goal.

Currently, one of the challenges of using VxLAN as a DCI solution is the behavior of the distributed controller across the two DCs following a DCI failure event (split-brain) and after DCI link recovery. In addition, considerations about the VxLAN gateway location (localized versus at a remote location) can impact the design. However, at the time of this writing, advancements in VxLAN are increasing rapidly. Therefore, it is quite possible that new features and capabilities will be available in the near future to make VxLAN a more robust DCI solution.

Note

The scope of each DC VxLAN can be contained within each DC locally, while the terminated VxLAN VTEPs at the DC/SP border nodes can be extended across the SP network that has EVPN enabled. Each VxLAN VNI is mapped to a unique EVPN instance (VNI-to-EVI 1:1 mapping) to maintain end-to-end path separation per VxLAN. This design offers a

scalable, integrated DCI solution that takes advantage of both solutions in scenarios where EVPN is already in place as a DCI and the DC operator is deploying VxLAN within each DC,⁵ as shown in Figure 8-33.

5. IETF draft, boutros-l2vpn-vxlan-evpn-xx, <http://www.ietf.org>.

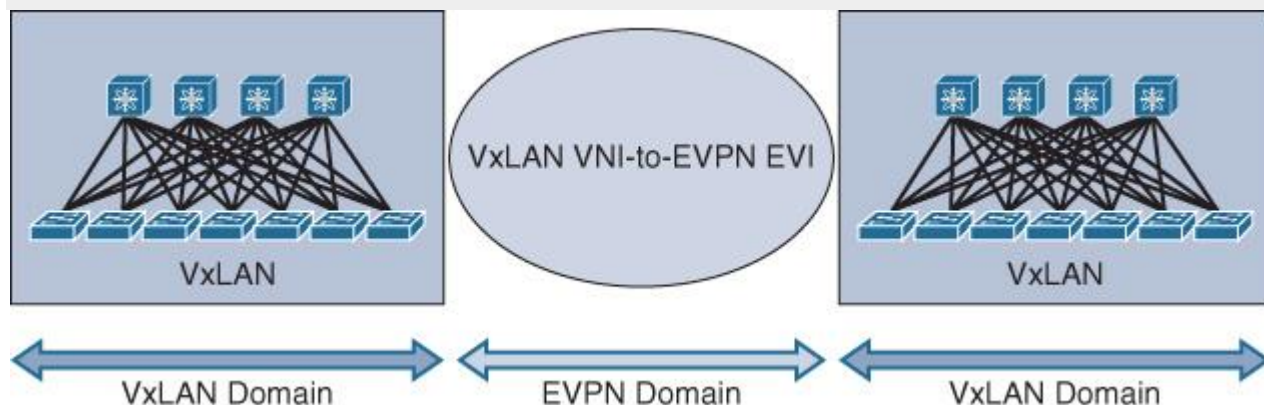


Figure 8-33 *VxLAN over EVPN*

Each of the preceding DCI solutions has different characteristics and strengths and weaknesses. Table 8-9 compares the DCI solutions from different design aspects.

Design Considerations	802.1Q over Dark Fiber				
	Fiber	E-Line	E-LAN	EoGREoMPLS (OTV)	VxLAN
Scalability (number of sites)	Limited (expensive)	Limited	High	Moderate	Limited
Transport independence	No	No (requires MPLS)	No (requires MPLS)	Yes	Yes
Multihoming and end-to-end loop prevention	Yes (limited between 2 sites with mLAG)	Yes (limited to 2 sites with mLAG)	Yes	Yes	Depends on the underlay DCI (for example, over mLAG)
Flow-based multipathing and load balancing	Yes (limited between 2 sites with mLAG)	Yes (PW/ EVPN based)	Yes (EVPN based)	No (load balancing VLAN based)	Yes - Depends on the underlay DCI (for example, over mLAG)
End-host detection	Flood and learn	Flood and learn (EVPN based will be localized flood and learn with Address Resolution Protocol [ARP] suppression.)	Flood and learn (EVPN based will be localized flood and learn with ARP suppression.)	Flood and learn locally and MAC routing across the DCI	Flood and learn (EVPN based will be localized flood and learn with ARP suppression.)

End-host reachability and distribution	Flood and learn	Flood and learn (EVPN based will be MP-BGP EVPN.)	Flood and learn (EVPN based will be MP-BGP EVPN.)	MAC routing (IS-IS based) over the DCI	Flood and learn (EVPN based will be MP-BGP EVPN.)
Localized L3 GW	No (requires manual filtering)	No (requires manual filtering)	No (requires manual filtering)	Yes (feature to turn on)	No - requires manual filtering
Location identifier	Port ID	VCID	VCID + VSI/ EVI	OTV edge node IP	VTEP IP
Host identifier	C-MAC	C-MAC	C-MAC	C-MAC	C-MAC
Site independence (STP, flood domain)	No	No	No	Yes	No
Multivendor support	Yes	Yes	Yes	Cisco proprietary	Yes
Operation complexity	High	High*	Moderate*	moderate	Moderate to high**
Commonly deployed***	Yes	Yes	Yes	Yes	No

* Operational complexity here has different layers. The first layer relates to the emulated solution itself, which depends on who is deploying and managing the solution. For example, if the VPLS solution is an enterprise self-deployed solution, the complexity is relevant to the enterprise and vice versa. The second layer is related to managing an extended LAN across two or more locations (large fault domain), which can add many complexities. Because of these complexities, the second layer requires careful planning and design for path optimization, as covered in the following section.

** The larger a VxLAN fault domain scope, the more complex to operate.

*** This point can be relevant to staff knowledge and expertise availability.

Table 8-9 DCI Connectivity Options Comparison

DCI Design Considerations

The complexity of designing data center interconnect solutions does not only pertain to the selection of the right DCI connectivity option (for example, Layer 2 or Layer 3). Several other components and considerations must be taken into account to achieve an optimal and reliable design. It is obvious from the design options discussed in the previous section that Layer 2 DCI has a wider range of connectivity options. At the same time, its design is the most challenging one. Therefore, it requires a careful and structured planning and design approach.

To achieve an optimal DCI design that is business driven and supports current and future requirements, network architects and designers must have a good understanding of the business goals and drivers with regard to BC. This is to be followed by a good understanding of the applications and services within the data

center and its architecture in terms of communications and network requirements (for example, multitier applications, data replication in use, and bandwidth and latency requirements).

For instance, if you are designing a data center that hosts public services, such as Internet banking services and applications, traffic spikes can increase rapidly because of an extremely large number of sessions, which are unpredictable to some extent. In contrast, designing a data center to host HR applications for a large-scale organization is more predictable in terms of the number of sessions. This relates to the expected design scale and bandwidth requirements to understand what these applications require (bandwidth, latency) and how they will behave during and after any DC failure scenario. In other words, the top-down design approach must be followed during the planning and design stages to achieve a successful DCI solution, as summarized in Figure 8-34.

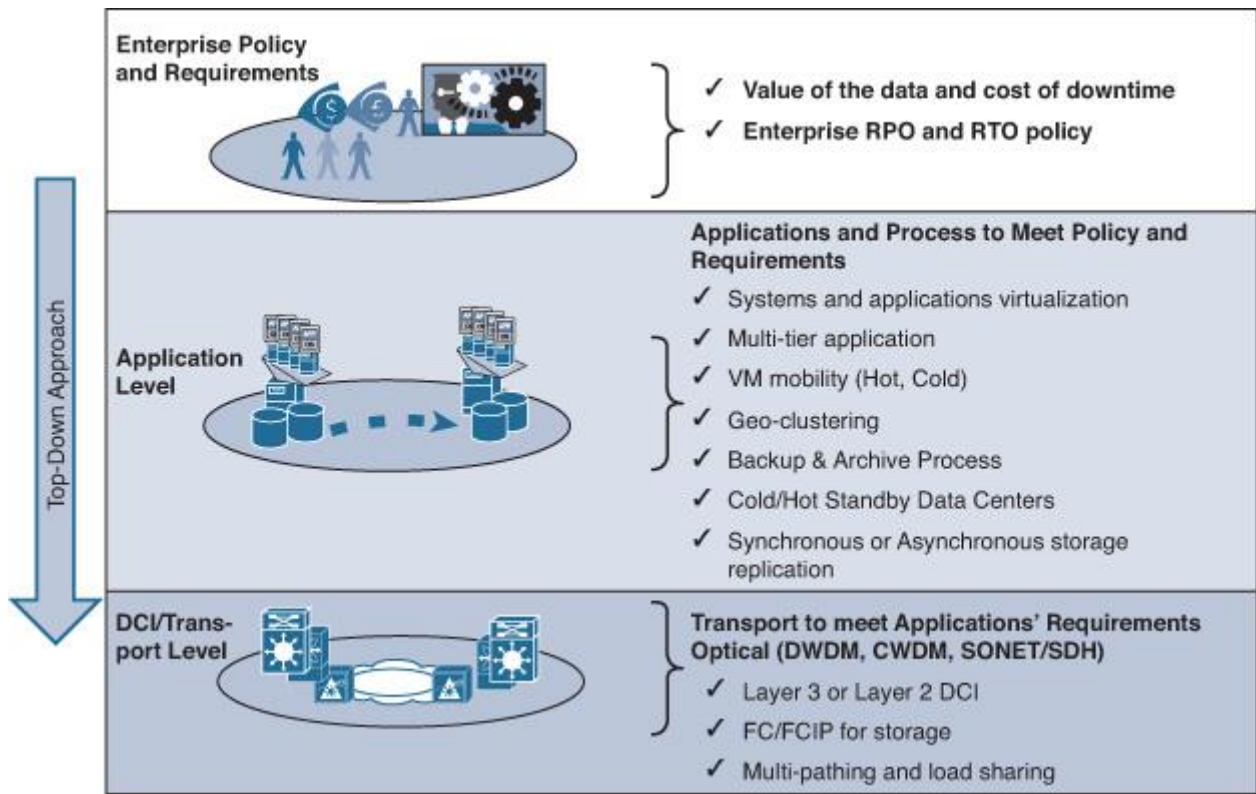


Figure 8-34 DCI Design: Top-Down Approach

The previous section focused more on DCI connectivity options and their characteristics (the focus was individually per DCI option). This section, however, focuses on the important design aspects of a DCI as a solution by highlighting the limitations and challenges and the recommended solutions that you can use to optimize the design. In general, an ideal data center interconnect design must consider the design aspects in Table 8-10 to achieve a scalable, stable, reliable, and flexible business-driven DCI solution.

Design Aspects	Design Considerations
Control plane hierarchy and federation within and across domains	Avoid merging large networks' DCN control planes for both Layer 2 and Layer 3 by clearly delineated fault domain boundaries. (For instance, STP must be constrained within the local DC, also IGP flooding domains should be contained within each DC boundary)
Storage/data protection	Storage-area network (SAN) extension method to facilitate data replication between the interconnected sites to maintain the desired RTO and RPO levels.
Reliable DCI solution*	Ideally to increase the DCI solution reliability and efficiency between two or more data centers, there should be loop-free redundant paths connected to redundant edge nodes/
Resiliency	Ability of the DCI to fail over to a redundant component following any failure event.
Path optimization (to support workload mobility and geoclustering)	Elimination or reduction of traffic trombone to the minimal, ideally by considering the following: Outbound: FHRP localization per DC for localized default gateway services Inbound: An intelligent path redirection such as LISP or GSS to localize ingress traffic
Flexible	Ability to adapt to business and technology changes such as support virtualization, multitenancy, and SDN.
Scalable**	Ability to support growth in terms of traffic volume, hosts, and the number of interconnect DCs through the selected DCI connectivity option and fault-isolation techniques

* The level of the DCI reliability and fault tolerance can vary from business to business based on different factors such as service availability, service level agreement (SLA), and the impact of application downtime on the core business functions.

** This also varies and depends on the targeted organization, goals, and scale; in other words, scalable DCI is not a must for each solution.

Table 8-10 DCI Design Considerations

This section covers the following design considerations with regard to DCI solution design:

- SAN extension
- Path optimization techniques with DCI
- Network and Security Services Placement

SAN Extension

The protection of the data via replication, caching, or backup is foundational to achieving the desired level of DR, BC, and business resilience (BR), and must take into account the RTO and PRO levels required by the business. Figure 8-35 describes the relationship between DR, BC, BR and the level of data protection.

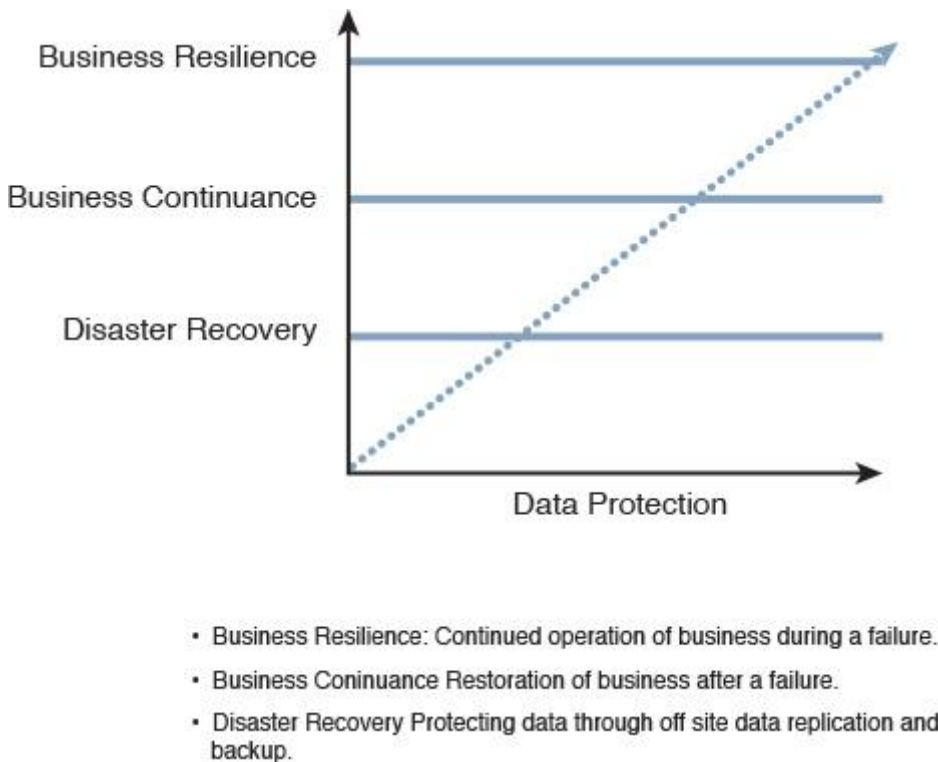


Figure 8-35 DR, BC, BR and Data Protection

Consequently, understanding the required or in-use data protection mechanism (for example, replication, caching) and how the application uses the storage is key to planning and designing a DCI solution. For example, some virtualization solutions are designed in a way that when a VM moves from one data center to another, the VM will continue use the storage located at the source/home data center.

In addition, moving application members across data center locations should not break the application. For instance, applications residing on top of the moved VMs need storage access (for example, database), as shown in Figure 8-36. Therefore, the storage design and replication can significantly impact the performance of the DCI. The dashed line in Figure 8-36 illustrates the scenario in which a moved VM continues to access data via the remote storage over the DCI link. The (nondotted) line shows the optimized solution in which the moved VM uses local storage when possible.

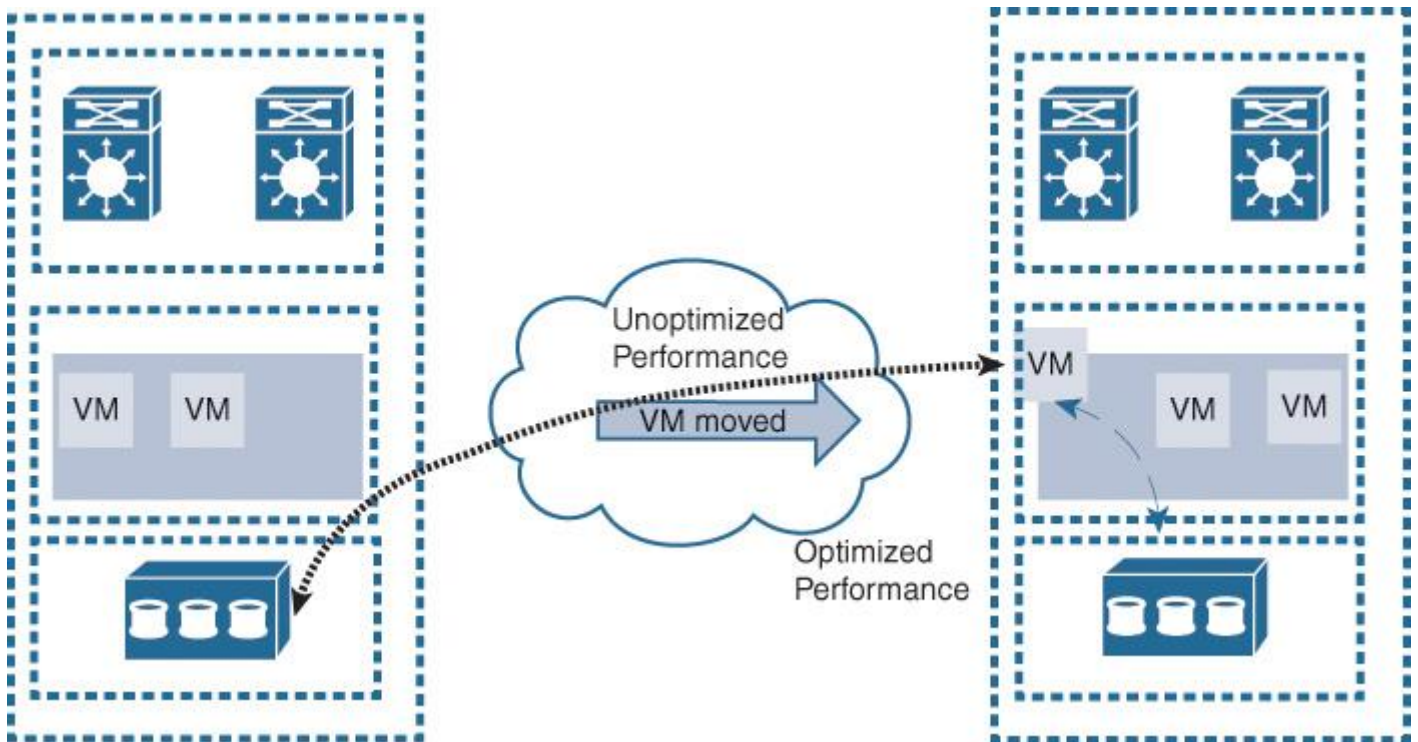


Figure 8-36 The Impact of Storage Access on DCI Design

Understanding this behavior can impact the DCI connectivity option selection in terms of distance (latency) and capacity.

Synchronous Versus Asynchronous Data Replication

Synchronous and asynchronous are the primary methods of data replications used in a majority of today's data centers; each has its own characteristic and influence to the overall DCI design:

■ **Synchronous data replication:** With this method, an application receives the acknowledgment of I/O completion when both the main and secondary sites are updated. This method offers almost zero data loss and very fast recovery time from failures that can occur at the main site. Appropriately, it is also known as the *zero data loss replication method* (zero RPO). A potential downside of this method is cost, because it can be an expensive choice with distance limitations because of its high-bandwidth and low-latency requirements.

■ **Asynchronous data replication:** Based on this method, an application receives the acknowledgment of I/O completion once the main disk is updated, while the copy continues to the remote disk array. Because the data is replicated to the remote array after the I/O is acknowledged as completed to the host, the remote disk array can be located anywhere, virtually or physically, at any distance. However, this method has a higher risk of data loss if the DC goes down before the data is completely replicated.

Furthermore, the distance between the interconnected data centers can influence to a large extent the selection of the data replication method that you can use on the disk arrays of the connected data centers, because the distance almost always impacts the response time (latency), which in turn will impact the maximum I/O per second (IOPS) performance across the interconnected data centers. However, if the application is designed to tolerate some IOPS, the distance impact may be reduced.

From the network and DCI design point of view, regardless of which method applications are operating with (synchronous or asynchronous), the design must identify the used method or methods [41,72]. Concurrently, it must take into account how the methods can impact the DCI design, especially with regard to factors such as bandwidth, latency, and distance. These factors must all collectively act as a solution to meet the DR and BC goals, from a business point of view.

Figure 8-37 summarizes the most common SAN extension options.

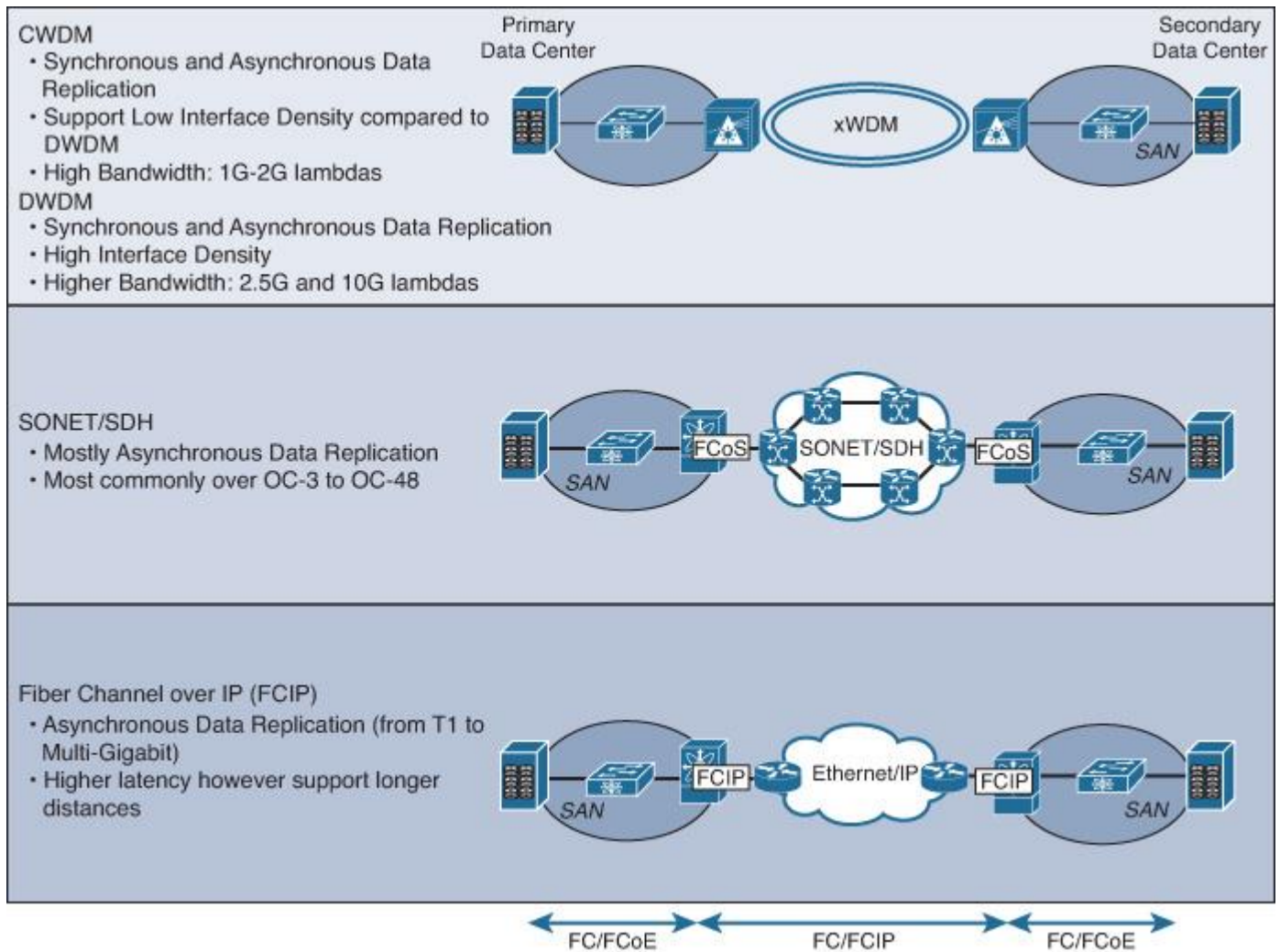


Figure 8-37 SAN Extension Options

Note

The highlighted data replication methods and each SAN extension option in Figure 8-37 are based on very generic link or application characteristics. However, these are not absolute, because as discussed earlier in this section, it can change based on various factors such as distance, latency, bandwidth, application attributes, and so on.

DCI Path Optimization Techniques

As covered earlier in this chapter, designing a DCI solution is one of the most challenging design tasks. For instance, selecting a DCI connectivity option, such as E-LAN, does not mean the network engineers can deploy it and everything will work fine to achieve the desired goals. In fact, selecting the DCI connectivity option is only the foundation of the DCI solution design. In other words, network designers should consider the behavior of the other components in the higher layers of the data center (storage, workload, applications, security services) when this DCI is enabled and when workload is moved from one location to another.

Throughout this process, several other considerations must be kept in mind. These include the targeted design goal of the DC (such as active-active versus active-standby) and the method through which external users should access the data center resources (ingress and egress traffic flow considerations). Therefore, the network designer needs to perform what is commonly known as *DCI path optimization*.

This section covers the different design considerations and techniques that must be taken into account when designing a DCI solution to optimize the traffic flow. In general, the traffic flow of DCNs can be categorized into three main categories:

■ **User to host:** This traffic flow usually refers to ingress traffic sourced from external or internal users to the local data center resources. The entry point of this traffic flow is normally controlled by the routes advertised by the DC network edge node.

■ **Host to user:** This traffic flow represents the return traffic or sessions from the server to the external users. The first exit point is controlled by the server's Layer 3 gateways, usually the FHRP VIP.

■ **Host to host:** This traffic flow represents the traffic flowing between hosts within the same physical DC or between interconnected DCs over a DCI link, such as a multitier application. (With workload mobility, this is also known as *east-west traffic flow*.)

In data center types of networks, packets often have to traverse multiple network services and security nodes in both ingress and egress directions (depending on the detailed design) before any given packet reaches its targeted destination (whether an internal host or an external user). These nodes are most commonly load balancers, firewalls, web application firewalls, intrusion prevention/detection systems (IPS/IDS), and DC edge routers. If you fail to take these nodes along the path into consideration, the entire data center design can be broken.

To a large extent, a good understanding and analysis of the three primary traffic flow types mentioned earlier can enhance the design by optimizing the path any traffic flow should take to avoid overloading the DCI link and breaking some application requirements. For example, asymmetrical routing can either lead to black-holing the traffic because of some security nodes in the path, or it may increase its latency. The subsequent sections discuss some of the impacts and possible solutions of these different traffic flows with regard to DCI path optimizations.

Understanding Multitiered Application Architecture

A thorough understanding of how multitiered applications and services within a data center work and communicate is vital for network designers during the planning of positioning or optimizing of a DCI solution. This is because with system virtualization and workload mobility, the physical host and IP localization is not associated anymore with a fixed physical location (data center).

For instance, several business applications, such as customer relationship management (CRM) and enterprise resource planning (ERP), consist of multiple elements interconnected in a sophisticated manner that construct the overall service or application. In addition, some of these elements may need to have a level of protection. For example, a web server might need to be placed behind a web application firewall, and the database behind a network firewall. As result, what is seen as a cohesive service or application is, in fact, a combination of multiple services and applications that communicate in the back end to provide the unified interface to the end user.

So to reiterate, a thorough understanding of how these services and applications communicate between each other can help the network designer make better decisions and provide a better DCI solution. What is meant by *decisions* here is not only related to the DCI connectivity type and routing design. It also refers to when an alignment needs to be done with other teams, such as server and application teams. With certain applications (multitiered), it is better to migrate them together rather than move only one element of the multitiered application. For example, in Figure 8-38, a web server (VM) that is part of a multitiered application consisting of a web server, application server, and database server was moved to the secondary DC.

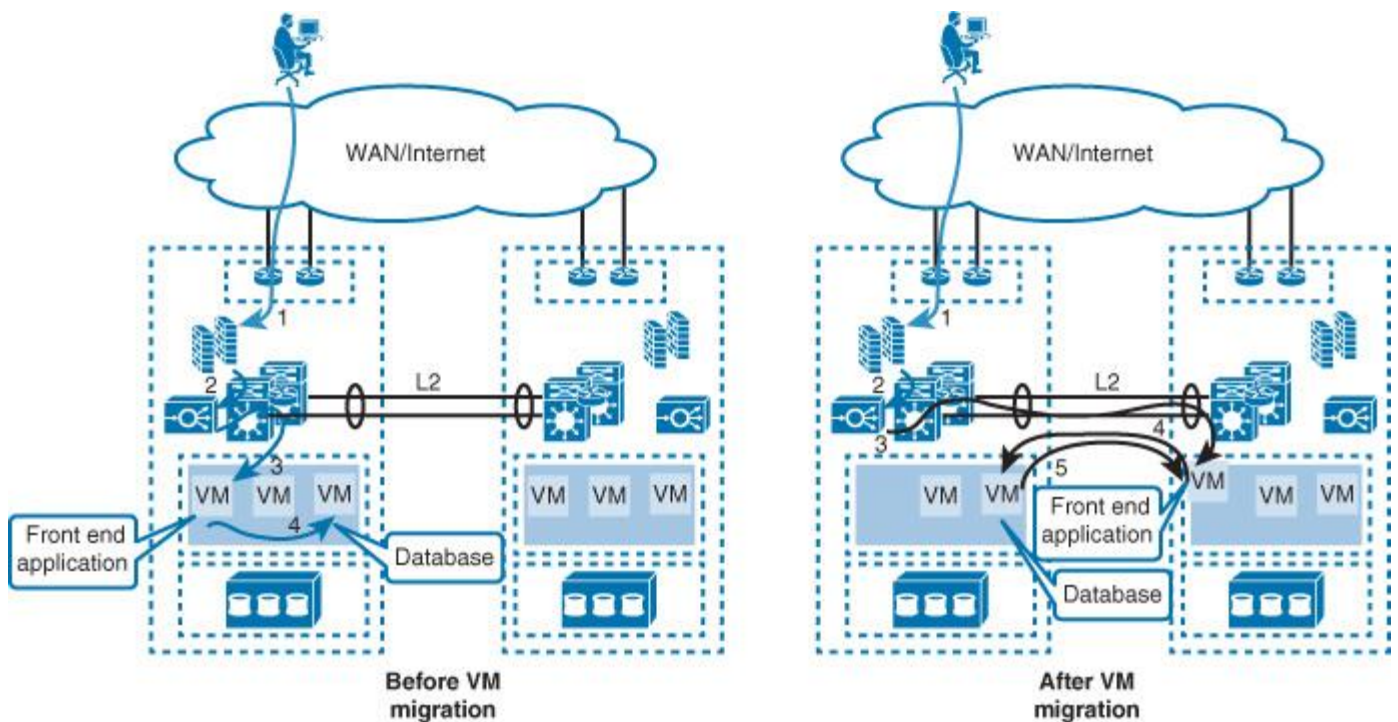


Figure 8-38 The Impact of Migrating Multitiered Applications on DCI

As shown in Figure 8-38 (the scenario on the left), before the front-end application is moved, VM traffic is routed and switched within the same physical DC (the traffic flow between different network and security components, also known as *ping-pong traffic*). In the scenario on the left, the effect of the ping-pong traffic has no impact on the latency because it is contained within the same DC. However, after moving the front-end VM traffic between the two data centers over the DCI, it can be increased. The ping-pong traffic, in this situation, can significantly impact the latency and efficiency of the DCI link. For example, traffic between each application tier will have to return to the primary DC where the active Layer 3 gateway and firewall are located. If additional network services are integrated between the different application tiers (which is the case in most of the deployments) such as load balancer, Secure Sockets Layer (SSL) offload, or web application firewall enabled, there might be about ten round trips (ping-pong traffic) for a simple web/DB query crossing the DCI. Consequently, by understanding how the multitiered application works, and to avoid the impact of this critical issue, it may be more feasible to move the entire multitiered application rather than only a single VM or application.

Note

To simplify the illustration, Figure 8-38 shows only one direction of the traffic flow.

Optimize Routed DCI with Stateful Security Services

As discussed in Chapter 4, this design must consider a direct interconnection between the edge routers connected to the external connections (physical or tunneled interface) to avoid any traffic black-holing in the event of WAN/Internet link failure of the active DC by the stateful nodes in the path such as the firewalls (usually each DC maintains its own independent pair of firewalls), which potentially leads to resetting the existing active sessions. This solution assumes that the time for the failover process to complete is less than the timeout duration of the active sessions. For a more detailed analysis, along with design considerations, see Chapter 4, the section titled “Asymmetrical Routing with Multihoming (Issue and Solution).”

Egress Traffic Flow Optimization Using FHRP Localization

One of the primary path optimization principles with regard to an extended LAN over a DCI is the localization of the Layer 3 gateway services (FHRP) of the outbound traffic flow by the moved workload between the different data center locations. With LAN extension, FHRP such as Hot Standby Routing Protocol (HSRP) will usually be active on the primary DC only, which will result in suboptimal (nonlocalized) egress traffic flow.

This might be acceptable in some scenarios where the interconnected DCs are located in the same building or campus and interconnected over high speed links, such as 10G or 40G dark fiber links. However, for large DCs with a high workload and interconnected over a private MAN, WAN, or over an SP-provisioned DCI, this can be unacceptable behavior. One simple and common solution is to use a VLAN access control list (VACL) or port ACL (PACL) to filter out FHRP over the DCI to maintain an active FHRP node per DC to process egress traffic flows locally, as shown in Figure 8-39.

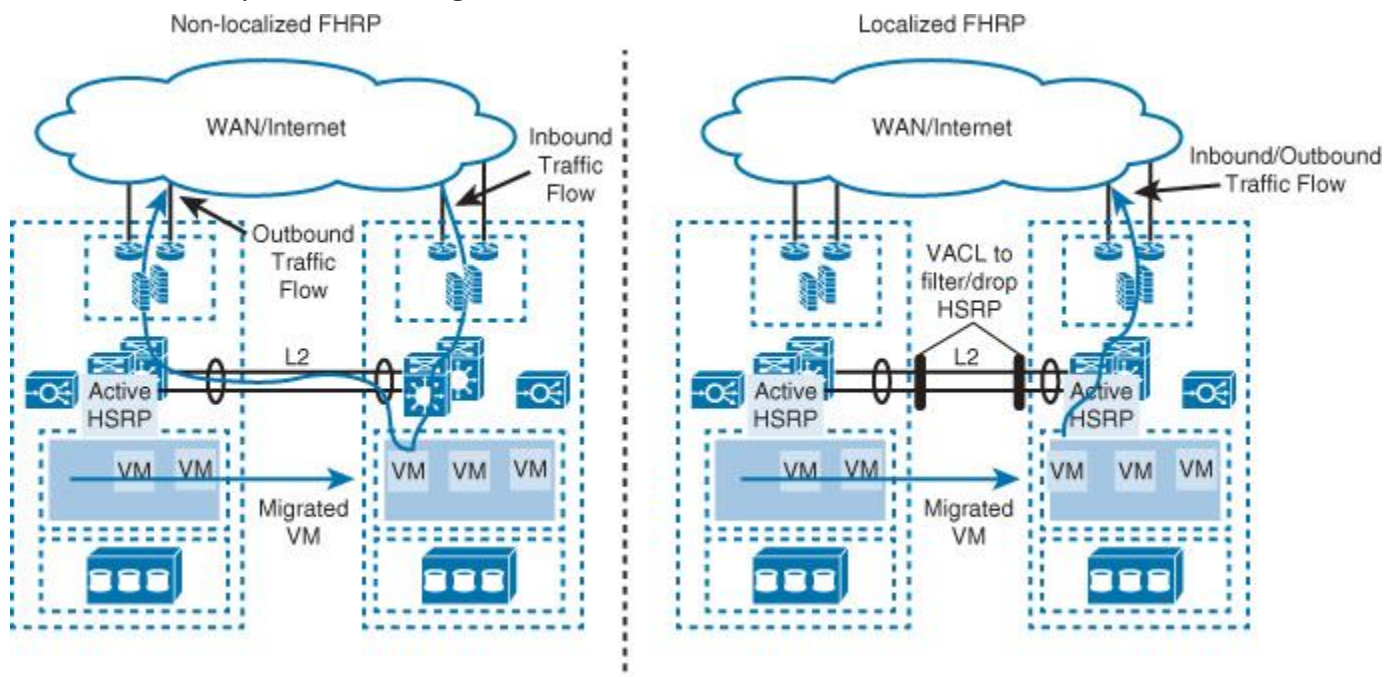


Figure 8-39 FHRP Localization

Ingress Traffic Flow Optimization

Optimizing egress traffic flow is only half the job in achieving acceptable and reliable path optimization for interconnected data centers. This section covers the most common techniques that you can use to optimize (localize) ingress traffic flow when the resources are moved or migrated between data center locations across a Layer 2 or Layer 3 DCI. Furthermore, these techniques can help achieve an active-active data center DR model:

- Domain Name System (DSN) based
- Route injection
- IP mobility (for example, LISP)

Figure 8-40 summarizes how ingress traffic flow optimization with interconnected DCs can decouple IP from its physical location.

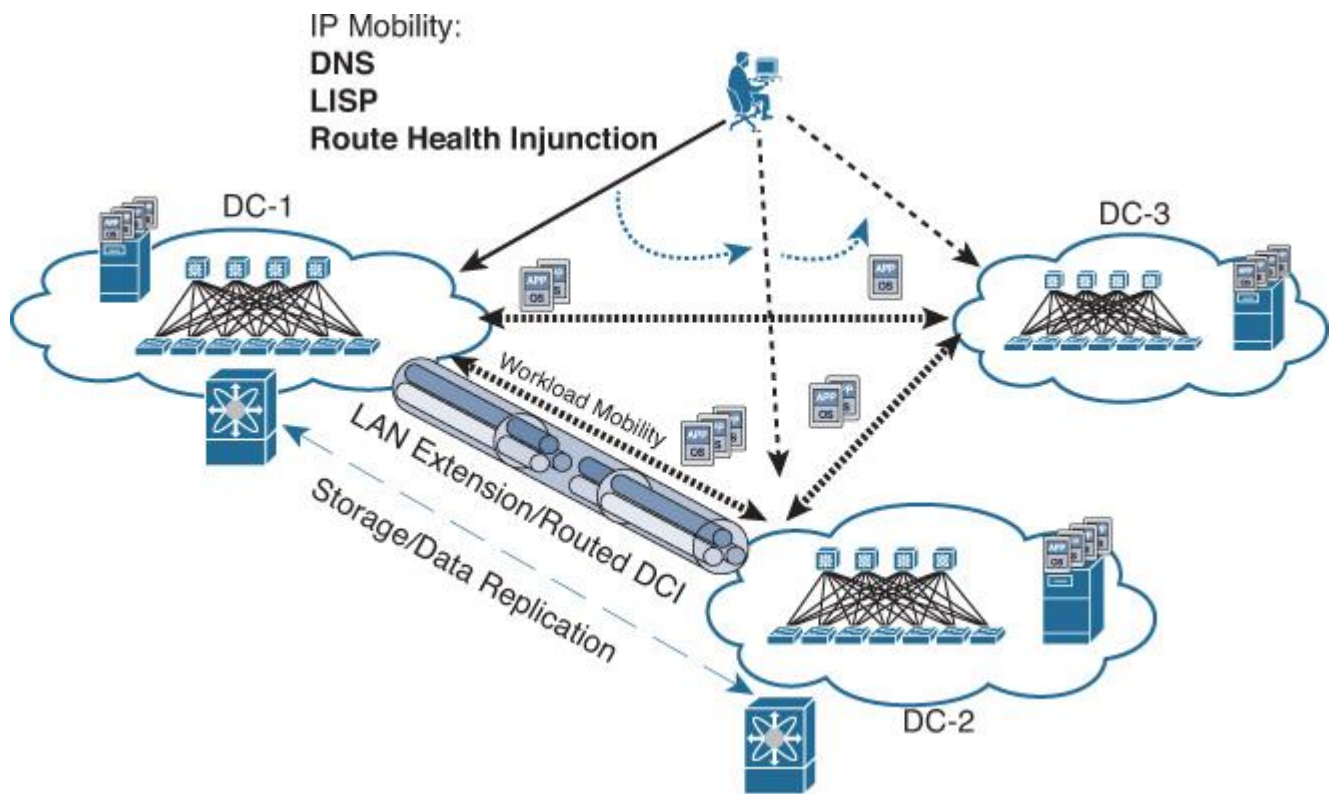


Figure 8-40 Ingress Traffic Flow Optimization with DCI

DNS Based

This design approach is one of the most common and proven approaches (but *most common* does not always mean the best). This solution relies on DNS (name to IP) redirection, most commonly deployed by using a global load balancer that acts as a DNS forwarder to resolve DNS queries to local (site-specific) load balancers per DC.

Normally, the global load balancer (also known as the *global site selector* [GSS]) performs keepalive checking with each DC load balancer while the local load balancers probe the real local servers. Hence, the global load balancer can populate the DNS entries with the appropriate IPs based on the keepalive response. In fact, the GSS can consider various criteria, such as response time and percentage of network utilization, in addition to weighting capabilities that can distribute traffic across different sites (active-active).

Therefore, this approach can achieve intelligent ingress traffic control and distribution across multiple DC locations facilitated by the GSS for the new ingress sessions. Existing sessions, however, cannot be redirected (even if the VM was moved) as long as the session is active, as shown in Figure 8-41. Although this behavior is most common with regard to GSS, load-balancing vendors may introduce new capabilities to overcome this limitation. Therefore, detailed traffic handling can vary while the overall design concept stays the same.

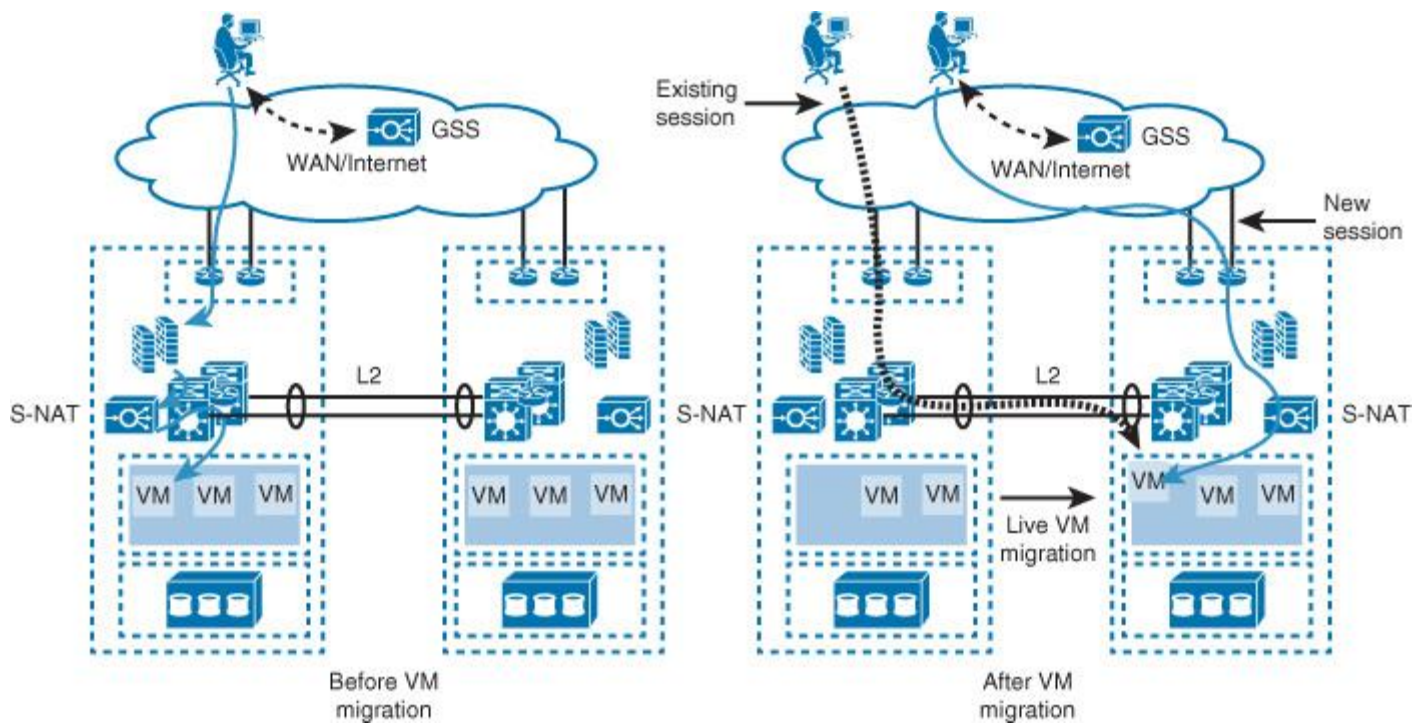


Figure 8-41 DN-Based Ingress Traffic Flow Control

Although DNS-based ingress traffic control is an easy and reliable solution, this design has the following limitations:

- Supports traffic/session that use DNS (name) only.
- Proxy cache of DNS entries can lead to traffic black holing until it is refreshed with a new DNS entry.

Note

The GSS concept described here is a generic concept that is available by leading load-balancing vendors.

With the DNS-based solution, Layer 2 extension over the DCI is not a prerequisite (unless it required by the applications or the distributed workloads). In addition, if a routed DCI in each DCI can maintain its own IP rang, this in turn will simplify the design by avoiding the egress traffic flow optimization considerations and other challenges related to the extended Layer 2 over DCI.

Route Health Injection

The route health injection (RHI) approach is also facilitated by load balancing (LB) nodes located across the interconnected data centers. With this approach, LBs are used in each DC to probe the local subnet/VLAN per host for an availability check with probe filtering over the DCI, as shown in Figure 8-42.

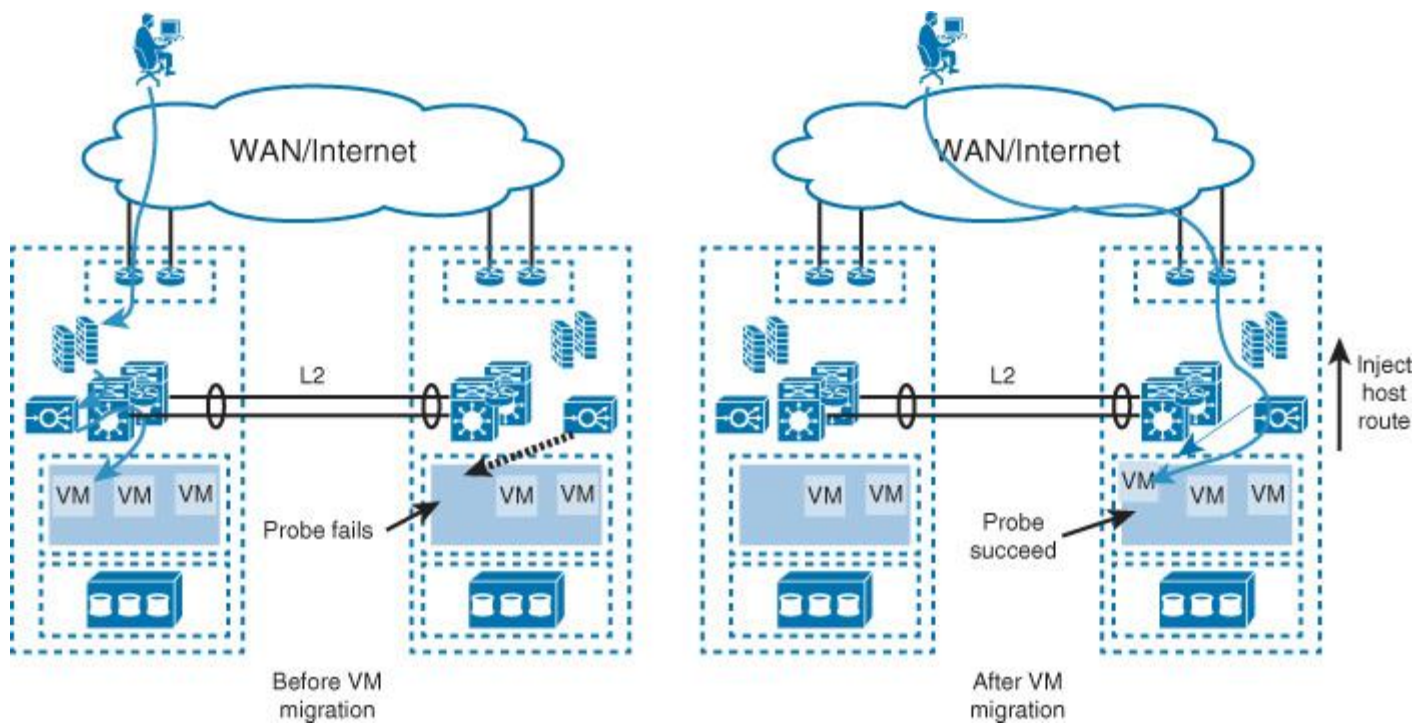


Figure 8-42 RHI

The scenario on the left in Figure 8-42 illustrates the DCs in a normal situation, in which the VM/server is located in the primary DC (DC-1) and the load balancer in DC-2 is monitoring the VM/host IP. At this stage, the probe should fail because the IP of the intended host is not located at the secondary DC. Once the VM has migrated to the secondary DC, the probe will succeed. This will trigger the load balancer to inject the host IP to the upstream edge routers (/32), which will redirect ingress traffic destined for this IP at the secondary DC (longest match principle).

Locator/ID Separation Protocol

LISP is developed to work in multihoming environments and facilitate communication in the IP mobility scenario in between LISP and non-LISP sites by separating IP addresses into two new numbering spaces: endpoint identifiers (EIDs) and routing locators (RLOCs), along with a dynamic endpoint tunneling encapsulation. Figure 8-43 summarizes the LISP architectural components.

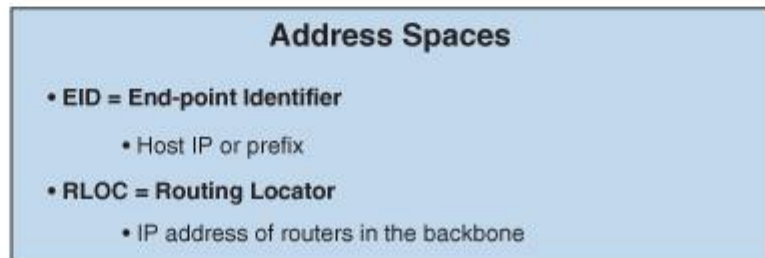
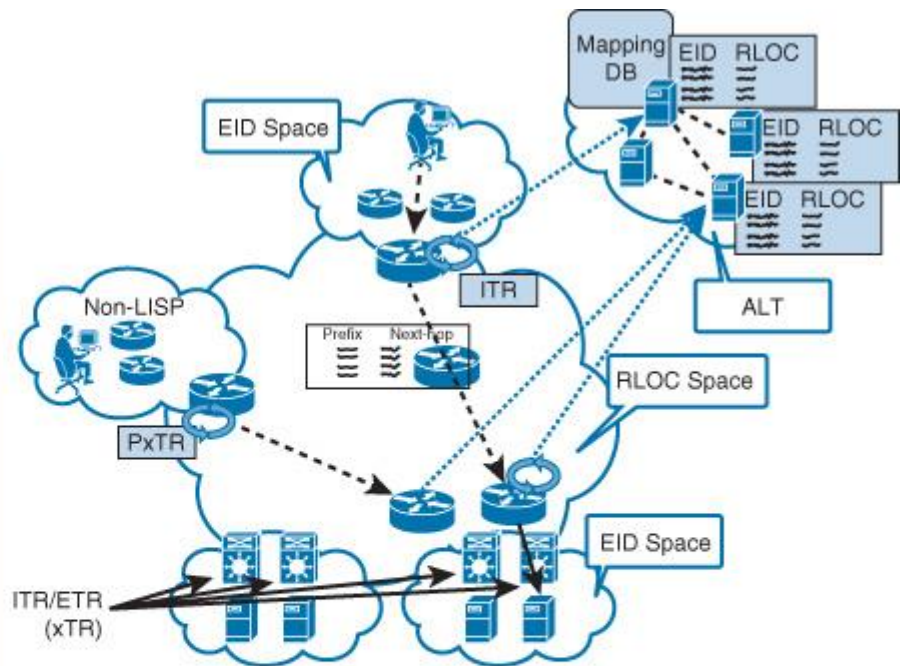
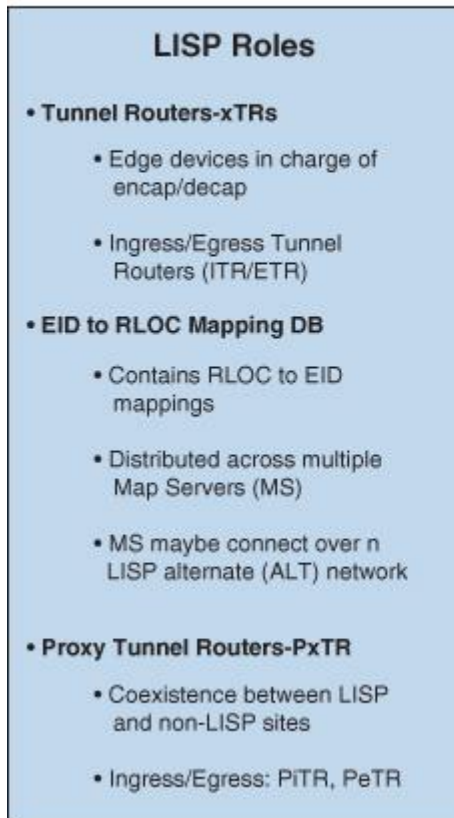


Figure 8-43 LISP Architectural Components

Note

Each DC edge node in Figure 8-43 of the LISP sites ideally should have both xTR roles (ITR and ETR) to correctly process both inbound and outbound traffic flows with regard to IP mobility to the internal EIDs. However, remote branches usually have only end users needing to access DC resources. Therefore, it is possible for the LISP edge nodes of the remote sites (benches) for only the ITR role to be deployed, in which return traffic or any traffic sourced from the DC LISP site will reach remote site subnets using the typical native routing. The LISP Mapping DB server role can also be deployed internally within the DC either by using dedicated nodes (in large-scale DC environments) or to be co-located at the DC aggregation nodes in smaller scale ones.

As an IP mobility routing mechanism, LISP offers data center operators the flexibility to either enable the transport over Layer 2 or routed DCI. In both scenarios, moved or migrated workloads (VMs) can maintain their original IP address. However, for LISP to deliver its promised value, remote sites must be LISP enabled, as much the nodes mapping between the non-LISP and LISP domain (PxTR), as shown in Figure 8-43.

Furthermore, although LISP offers data center operators the flexibility to use either the Layer 2 or routed DCI connectivity model between the different data centers, they do not operate exactly in the same manner with regard to workload mobility. That is why when migrating or moving VMs over routed DCI it is recommended (at the time of this writing) to target cold VM migration and not live migration. This avoids some ARP issues that can break the communications between the moved workload and other nodes (end users or hosts) after the migration. In addition, LISP supports multitenant environments that require path separation between networks within LISP sites and between non-LISP sites, as shown in Figure 8-44.

	Strengths	Design Concerns
DNS based	Supports multiple data centers No core networking design changes Proven and commonly used by many DC operators (staff knowledge) Provides flexible load balancing based on the traffic sources (for example, geographic load distribution) Reasonable operational complexity Supports multitenancy	The already established sessions to the “live moved” workload will traverse the DCI link. Failover time (DNS failover/flush) may not be desirable for some data centers with fast failover time requirements. Does not support DC applications and services that are reachable by IP only. Requires other network components to function (load balancers, global load balancers).
RHI	Simple to design and implement Offers an automated mechanism to detect workload move No or minimal core networking design changes Supports multitenancy	Because this solution is based on injecting the host route, it will probably impact Layer 3 node routing tables considerably and may introduce significant amount of churn in the routing protocol in large virtualized or cloud-based DC environments. Because of this, it can lead to instability or increased latency. Increased operational complexity, Requires other network components to function (load balancers). Limited scalability.

Mobile IPv4	IPv4-based solution Supports multitenancy	Increased design and operational complexity because there will always be a triangular traffic pattern combined with IP tunneling as one mobile IP moves between home and the foreign agent. Endpoints need an IPv4 mobility agent. Core network changes. Can lead to suboptimal traffic routing. Multicast is an issue with IPv4 mobility. Because of this described limitation, this solution offers limited flexibility; scalability will be associated with increased design and operational complexity. Limited to IPv4.
Mobile IPv6	IPv6-based solution Provides optimal data paths between server and client compared to mobile IPv4 Supports multitenancy	Supported IPv6 only (for both endpoint and the end-to-end data path). Core network changes. Endpoints must have IPv6 mobility agents installed on them. Increases design and operational complexity in large deployments.
LISP	No core network changes. Supports IPv4, IPv6, and hybrid mode as well (IPv4-IPv6 communication) Supports live and cold VM migration. Supports both Layer 2- and Layer 3-based DCI Supports communication with non-LISP sites Offers a scalable and flexible solution that can interconnect multiple sites Supports multitenancy	For this solution to work properly, the remote site should be LISP enabled (edge node to be xTR). Requires additional network components such as a mapping server. Network nodes need to support LISP (at least the participating nodes such as the edge xTRs). Increases the design complexity. Adds operational complexity to configure and troubleshoot. With L3 DCI, cold migration is supported. (Live is technically doable, but it will be associated with operational limitations.)

Table 8-11 Comparison of Ingress Traffic Flow Optimization and IP Mobility Techniques

Network and Security Services Placement

The main concern with these devices in the data center traffic flow is that these devices normally operate in a stateful manner by maintaining information about the *state* of each specific traffic flow that passes through them. Therefore, to achieve smooth integration between data centers over a DCI without adding additional complexity, it is recommended to avoid stretching network service and security devices such as load balancers and firewalls, for the following reasons:

- Redundancy is usually controlled by two devices, whether they operate in active-standby or active-active modes. Stretching these devices across two different locations limits the maximum number of data centers to two.
- Maintaining a reliable connectivity for the failover link between the redundant devices (the physical interconnect between devices used for health check and session state synchronization) is difficult because of its sensitivity to latency. Any failure in this link can lead to a split-brain situation, which can bring both data centers down.
- Stretching the redundant devices can increase traffic trombone over the DCI.
- In addition, when considering LISP to steer flows between data center locations, special attention needs to be paid to the following concerns:
 - Because the LISP xTR nodes performs the User Datagram Protocol (UDP) encapsulation for the original IP packets, firewalls must be deployed *south* to the LISP xTR nodes to permit the application of security policies on the original IP packets.
 - If a firewall needs to be placed between the host default gateway and the xTR, LISP multihop host mobility functionality is required.

Summary

It is obvious that data center design combined with data center interconnect is one of the most challenging network design tasks. It requires the involvement of several teams and has to consider the different components collectively forming a data center environment. This chapter also covered how the workload and applications of today's data centers are driving the change of modern DCN architectures. In addition, various approaches to designing modern data centers were discussed and compared, with regard to the underlay and overlay protocols. Furthermore, this chapter stressed the importance of BC and DR models on the overall DCI design, while considering the different application requirements and how they can influence design decisions about DCI connectivity and traffic path optimization.

Further Reading

"Software-Defined Networking: A Perspective from Within a Service Provider Environment," <https://tools.ietf.org/html/rfc7149>

"VxLAN DCI Using EVPN," <http://tools.ietf.org/id/draft-boutros-l2vpn-vxlan-evpn-04.txt>

IETF draft, "Extending TRILL over WAN," <https://tools.ietf.org/html/draft-xl-trill-over-wan-01>

"VMDC 3.0.1 Design Guide," http://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/VMDC/3-0-1/DG/VMDC_3-0-1_DG.html

"Data Center Access Layer Design," http://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/DC_Infra2_5/DCInfra_6.html

"Cisco Application Policy Infrastructure Controller Data Center Policy Model," <http://www.cisco.com/c/en/us/solutions/collateral/data-center-virtualization/application-centric-infrastructure/white-paper-c11-731310.html>

"Routing Bridges (RBriges): Base Protocol Specification," <https://tools.ietf.org/html/rfc6325>

Part V: High Availability

Chapter 9 Network High-Availability Design

Chapter 9. Network High-Availability Design

Business application and service availability considerations in today’s modern converged networks have become a vital design element, especially now that many businesses understand the IT network’s availability impact and cost to the organization, whether it is an enterprise or service provider network. Practically, the cost of a service downtime due to a network outage can take different forms: tangible, intangible, or a combination of both. Some of the common losses that can be caused by network downtime (which normally impact the business either directly or indirectly) are as follows:

- Tangible such as data loss, revenue loss, and reduced user productivity
- Intangible such as degraded reputation, customer dissatisfaction, and employees dissatisfaction

Consequently, the availability of the different IT infrastructure and services becomes one of the common primary requirements of today’s modern businesses that rely to a large extent on technology services to facilities achieving their business goals, such as mobility, cloud-hosted services, e-government services, Internet of Things (IoT), and smart connected cities. Therefore, many of these businesses with converged next-generation networks are aiming to achieve the *five nines* of network and IT services availability when developing their network designs.

Note

The term *five nines* usually refers to the degree of service availability. Literally, it refers to 99.999 percent of the functional time of a system, which can be translated to time of downtime per month/year as listed in Table 9-1.

9's Designation	Downtime per Year	Downtime per Month
9.99 percent (three nines)	~ 8.76 hours	43.8 minutes
9.999 percent (four nines)	~ 52.56 minutes	4.38 minutes
9.9999 percent (five nines)	~ 5.26 minutes	25.9 seconds

Table 9-1 Service Availability and Downtime

Although it is hard to achieve the absolute five nines, it is becoming common nowadays for it to be targeted as an availability requirement in network service level agreements (SLAs). Achieving a true end-to-end five nines level of availability for a given service or application is not an easy goal to achieve because there are multiple influencing factors. For example, multiple factors impact the overall level of availability of any given application in a data center, such as network infrastructure availability, server component availability, storage availability, and all other components that are part of this application architecture on higher layers such as database, front-end application, and so on. Therefore, this chapter focuses only on the network infrastructure part and the design aspects and considerations that pertain to the network elements only (elements that helps to optimize the overall availability of the other elements carried over the network such as critical business applications).

Note

As discussed earlier in this book, the need for network availability is increasing with the increased adoption of converged types of network and service consolidation. However, several variables dictate the level of network availability. It is not always the case that every part of the network requires the same level of network availability. In other words, the high level of availability is not always a common requirement.

Although network availability is not always a common requirement, it must be evaluated based on different criteria such as level of criticality and business priorities, which can specify the level of impact of the downtime on the business. In addition, sometimes it is too difficult to define when the network is down. For instance, is network downtime based on a specific application's downtime, or is it based on specific part of the network, such as the data center? In fact, business requirements, goals, and the level of impact caused by the downtime with regard to these business priorities and goals can determine the standards of how network downtime is defined.

In general, the term *network availability* is an operations parameter reflected by the ratio of the time a network can be considered functional and performing as expected. Moreover, a system is seen as reliable when functioning without failure over a certain period of time. In fact, a system's reliability can be measured either by the level of its operational quality within a given time slot (when the system was up and running) until the system went to a down state, which is also known as *mean time between failures* (MTBF = Total operational time / Number of failures), or by calculating a system's failure rate (Number of failures / Total operational time).

The term *operational quality* here refers to some scenarios where a system might be technically up, yet is not performing its functions at the minimum required level. This means that the system not delivering the intended service or performing a function reliably even if is technically up. It is important to note that system reliability is one of the primary contributing elements to achieving the ultimate level of system or network availability.

The importance of system reliability is illustrated in the following case, where there is a network designed with a high level of availability (including redundant components). However, the interconnections between the core components of this network have low reliability. Examples of low reliability include the following:

- Unreliable line cards used at the core node.
- The transport medium is unreliable, and the links keep flapping.

Despite the fact that this network was designed with high-availability considerations, the low level of reliability of the core components in this network will impact the overall service availability.

However, the mean time to repair (MTTR) reflects the actual time required to repair a system from a failure condition. The following standard formula shows the correlation between availability, MTBF, and MTTR to calculate a system's availability:

$$\text{Availability} = \text{MTBF} / (\text{MTBF} + \text{MTTR})$$

Based on this formula, it is obvious that increasing the MTBF and decreasing MTTR can offer an improved overall level of availability. Thus, you can improve the availability of a network by optimizing the capability of its elements, such as the interconnecting transport links, to function without failures. A reliable network must be designed to tolerate at least a single component failure. Some designs may require the network to cater for multiple failures to meet certain design requirements. This usually translates into more redundant components, higher cost, and probably higher operational complexity.

Hypothetically, the more redundant paths we add to the network design, the higher the MTBF we can achieve. However, increasing MTBF is only half of the job. Therefore, network designers must consider how to reduce the repair time as well. Technically, adding more redundancy, which means increasing the parallelism here (redundant paths/links) leads to a higher MTTR at the network control plane layer. This will result in an increased control plane complexity and probably slower convergence time. In sum, for network designers to achieve the desired level of network availability, there must either be very highly reliable components end to end that reduce the probability of failure to a minimum or there must be redundant

components that can take over in the event of a failure (reducing the possibility of failures versus reducing complete network outages).

Many recent industry studies and reports show that the majority of system downtime occurs because of either a deficiency in design analysis, testing (proof of concept [POC]), or operational mistakes. The minority of system downtime results from hardware or software failures. Therefore, decreasing the possibility of service or network outages can be more efficient and achievable than focusing only on eliminating any system or network component failure possibility. Consequently, this chapter always focuses on how to balance between having a reliable and fault-tolerant design to achieve the intended goal in a resilient and effective manner.

Fault Tolerance

In general, network *fault tolerance* can be defined as the capability of a network to continue its service (no downtime) in the event of any component failure, with minimal service interruption using a redundant component that can take over in transparent fashion. Ideally, this should be unnoticeable by the end users or applications. However, the operating quality may decrease, depending on the considerations for the post-failure situation, with regard to whether the redundant component can fully handle the load or partially handle it.

In addition, some businesses deem it acceptable for their networks to stay operational with degraded quality, rather than facing a complete network failure. In other words, a network can be considered fault tolerant when it offers the capability to function (fully or partially) for the duration of any network component failure, usually over a redundant component. Commonly, the more network nodes and components that are added, the more likely it is for network high availability to be undermined. Therefore, network designers need to consider how to optimize these types of designs and reduce single points of failure to achieve the desired level of the network uptime. To achieve the desired level of network availability and a high level of resiliency to cater for different types of failure conditions, the following fault-tolerance design aspects must be considered:

■ **Network level redundancy:** It is also referred to as *network path redundancy*, and is essential to catering to link failures, such as fiber cuts, incorrect cabling, and so on. It is uncommon in LAN environments that end hosts connect with dual links at the access layer. Even though this will reduce the end-to-end reliability level, it is almost always acceptable as long as path redundancy is from the access layer onward (excluding data center high-availability [HA] considerations, because redundant connections are critical at the server access layer), as illustrated in Figure 9-1.

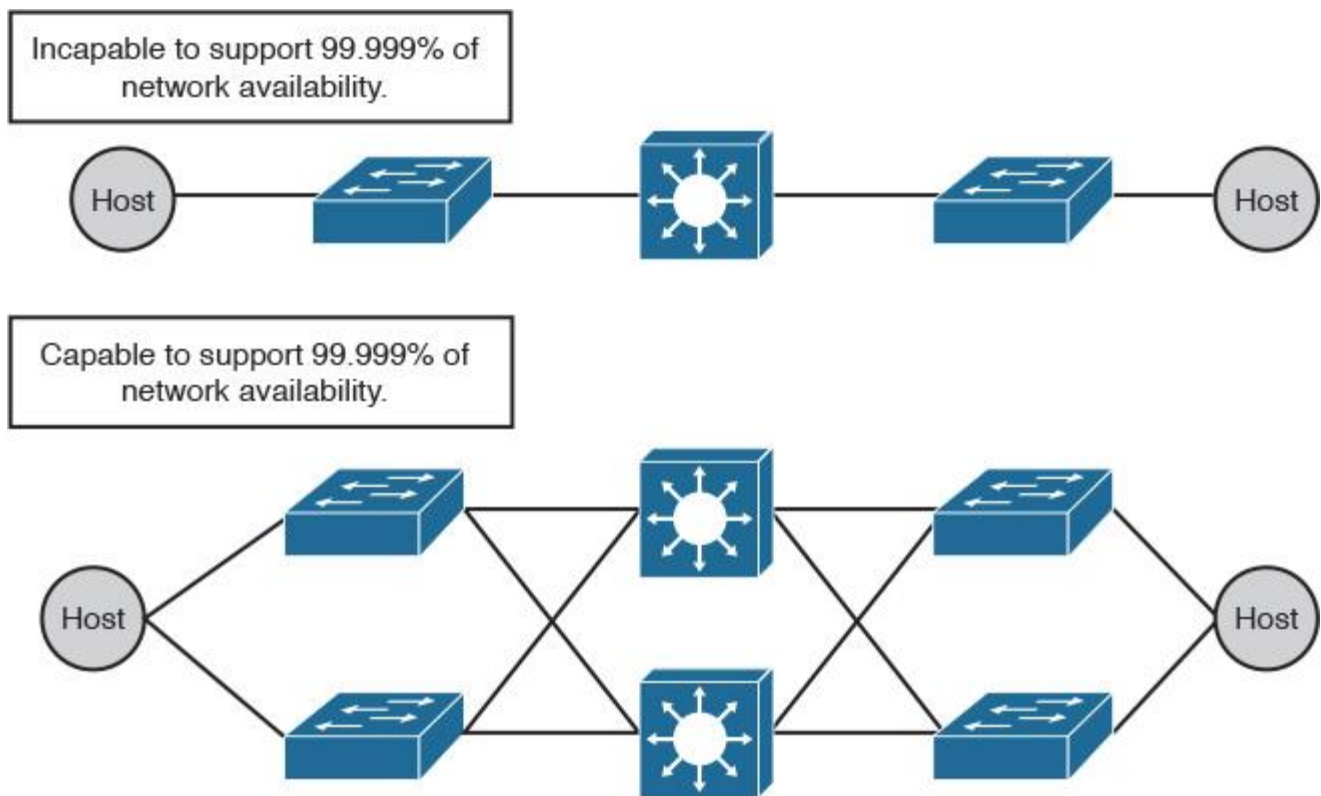


Figure 9-1 Network Path Redundancy

■ **Device and component redundancy:** Typically, redundant nodes provide redundancy during a node failure triggered by hardware or software. There are two types of node-level redundancies: redundant nodes and redundant components within a single device. You can use one or both types, based on the level of availability required and the targeted network environment (criticality). Considerations regarding passing traffic through the failure (for example, using redundant supervisors) versus passing traffic around the failure (using redundant devices) are primary influencing factors to the choice here as well. In addition, remember that redundant components (for example, redundant line cards for the uplinks) help to increase the overall level of system reliability. In general, redundant devices with diverse network paths offer better resiliency compared to a single device with redundant components. That said, a device with redundant components is more feasible for nodes that represent a single point of failure, such as a provider edge (PE) node in service provider networks.

■ **Operational redundancy:** Although the focus is almost always on how to optimize network redundancy and availability during unplanned failure scenarios, it is also important to consider how to provide redundancy during planned network outages, such as using in-service software upgrade (ISSU) or software maintenance update (SMU) capabilities.

■ **Network architecture:** The network architecture itself can facilitate, to a large degree, achieving an optimized level of availability. For instance, a hierarchical network design stresses redundancy at multiple layers to overcome any possible single point of failure. The other good example is "how" Clos architecture (TRILL or VxLAN based) helps to achieve large scale-out expansion in data center networks with several redundant components at the spines level without introducing complexity to the overall architecture. Furthermore, design redundancy at site or module level can help achieve optimized business and service availability for critical sites. A typical example of this is a disaster recovery (DR) data center with data center interconnect (DCI), as illustrated in Figure 9-2.

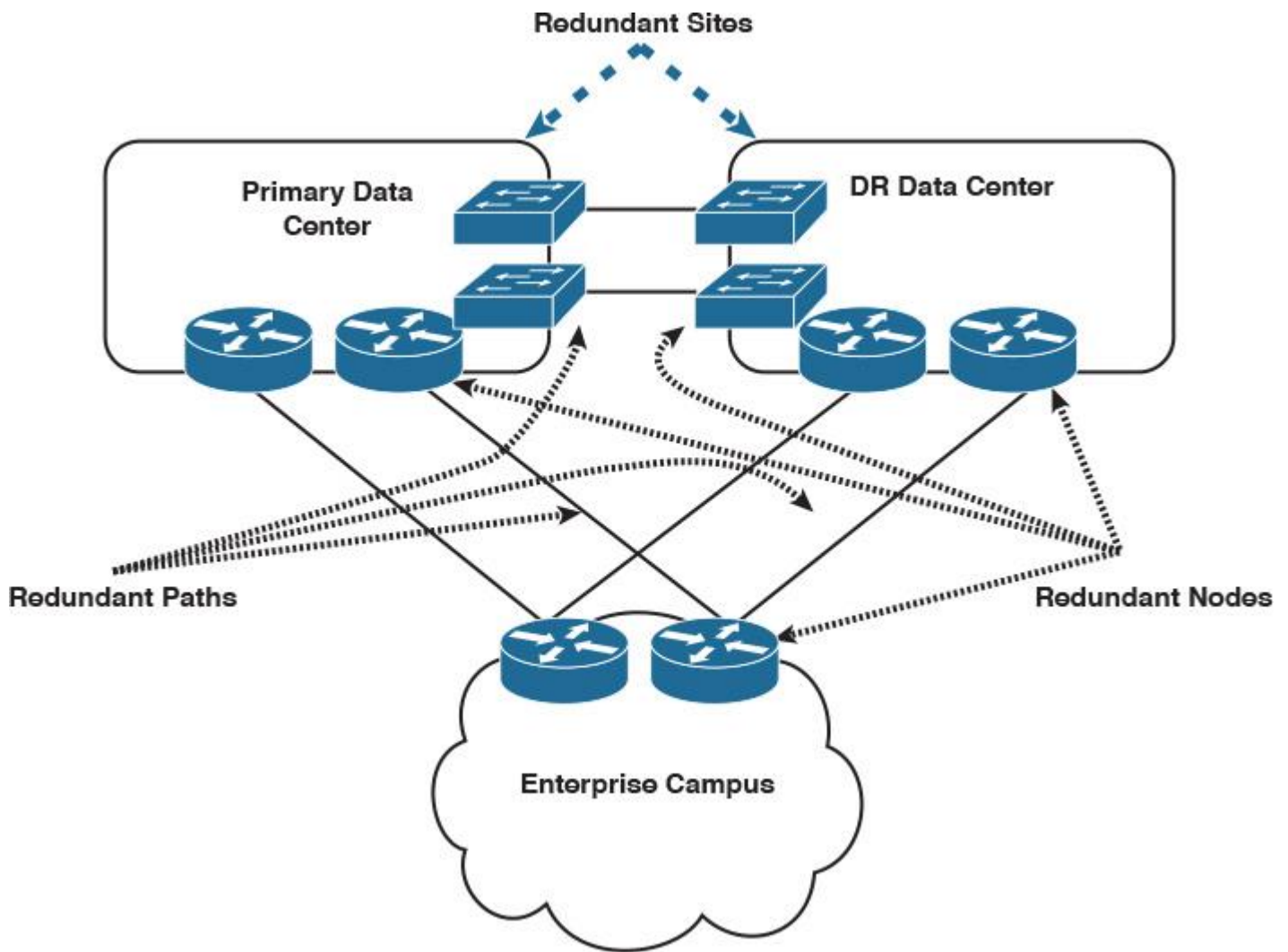


Figure 9-2 Redundancy at the Network Architecture Level

Note

In Figure 9-2, the network can operate in an active-standby or active-active manner using the available redundant paths/components.

Note

In any network, fault tolerance relies on different process and mechanisms to achieve the desired level of switchover to the backup component, as discussed in subsequent sections.

Fate Sharing and Fault Domains

It is important that network architects and designers understand the difference and relation between fate sharing and fault domains. *Fate sharing* refers to the philosophy of system design in which a single system is constructed of multiple subsystems or components, with a failure of one subcomponent leading to a complete system failure (sharing the same fate). For instance, a failure of the CPU of a smartphone will normally lead to a complete device failure as a result of a single subcomponent failure. Moreover, in the networking world, fate sharing also refers to the scenarios where multiple virtual networks share a single network infrastructure (more specifically, single network element such as a physical link with multiple subinterfaces where each subinterface carries a different virtual network).

However, *fault domain*, as highlighted earlier in this book, refers to a domain where multiple systems reside under a single domain (physical or logical) and a failure in one component can impact all other components within the same domain, such as flooding of updates or error messages. Let's consider the following example to demonstrate the impact and relationship between these concepts.

The scenario illustrated in Figure 9-3 shows two data center networks interconnected over a Layer 2 DCI. In addition, some virtualized networks are interconnected over the same DCI over a generic routing encapsulation (GRE) tunnel (with Multiprotocol Label Switching [MPLS] Layer 3 virtual private networking [L3VPN] enabled over the GRE tunnel). In this scenario, if any failure happens to the MPLS labels inside this tunnel, all the virtual networks passing through this tunnel will be impacted (sharing the same fate) as a result of a subcomponent failure of the entire MPLS L3VPN over GRE solution.

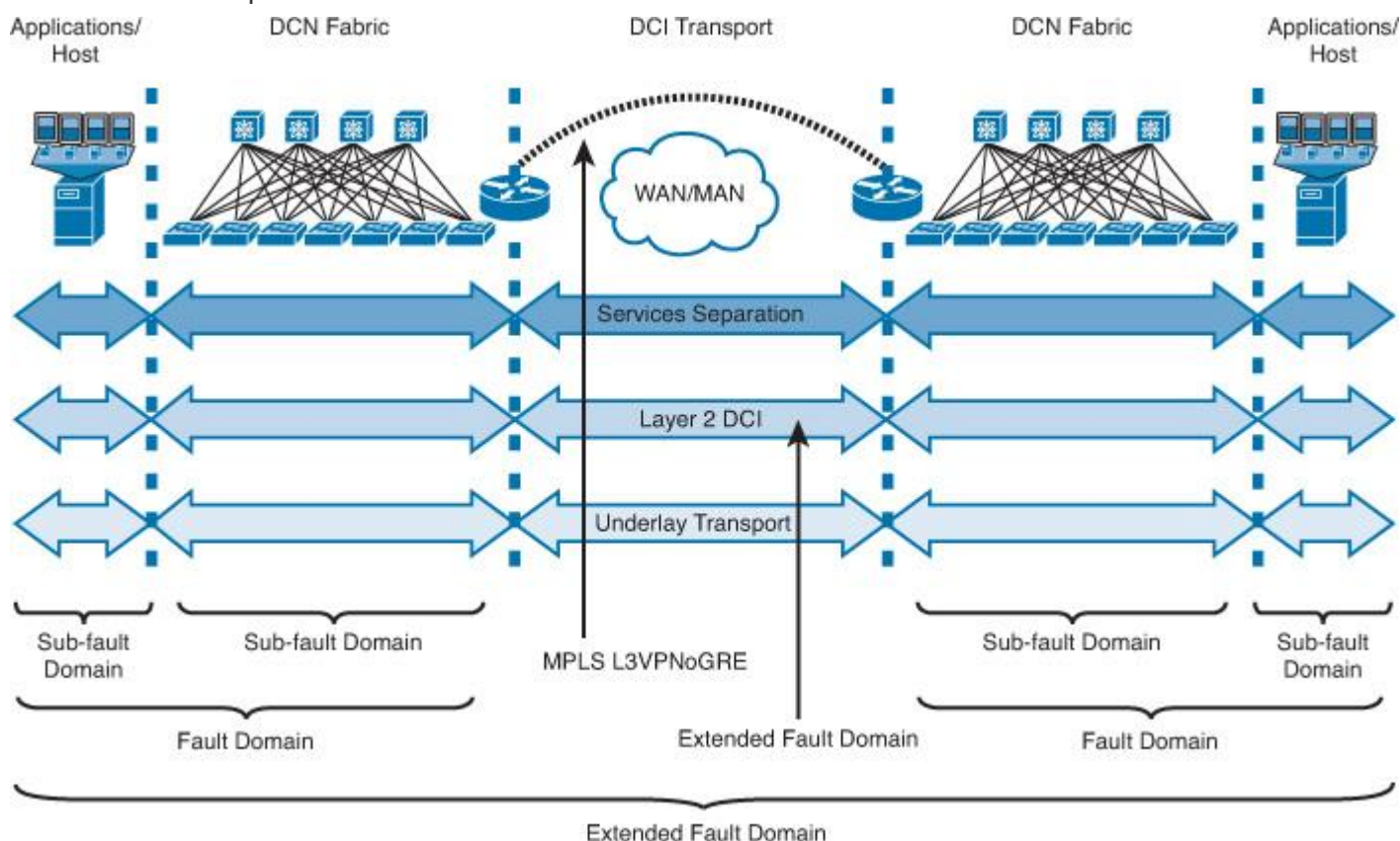


Figure 9-3 Fate Sharing and Fault Domains

Similarly, the failure of the DCI in this particular scenario will lead to the failure of the multiple virtual networks carried over the GRE tunnel and of the extended VLAN over the Layer 2 DCI, resulting in a fate sharing situation as well. Another common terminology applicable here is the shared risk link group (SRLG), which is commonly used in traffic engineering designs where multiple links share a single point of failure, in which all the links pass through one network node (for example, all the cables are going into a single device on the same line card). A failure of this node or any subcomponent (such as the line card where the links terminate) will usually lead to complete failure for all the networks that have their links passing through this node (the SRLG point).

Moreover, as shown in Figure 9-3, each Layer 2 segment in each data center (DC) represents a fault domain. Extending the LAN over the DCI, however, will in turn lead to extending the fault domain of each DC to the other one. For instance, any host MAC flooding in an extended VLAN in one DC will be propagated across the DCI to the other DC. Layer 2 flooding may lead to DCI congestion, which will usually impact all other traffic passing over this path, including the GRE tunnel. For instance, if a host or node fails in one of the DCs, in some situations this may lead to a significant amount of flooding within a single failure domain (in this scenario, the failure domain extended across the DCI), which most probably will lead to congestion of the DCI link. As a result, all the virtual networks passing over the MPLS L3VPN over GRE will be impacted as well. In other words, the impact of a failure in a fault domain can affect all the systems in that domain, unlike fate sharing, which is normally limited to a given system or subcomponent of the entire network such as the GRE tunnel (described in the example), or it can be a group of MPLS Traffic Engineering (MPLS-TE) tunnels passing through an SRLG point.

One of the common design principles used to generate a design that is reliable and resilient enough and that caters for different failure scenarios (like the one described in this section) is called *design for failure*. This principle is discussed in the following section.

Network Resiliency Design Considerations

Today's converged networks that carry voice and other real-time applications must recover in the event of network component failure in a few seconds (and sometimes within milliseconds) to effectively transmit traffic without affecting the end-user experience. Convergence times of less than one second could be highly desirable or a requirement for some networks, such as MPLS VPN service providers that offer virtual leased line (VLL) services. Therefore, network designers who aim to achieve the desired level of HA of network services must eliminate any single point of failure throughout the network. The network must recover during a component failure, such as link or node failure, to a redundant component that can continue the service operation. In other words, a redundant component that takes over in the event of a node or link failure must offer a fast network recovery (self-healing), also known as *fast convergence*, to maintain the desired level of network and service availability.

Furthermore, the redundant component should ideally take over within an acceptable period (for example, before mission-critical business application sessions restart) and handle the required functions reliably during the failure event. For example, you may have link- and device-level redundancy in your network that eliminates any single point of failure along the network path. However, in the event of any component failure, the network could take up to five minutes to recover from a simple failure to start using the redundant component. This may not be an acceptable resiliency level by the business, and this downtime in a next-generation converged network will impact many other services, such as mission-critical business applications and IP telephony services. Therefore, in this case, the availability goal from the business point of view is not optimally achieved, and so this does not justify the cost of buying and deploying redundant components.

The other factor that needs to be considered is how reliably the redundant component can function following a network failure event to provide a truly resilient network design. For example, one common scenario where the increased parallelism (redundancy) may not lead to optimized service availability is with an enterprise Internet edge designed with multiple paths (links) with the intention to maximize the overall performance and provide a reliable solution.

However, the concern in this type of design is when the business requires maximizing the utilization of the available Internet links to increase the return on investment (ROI) of these links and to improve application and Internet service performance, as illustrated in Figure 9-4. If one of the Internet links goes down, the second link, with a maximum of 10 Mbps of bandwidth, will start handling 20 Mbps during the link 1 failure, which will lead to an almost 50 percent degraded performance (assuming both links are fully utilized). Therefore, unplanned redundancy probably leads to an undesirable outcome following a failure event.

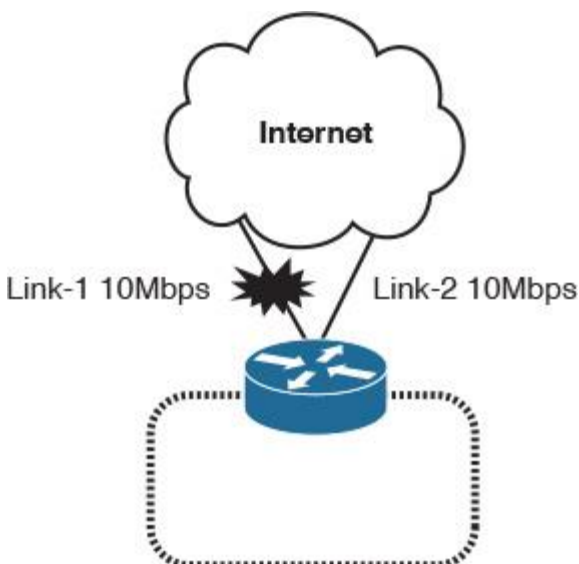


Figure 9-4 *Unplanned Failover Design*

For this enterprise to resolve this issue, it should add a third link to offload some of the traffic during any link failure. However, this solution will increase control plane and operational complexity and will add additional cost from the business point of view (capital expenditure [capex]). Alternatively, this enterprise needs to consider accurate link capacity planning, where each link utilization should not reach more than ~45 percent in a normal situation to cater to both links' bandwidth requirements during a link failure scenario. Although this solution will incur additional cost, it adds no operational and control plane complexity to the existing design.

Note

Both solutions are valid, and the goal of this scenario is to highlight the importance of the planning and the analysis of the different failure scenarios and its impact on achieving a true resilient network design.

Network designers must aim to create a network design that offers resistance to failures (reliable) and that can recover quickly and reliably in the event of a failure (resilient). In this regard, network designers should always consider the analytical principle used today by public cloud providers such as Amazon AWS Cloud, commonly known as *design for failure*, in which network designers ensure that the possible failure scenarios (including root causes and the failure scenario's impact on the business) have been analyzed and addressed within the design. Applying this design for failure principle facilitates the desired level of availability. It also helps network architects and designers identify and understand the impact and risk of different failure events on the business, such as the estimated outage period and to what degree this failure can disrupt business operations.

Consequently, when you design a network for failure with the objective of offering a highly fault-tolerant, reliable, and resilient network design, consider the following as you construct the foundation of the design architecture:

- Perform analysis of different downtime failure scenarios and their impact on the business's critical functions and data to be aligned with the business's expectations, such as recovery time objective (RTO) and recovery point objective (RPO) levels.
- Optimize network services availability on higher layers to the desired level that meets business and user expectations, such as web-application firewalls, Domain Name System (DNS), and load balancers.
- Every network component should be redundant (when possible) to eliminate any single point of failure across the network path.
- The network should be adequately capable of recovering in the event of a node or link failure with minimal or no service interruption during the recovery process, and without any operator intervention (resilient).
- The redundant component can continue to function at the desired level during a failure scenario (reliable resiliency).
- Classify and set recovery priorities from both business and technical points of view. (Critical functions and prerequisite services for other functions must always recover first.)

Note

The desired level of a network's availability or convergence time will vary from business to business. Therefore, for the purpose of the CCDE exam, only consider fast network recovery time if it is explicitly required or implied through other requirements.

When designing a highly available and resilient network that can converge fast enough without impacting any business-critical activities or applications, network designers need to consider a number of things. In other words, it is not limited to the scope of tuning control plane timers; in fact, it is an end-to-end process. Network designers need to consider all the network layers involved in this convergence process to achieve a true and effective convergence.

As a network designer, start by asking the following questions to understand whether the network is performing at the desired level, and to meet application requirements and business expectations:

■ **The characteristics of business-critical applications:** How much time does each business-critical application or service running on the network require before it loses its state or session? For instance, the CEO's voice call might be dropped because of minor packet loss (over 3 percent of packet loss) during network convergence time.

■ **Network reaction:** How long does the network take to converge following a component failure? You should consider end-to-end recovery time because the network might reconverge in some parts quicker than other parts and introduce microloops.

■ **Network security components:** Are there network security devices in the path, such as firewalls? If yes, are they redundant? How quick is its failover process? Is it a stateful or stateless failover?

Note

It is important to note that the target here is not only routing convergence, but the overall network convergence. As a network designer, always ask the question, "What can be done apart from the typical control plane timers to minimize loss in the event of a failure?"

This section discusses various mechanisms that network designers can consider to optimize network availability and convergence time to achieve the optimal targeted level of resiliency at different levels:

■ Device-level resiliency

■ Protocol-level resiliency

Note

As mentioned earlier, this chapter focuses only on network availability. In reality, the overall service and application availability must be measured end to end (at all layers; instance, if a voice control system [such as Cisco CallManager] is running on a cluster distributed across two data centers). In the event of the primary (active) data center failure, the network may reconverge in a few seconds, whereas Session Initiation Protocol (SIP)-based IP phones may take up to three minutes to register with the new active call control system because of a DNS failover delay (timeout duration).

Device-Level Resiliency

Device-level resiliency (also referred to as *component resiliency*) aims to maintain or keep traffic forwarding intact during a component failure at the network node level, such as routing processor failure. Therefore, it is critical for network designers to not get this concept confused with protocol-level resiliency, which usually aims to route traffic around the failed component. If routing protocols are tuned to converge quickly (subseconds), this will demolish the value of deploying device-level redundancy. Therefore, a balance between the device-level resiliency and the protocol-level must be considered when the device-level resiliency is a requirement.

Different protocols and mechanisms aim to offer a network design that can continue to forward traffic through a network node with a failure (such as a primary routing processor failure), with the assumption that a redundant routing processor is available on the same device. The following are the primary mechanisms used to achieve device-level resiliency:

■ **Stateful switchover (SSO):** Allows the standby route processor to take immediate control and maintain connectivity protocols

■ **Nonstop forwarding (NSF):** Continues to forward packets until route convergence is complete

■ **Graceful restart (GR):** Reestablishes the routing information bases without churning the network

■ **Nonstop Routing (NSR):** Continues to forward packets and maintains routing state

GR and NSR suppress routing changes on peers to SSO-enabled devices during processor switchover events (SSO), reducing network instability and downtime. In addition, the following protocol-specific considerations must be taken into account with regard to device-level resiliency design:

■ NSR is desirable in cases where the routing protocol peer does not support the RFCs necessary to support GR. However, it comes at a cost of using more system resources than would be used if the same session used GR.

■ For MPLS-enabled networks, you can use the following features to enhance network resiliency during a component failure:

■ Label Distribution Protocol (LDP) GR (RFC 3478)

■ LDP-IGP synchronization (RFC 5443)

■ LDP session protection (based on the LDP targeted hello functionality defined in RFC 5036)

■ Because NSR is a self-contained solution, it is more feasible for when the peer routing node is not under the same administrative authority, as is the case with PE-CE (customer edge) routers. In addition, NSR improves the overall system's reliability.

Network edge nodes such as PE devices in a service provider (SP) environment represent a single point of failure; therefore, features such as NSF and SSO, along with redundant components (redundant supervisors), are more common with these devices to provide redundant intra-chassis route processors and network resilience. Backbone network nodes, however, are more commonly deployed with protocol-level resiliency, in which network convergence depends on protocol convergence to an alternate primary path such as using tuned interior gateway protocol (IGP) for fast convergence or MPLS-TE fast reroute (FRR).

Note

Operationally, a major consequence and benefit of SSO is that adjacent devices do not see a link failure when the route processor switches from the primary to the hot standby route processor. This applies to route processor switchovers only. If the entire chassis were to lose power or fail, or a line card failure were to occur, the links would fail, and the peer would detect such an event. Of course, this assumes point-to-point Gigabit Ethernet interfaces, Packet over SONET (POS) interfaces, and so on, where link failure is detectable. Even with NSF enabled, physical link failures are still detectable by a peer and override NSF awareness.¹

1. http://www.cisco.com/en/US/technologies/tk869/tk769/technologies_white_paper0900aecd801dc5e2.html

Protocol-Level Resiliency

As highlighted earlier in this chapter, a resilient network is a network that can recover following any network failure event. Network recovery, however, refers to the overall required time for all network nodes participating in path calculation, and traffic routing and forwarding, to complete restructuring or updating network reachability information following a network component failure (for example, a link or node failure). This is usually performed at the control plane “protocol” level. The faster the convergence time, the quicker a network can react to any change in the topology. There are two primary network recovery approaches:

■ **Network restoration approach:** A reactive approach that computes an alternate primary path to reroute traffic over it after detection of a network component failure. Normal Open Shortest Path First (OSPF) behavior after failure of the primary path is a good example of this approach.

■ **Network protection approach:** This approach is a proactive concept that precomputes (or signals) a backup loop-free alternate path in advance before any failure event. In the event any failure is detected, the protected traffic is moved over to the backup path transparently (commonly within 100 ms or less).

In general, at any given layer in the network, network protection approach offers faster convergence than restoration. However, in some scenarios, it can be an expensive option, such as with optical layer protection. It may also introduce an added layer of complexity to the design, control plane, and the manageability of the network, such as using a full mesh of MPLS-TE tunnels among a large number of nodes.

That said, there is no single best approach; it always depends on the business goals and functional and application requirements to be satisfied. For instance, a five-minute recovery time might not be an issue for a retail business, whereas in a financial services environment, this duration may cost the business hundreds of thousands of dollars. Therefore, the cost of a failure here can justify the expensive but fast and reliable network recovery solution. The subsequent sections cover the two primary approaches of protocol-level resiliency in more detail.

Network Restoration

As described earlier, the concept of network restoration or convergence refers to the time required for network nodes to recompute their routing and forwarding tables and calculate a new primary path reactively following a network component failure. Convergence time is an essential measurement of “how long” routers within a routing domain can take to reach the state of convergence. In addition, network and protocol convergence can be seen as a primary performance indicator. However, achieving the desired level of convergence time is not as simple as tuning a few protocol timers. In fact, multiple elements determine how quickly and reliably a network or routing protocol convergence can occur at different network layers. Hence, any design that aims to optimize a routing protocol convergence time has to take all of these elements into consideration. These elements are categorized as follows:

- Event detection
- Event propagation
- Event processing
- Update routing and forwarding tables

As depicted in Figure 9-5, each of these elements contributes to the overall convergence time. Therefore, the optimization of these elements collectively reduces the time a network needs to recover following a failure event.

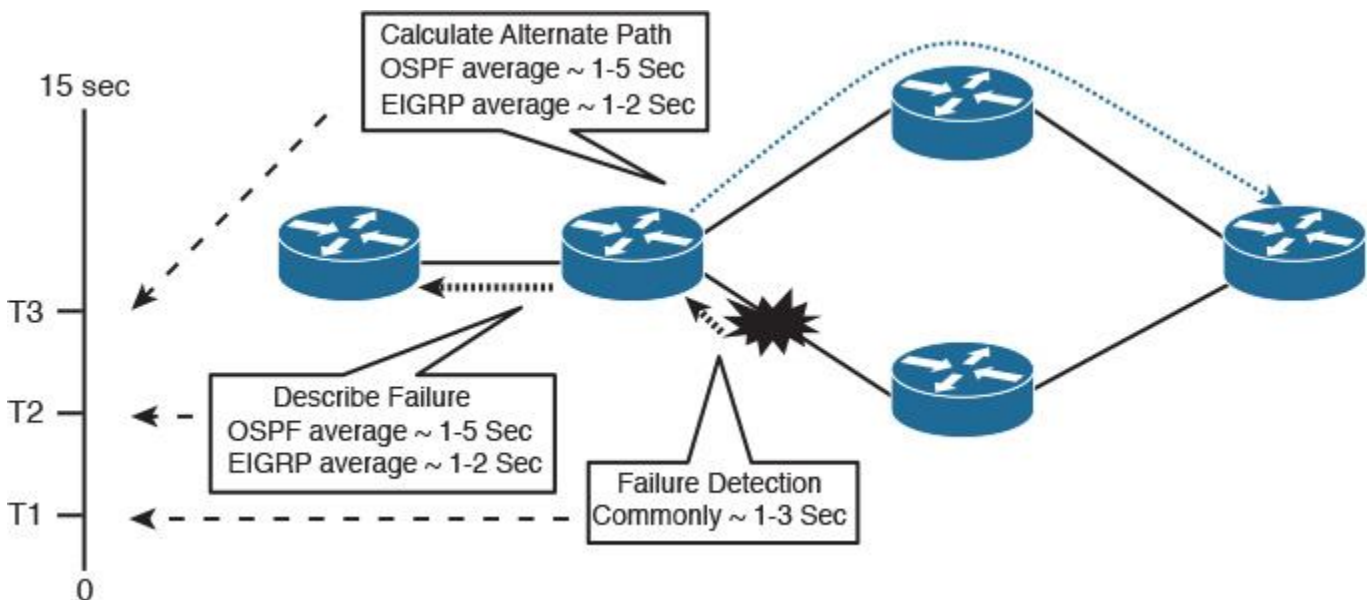


Figure 9-5 Network Convergence Influencing Factors

To achieve the desired convergence time, all these elements must be considered. In other words, the optimization of one or a subset of these elements only does not necessarily lead to an optimized convergence time at the desired level. The following subsections, considering the different characteristic of each routing protocol, discuss how each of these elements influence the overall convergence time.

Event Detection Considerations

The declaration or detection of a peer failure is a primary element in optimizing network convergence, simply because it is the trigger to starting the network recovery process. The detection of a failure in networking can happen at different layers, such as the physical layer (IP over dense wavelength-division multiplexing [IPoDWDM]), data link layer (Point-to-Point Protocol / High-level Data Link Control [PPP/HDLC] keepalive messages), network layer (routing protocols), and up to the application layer. This section focuses on the network convergence at the network layer and below.

The most commonly used mechanism to detect a link failure and to check a control plane neighbor's reachability is a “poll-driven” mechanism that relies on the Layer 3 control plane hello messages and hold timers, also commonly referred to as *fast hellos*. Due to the deployment simplicity of fast hellos, it is one of

the most common mechanisms used in many networks today. However, in large-scale networks and networks with strict convergence time requirements, this mechanism introduces several limitations that may inhibit achieving the desired level of network convergence, such as the following:

■ **Scalability limitations:** Typically occur in a network with a large number of Layer 3 control plane peers

■ **Hardware resource consumption overhead:** Normally, when the number of Layer 3 control plane peers increases, there will be potential CPU spikes that may introduce what are called *false positives*, where links might be incorrectly declared to be broken

To overcome the limitations of this poll-based failure detection mechanism, you can use an event-driven mechanism instead to add more reliable and faster detection capabilities with low overhead. The most common and proven protocol used for this purpose is the bidirectional forwarding detection (BFD). BFD operates independently of the Layer 3 control plane protocols based on the concept of hello or heartbeat type protocols. Moreover, BFD supports several control protocols such as Hot Standby Routing Protocol (HSRP), link-state, Border Gateway Protocol (BGP), and Enhanced Interior Gateway Routing Protocol (EIGRP), where each protocol can register with BFD to be notified when BFD detects a neighbor loss. As a result, these routing protocols can overcome the delay and limitations of the poll-based failure detection mechanism that is mainly based on protocol hello and hold timers. BFD offers a more simplified, reliable, and faster failure detection mechanism (within tens of milliseconds).

Note

Typically, routers connected over direct point-to-point links can be notified by the physical layer when a link fails, which accelerates failure detection by the Layer 3 nodes. BFD, however, enables network operators to achieve fast failure detection in scenarios where two or more Layer 3 nodes are interconnected over a Layer 2 switched network. In this scenario, these Layer 3 nodes do not need to rely on control plane fast hellos and hold timers to detect a Layer 3 peer link or node failure.

Note

Besides the BFD-supported control protocols highlighted earlier, BFD is also supported with static routing, and based on the status of the associated BFD session, the static route is either added to or withdrawn from the Routing Information Base (RIB) table, which makes static routing more reliable.

In BGP environments, however, the IGP fast failure detection and subsecond convergence time may lead to temporary traffic black-holing because of the typical significant convergence time difference between IGP and BGP by default. For instance, in a typical enterprise network with BGP in the core or an MPLS VPN SP network, such as a Carrier Ethernet (MPLS L2VPN Ethernet VPN [EVPN] based), the operator may deploy its backbone IGP to detect a link's failure and recover within one second to achieve fast-converging next-generation L2VPN transport. As a result, in this scenario after a link or node failure, IGP will converge in one second, whereas Multiprotocol BGP (MP-BGP) continues to use the same next hop to reach a MAC addresses advertised in BGP by a remote PE for a given EVPN-EVI (EVPN instance) that is no longer a valid next hop after the last failure event. This is because MP-BGP, in this situation, is still relying on the poll mechanism to deal with the changes in the RIB, where "every 60 seconds, the BGP scanner recalculates best path for all prefixes." In other words, MP-BGP in this case may take up to 60 seconds to realize that the next hop is not anymore useable, and during this 60 second time period between scan cycles, IGP instability, invalid BGP next-hop, or other network failures can cause black holes and routing loops to temporarily form².

2. "BGP Support for Next-Hop Address Tracking", <http://www.cisco.com>

However, with BGP next-hop tracking (NHT) and the address tracking filter (ATF), if the next-hop IP of MGP-BGP routes changes because of link failure followed by fast failure detection and IGP convergence, BGP now can respond faster (by lowering the response time to next-hop changes for prefixes installed in the RIB) without the need to wait for the initial 60 seconds of the "scanner." Therefore, it is always important to understand what other protocols are running over the network besides IGP, and to understand how fast failure detection and IGP fast convergence (for example, subsecond convergence time) will impact the performance and behavior of these protocols or applications.

In spite of the benefits of deploying fast failure detection (subsecond), the following are the main design concerns that must be considered when designing a fast convergence network with subsecond failure detection:

- **Instability:** In many situations, very fast failure detection can lead to flapping and can add high overhead on the network hardware and routing protocols because of unnecessary processing and calculation.

- **Break device-level resiliency:** Very fast failure detection breaks the concept of device-level resiliency described earlier (SSO, NSF, NSR). Therefore, the design cannot mix these two approaches (passing traffic through the failure versus passing traffic around the failure). This is because it may take about one to two seconds for the secondary route processor (supervisor) to start the failover process. Therefore, detecting a failure too fast (such as using BFD or too-low-tuned fast hellos) will usually override the entire failover process.

- **Traffic black-holing:** If the fast failure detection was not aligned properly with the other protocols and overlay technologies running over IGP, such as MP-BGP or IP tunneling keepalives, the network could possibly face temporary traffic black-holing after a failure.

Event Propagation and Processing Considerations

The propagation of a network failure event and the computation process after detecting the failure refers to the reaction of the control plane to describe the failure event to other peers and finding an alternate primary path, respectively. With regard to route convergence design considerations, what is important is how long each of these processes takes and how they can be optimized. In general, there are two primary influencing factors to optimize these processes:

- **Protocol characteristic and its timers:** This varies based on the control plane protocol in use.

- **Network design:** This is common for all protocols, to a large extent.

As discussed earlier, each protocol has different characteristics and its own philosophy for propagating and processing routing changes and updates, whether its link state, distance vector, or path vector. Typically, when a failure is detected by a Layer 3 node running a link-state routing protocol, this node and any other node connected to the failed component will start propagating this event by flooding OSPF link-state advertisements (LSAs) or Intermediate System-to-Intermediate System link-state packets (IS-IS LSPs) to every node within the link-state flooding domain. The overall event propagation duration of each LSA/LSP is normally governed by three primary timers: generation delay, reception delay, and processing delay.

Both IS-IS and OSPF, however, can use the incremental shortest path first SPF (iSPF) algorithm to optimize SPF computations and converge faster in response to network events and topology changes, simply because iSPF is more efficient than the full SPF algorithm. In contrast, EIGRP uses a completely different logic and sequence. EIGRP finds an alternate path (usually precalculated using EIGRP feasible successor, thus avoiding the entire EIGRP query process, which significantly improves EIGRP convergence following failure event detection). Then, EIGRP propagates the failure to the peers. Table 9-2 highlights the fundamental differences between link state and EIGRP with regard to each protocol's convergence characteristic.

	Within a Flooding Domain	Between Flooding Domains
Link State	By default, can converge within ~3 to 10 seconds With tuned times, such as fast hellos, can converge within milliseconds	Depends on the flooding domain design with regard to reachability information hiding and topology information hiding (acts like a typical distance-vector protocol)
	With Feasible Successor	Without Feasible Successor
EIGRP	Usually within tens of milliseconds	Primarily depends on the network design (for example, number of nodes to handle queries, EIGRP query scope, EIGRP stuck-in active [SIA] delay)

Table 9-2 Protocol Convergence Characteristics: EIGRP Versus Link State

In contrast, with BGP as a path-vector protocol, its propagation philosophy of a failure event is based on the propagation of withdrawal update messages that normally can be tuned to reduce the overall duration. In addition, the tuning of the following elements that technically influence BGP transport offer an optimized BGP performance to handle a large number of prefixes during the propagation and processing following a failure in the BGP path:

- TCP maximum segment size (MSS)
- TCP window size
- Path maximum transmission unit (MTU) discovery (RFC 1191)
- BGP minimum route advertisement interval (MRAI)
- Router queues (for example, input queue hold)

Note

BGP prefix-independent convergence (PIC), described later in this chapter, helps to significantly enhance BGP convergence time.

From a network design perspective, the following common design considerations can help to optimize the convergence time and improve network stability:

■ **Fault isolation:** As discussed earlier, fault isolation has a significant influence on enhancing the operation of routing protocols and reducing the amount of information that needs to be propagated. It also enhances the scope of the topology information shared between logical routing domains, which ultimately reduces hardware overhead utilization and the time required for the Layer 3 nodes to complete the propagation, processing, and finding an alternate primary path.

■ **Number of routes:** One of the primary influencers on the overall time required for propagating, processing, and updating routing and forwarding tables is the number of the effected prefixes with a network change. Therefore, the information reachability hiding techniques discussed earlier in this book will help considerably in optimizing network convergence time and at the same time reduce the impact on its stability. For instance, with EIGRP, in addition to route summarization, stub routers help to reduce the processing of the DUAL algorithm and the scope of query messages. This will enhance the overall EIGRP convergence time while keeping the network as stable as possible (minimal to zero impact on the remote stub networks). Similarly, with link-state protocols, such as IS-IS (L1) or OSPF, in stub areas or totally stubby areas, the router's LSA database size can be reduced, which will enhance the overall convergence time as well.

■ **Multiple available paths:** The availability of a second path can significantly enhance convergence time. Therefore, with EIGRP, it is essential that the design supports the precomputation of EIGRP feasible

successor. In addition, apart from EIGRP feasible successor, if a BGP speaker has two available paths to a given destination network, the time required to process and find an alternate path can be eliminated because there is already an alternate path. However, unlike IGP, achieving this with BGP is sometimes challenging, especially in large networks where there is a route reflector (RR) in the path. The features and techniques covered in Chapter 5, “Service Provider Network Architecture Design,” are designed specifically to overcome these limitations in some BGP topologies. Similarly, availability of a second route in link state will offer an almost immediate failover with minimal to zero service interruption.

Let’s consider the scenario illustrated in Figure 9-6 to demonstrate the impact of link-state design on convergence time and the overall network stability following a network failure event. In this scenario, if PE1 needs to calculate the possible paths to reach PE3 loopback IP, there is a high number of possible paths to be considered part of the path computation process. It is obvious that this design has an increased MTTR because of the high degree of parallelism (redundant paths). Therefore, when network designers evaluate a design like the one depicted in Figure 9-6, they should consider the following fundamental questions when there is a link or node failure:

- How many possible paths must PE1 calculate and process?
- How many routers will be notified about a single link failure?
- How many routers will not be effected by this failure, but yet will unnecessarily receive and process the update messages?

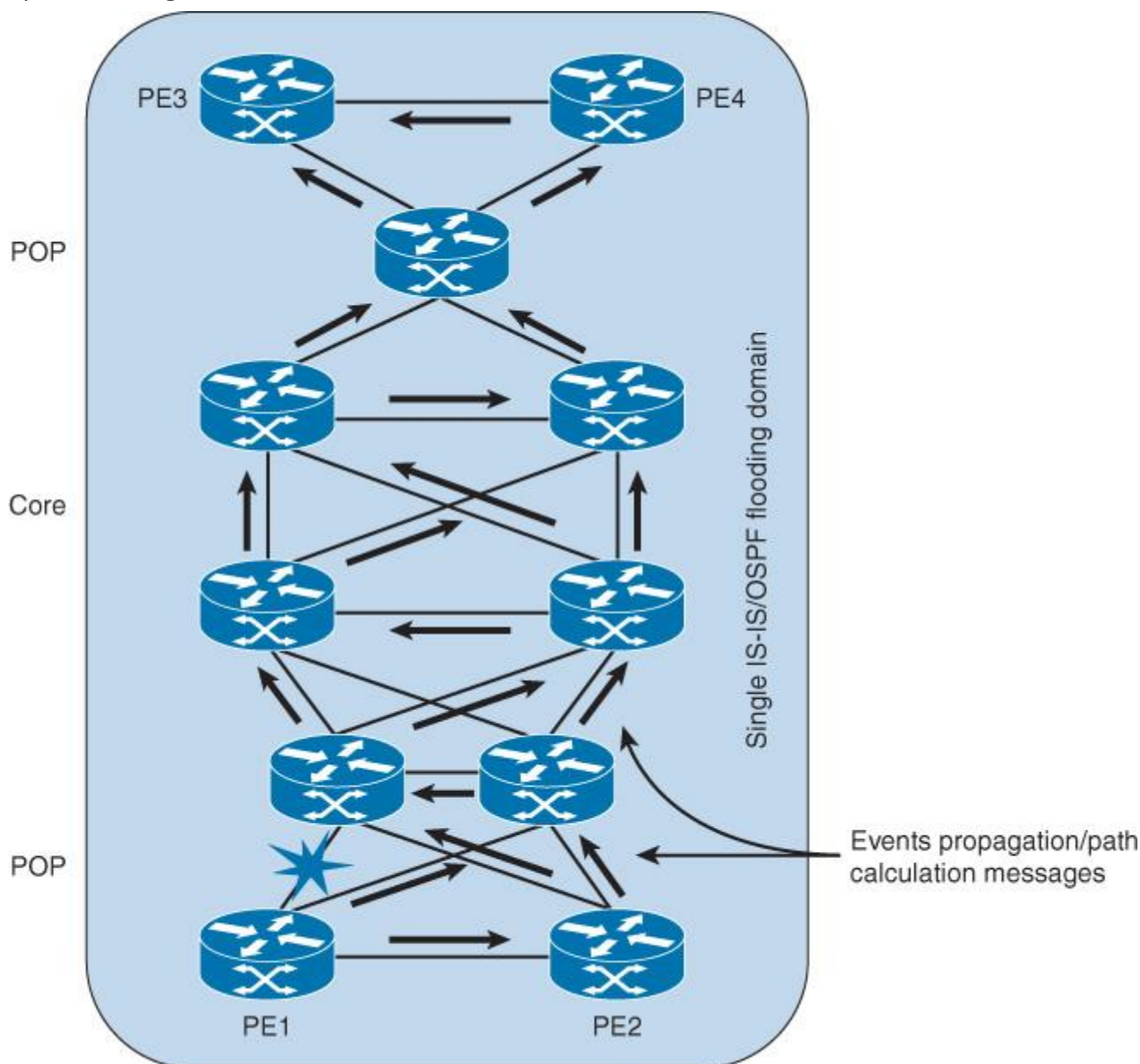


Figure 9-6 Impact of Flat Control Plane Design on Convergence Time

Note

Even if EIGRP were used in this design, the question is, “How big should the EIGRP query scope be?”

In conclusion, the design presented in this scenario will slow down the convergence time, stability, and the performance of the network, as well, because of the flat routing design, especially if there is a large number of prefixes carried by the control plane.

If you introduce the principle of fault isolation with IGP logical flooding domains (using IS-IS levels, OSPF areas, or EIGRP summarization) depending on the used control plane protocol, PE1 will have a significantly optimized topology because it will have a fewer number of paths to calculate to reach PE3 (the calculation and updates flooding will usually be up to the border of the flooding domain), as illustrated in Figure 9-7. As a result, the design offers faster network convergence time with a higher level of stability (fault isolation).

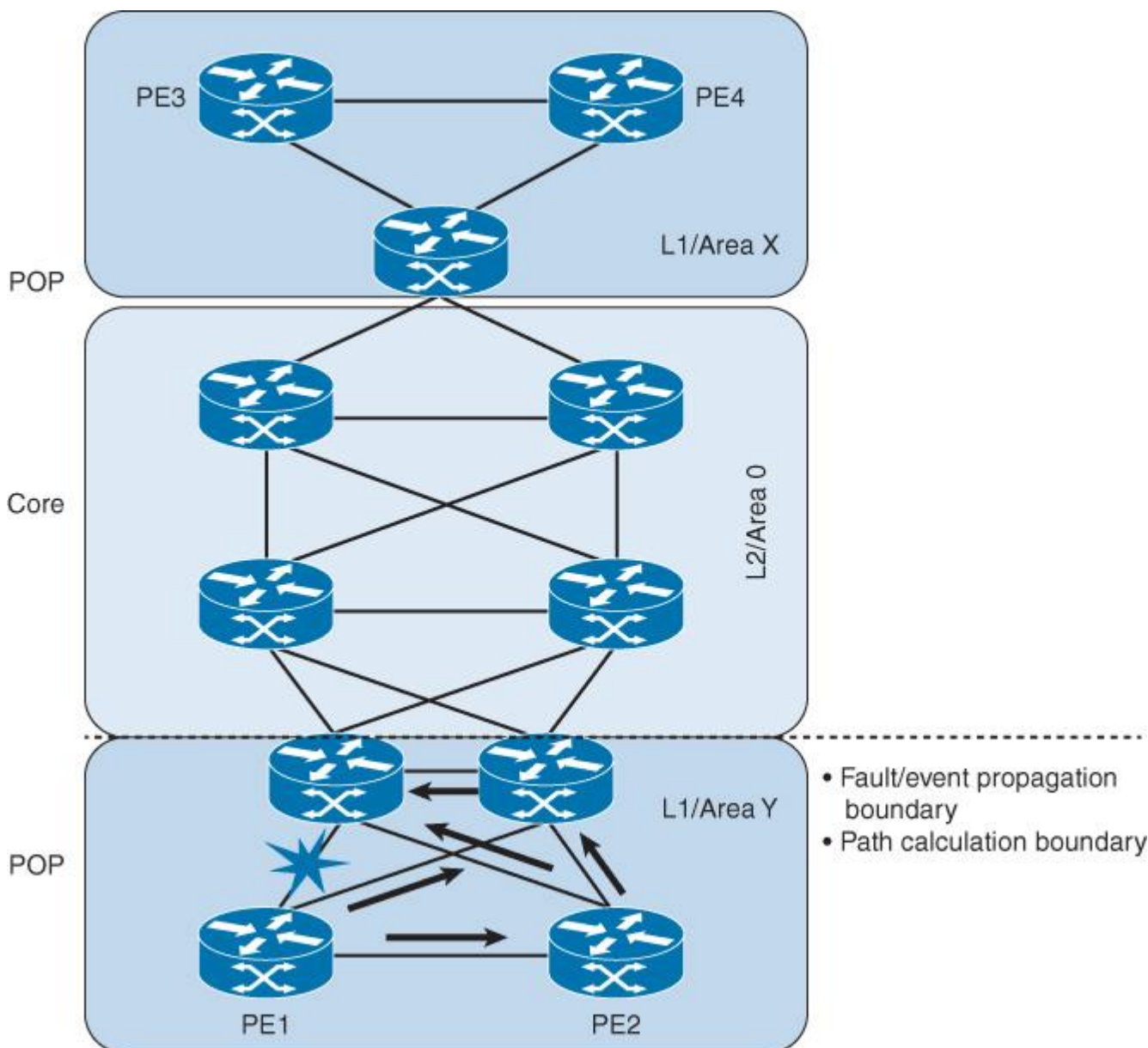


Figure 9-7 Optimized, Structured Routing Design

Note

It is important, from a design point of view, to understand the location and role of the node that needs to be optimized for network convergence (for instance, if it is a core router or edge [PE] router) because the impact of applying some

optimizations techniques can differ. Typically, PE devices handle more customer routes; consequently, tuning IGP timers aggressively down may introduce noticeable stability and performance issues.

Updating the RIB and FIB

Updating the Routing Information Base (RIB) and Forwarding Information Base (FIB) are other influencing factors within the network convergence process. Typically, after the control plane updates the Layer 3 node RIB table, the updates are propagated and reflected to the FIB table (either centrally or distributed depending on the hardware platform architecture). Normally, any process containing prefix installation will first be impacted by the number of prefixes to be installed (the larger the number of prefixes, the longer the duration of installation). Therefore, for networks with a significantly large number of prefixes (thousands or hundreds of thousands of prefixes), this process will be impacted at both levels:

■ Updating the RIB table

■ Populating the updates to the FIB table or tables

One common mechanism to optimize the prefix installation process is prefix prioritization, in which network operators can assign a higher priority for certain prefixes, such as loopback IPs, to establish an LDP session quickly (in MPLS networks) before other prefixes, such as BGP prefixes. Similarly, the BGP event-based VPN import helps to accelerate route installation, where (PE) nodes in an MPLS L3VPN environment can propagate VPN routes to the CE nodes without relying on the scan time delay. Furthermore, Link-state offers the flexibility to network operators to avoid packets loss (instability) when a network node reloads or recovers from a failure. In this particular scenario, IGP most probably will converge (build its routing table) before BGP, and some routers may start forward traffic through the just reloaded node, which may lead to packets' loss. Because at this point of time, BGP is still rebuilding its routing tables (not fully populated with all IP prefixes). Link-state overcomes this issue by advertising routes with higher metrics "for a fixed period of time" after a node reload or a recovery from a failure. This ultimately will help to avoid impacting transit traffic that is currently using other alternate routes and gives BGP more time to rebuild its routing table. Technically to achieve this, OSPF uses the "wait-for-bgp" feature to send route-LSAs with infinity metric after a reload until BGP is converged or the predefined timer is expired, in the same way, IS-IS uses "set-overload-bit" to wait until BGP converged or the predefined timer is expired³.

3. "Uses of the Overload Bit with IS-IS", "Using OSPF Stub Router Advertisement Feature", <http://www.cisco.com>

Network Fast Restoration Versus Network Stability

Today's converged networks carry sensitive real-time traffic (such as Voice over IP [VoIP] and business video) and should offer a relatively fast restoration time following any network failure event (usually by tuning routing protocol timers aggressively low). However, the very fast reaction may sensitize the network, causing it to potentially take an action and converge and reconverge again because of the very fast failure detection and protocol reaction for any simple link flap or packet loss. As a result, this will impact the overall stability of the network and probably will outweigh any benefit of the fast convergence.

Therefore, network designers must compromise between both (fast convergence and stability) to achieve a stable fast converging design. There are multiple ways at different layers to optimize the overall network stability when the network is tuned to converge quickly. For instance, a dampening mechanism can relatively mitigate the effect of network instability. Dampening mechanisms are applicable at the protocol and link level, such as link-state exponential backoff and IP interface dampening, respectively. In addition, link-state protocol "throttling" of SPF and LSA/LSP generation mitigates the impact of network instability, such as link flapping, to some extent.

Furthermore, the design considerations discussed earlier in this chapter and previous chapters in this book, such as fault isolation, can play a primary role in optimizing the overall network stability. This is because the flooding and bad behavior caused by an unstable component in one domain will be contained and will not be propagated across the entire network (multiple flooding domains concept). As a general rule of thumb,

tuning control plane timers aggressively low should be done with very careful planning, considering the different design aspects and influencing factors, along with a thorough testing of multiple failure scenarios.

Microloops

Microloops, also known as *transient loops*, occur when different nodes in the network calculate new alternate paths following a failure event independently of each other and at different times. This behavior is typical in a ring full- or partial-meshed topology running the link-state control plane. For instance, in Figure 9-8, if host A is communicating with host B and the link between R3 and R5 fails, R5 will reconverge and send the traffic to R4. Assuming R4 is converged as well, it will usually then forward the traffic toward R1. If R1 has not converged yet, traffic will loop between these two nodes (R4 and R1). If the duration of this microloop is small, where the network can converge quickly, packets will usually loop for a short duration before their time to live (TTL) expires. Eventually, the packets from host A will reach host B in this scenario. However, if the application is very sensitive, such as voice traffic, this might lead to a call dropped or low voice quality. In other words, application characteristics and their level of tolerance must be considered here as well.

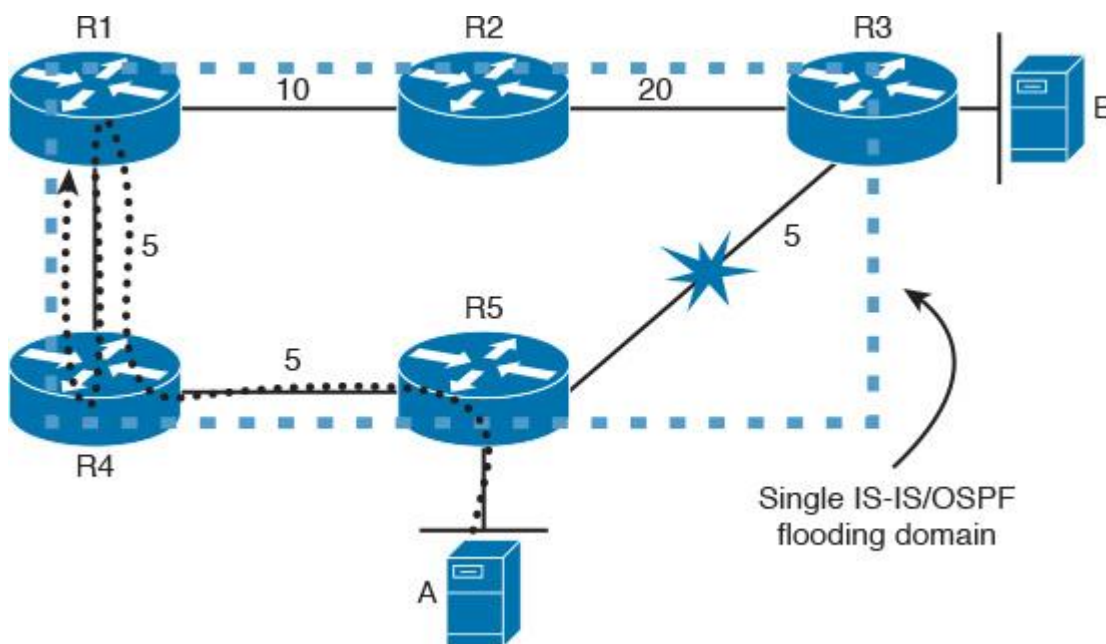


Figure 9-8 Link-State Microloop

In contrast, if the duration of the microloop is long, in this scenario it may be because of the slow R1 convergence time. In this case, a packet's TTL will normally expire, or the packet's rate may exceed the available bandwidth. In both cases, it will result in packet drop. Practically, this type of microloop is not always an issue and is acceptable by many businesses. However, it might not be an acceptable behavior for other businesses. In these cases, where this type of microloop is unacceptable, network designers need to either use a triangle topology or a different approach to control network convergence in fully meshed, partially meshed, square, or ring topologies following a network component failure. This limitation can be resolved with the network protection approach, which is covered in the following section.

Note

R1 prefers the path via R4-R5 to reach R3 before the failure event in the scenario described earlier, because of the lower path cost.

Note

Network designers may use the microloop prevention feature described in IETF draft “litkowski-rtgwg-uloop-delay-xx” to mitigate the impact of link-state routing protocol transient loops during topology changes when possible.

Network Protection Approach

Although the network restoration approach is the most commonly used approach to optimize network recovery duration, this approach has the following limitations, which may impact the requirements of some networks:

- With the restoration approach, large-scale networks may face microloops (as discussed earlier) for a few seconds until the entire core and network has converged properly, which will usually lead to some packet loss within this duration.

- Tuning routing protocol timers to react and converge very quickly is required to achieve subsecond network convergence time. However, this will result in reduced network stability, which increases the likelihood that the network may react to any simple packet loss. It will also add operational complexity in large networks with a large number of nodes.

In contrast, the network protection approach may be more suitable for satisfying the requirements of large-scale networks that need very fast convergence time (subsecond) without sacrificing network stability. Typically, with the network protection approach, the alternate path is always preestablished and does not need to wait for a protocol calculation across several devices throughout the network following network component failure detection. However, regardless of how fast the switchover method is, if the failure detection mechanism is slow it will outweigh the benefits of the FRR. Therefore, the issues and mechanisms discussed earlier in this chapter regarding failure detection must be considered to achieve a true fast convergence.

This section discusses the primary and most common technologies used to achieve the network protection approach. Each of these technologies can be deployed solely or in combination with other protocols, based on the goal to be achieved, supported features, targeted topology, and network environment.

MPLS-TE Fast Reroute

MPLS-TE fast reroute (MPLS-TE FRR) is commonly used by service provider networks and large-scale enterprise networks where fast recovery after failure detection is a fundamental requirement. It is normally driven by application requirements, such as real-time applications (VoIP), where a network has to reconverge following a network component failure in a relatively undetectable time from the application perspective. However, designing an MPLS-TE with FRR can be a complex task, especially when the network is large and has a large number of paths and nodes. Therefore, to select the right design approach, you must identify the goal to be achieved. There are two primary MPLS-TE FRR approaches:

- **Path protection (end to end):** With MPLS-TE FRR, there are usually two LSPs used. The first one is the primary, and the second one is the presignaled backup LSP. The end-to-end FRR approach has some limitations with this behavior that network designers must consider [115]:

- **Resource overutilization:** When the secondary path is preestablished, it will lead to what is known as *double booking* for the network resources, in particular bandwidth booking, with Resource Reservation Protocol (RSVP) to maintain the same bandwidth constraints as the primary LSP (TE tunnel) to keep the same level of the offered service level agreement (SLA) (1:1 protection).

- **Overprotection:** This can occur when other mechanisms are used to protect a given set of links within the path, because this approach cannot protect links selectively along the path.

■ **Longer restoration time:** The path protection approach is usually based on a headend-controlled switchover; therefore, this behavior may lead to unexpected delays because it relies on the propagation time of RSVP error message to the headend node to perform the switchover following a failure event.

However, the path protection approach may offer lower operational complexity when a small number of LSPs need to be set up with a more controlled end-to-end LSP backup path.

■ **Local protection:** Both local and path protection rely on the concept of a preestablished or signaled backup LSP. However, with the local protection MPLS-TE FRR, the protection is based on a segmented approach rather than end to end. In other words, the aim of MPLS-TE FRR is to locally repair the protected LSPs by rerouting them over a preestablished backup path (MPLS-TE tunnels) that bypasses either failed links or nodes. This offers an improved scalability (fewer network states) and recovery time compared to the path protection approach. The two primary MPLS-TE FRR local protection approaches are as follows (shown in Figure 9-9):

■ Link protection

■ Node protection

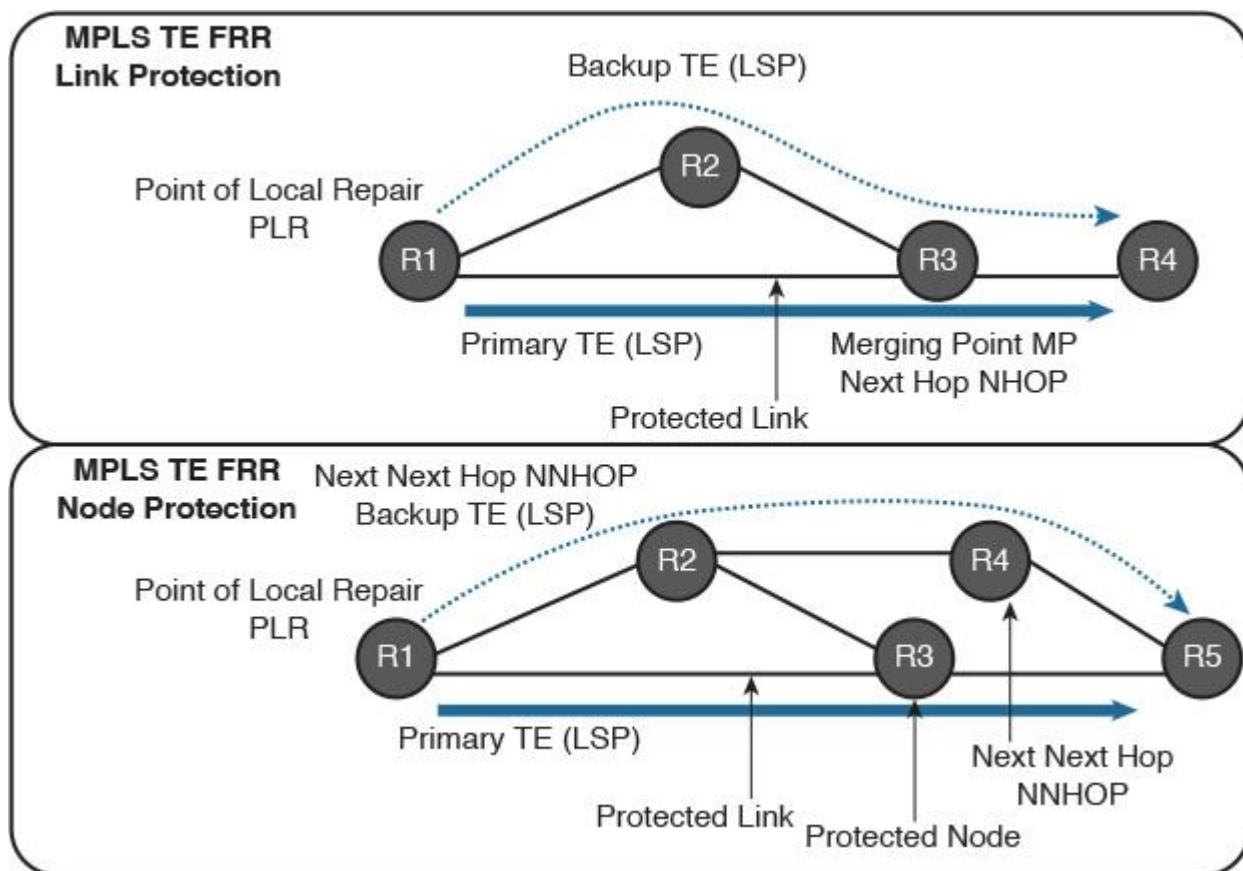


Figure 9-9 MPLS TE FRR Link Versus Node Protection

The decision of whether to use link protection or node protection, or both, is mainly based on the network architecture, physical topology layout, level of protection, and the ultimate goal to be achieved by the FRR. However, under general design principles, it is always recommended to keep it as simple as possible. For instance, consider protection for one failure rather than double failure scenarios to avoid situations where you design MPLS-TE FRR to cater for both link and node protection. In addition, if the network node has a reliability level of 99.999 percent, while the line cards or the interconnecting optical links have 99.5 percent of reliability, it is more reasonable to use link protection rather than node protection. Similarly, if a high level of end-to-end LSP availability is required and the next hop (NHOP) passes through an SRLG but the next-next hop (NNHOP) does not, it might be more feasible to use NNHOP to satisfy this design requirement. Again, these are only general design decision considerations. The actual decision has to be driven by the design requirements in terms of business requirements, functional requirements, and application requirements.

MPLS-TE FRR Design Considerations

In addition to the points highlighted with regard to the different approaches of MPLS-FRR, some technical considerations must be taken into account during the planning and design phases of any MPLS-TE FRR to achieve a successful and effective solution that can deliver its promised value. This section discusses the most common considerations with regard to MPLS-TE FRR design.

Multiple Backup Tunnels and Tunnel Selection

The only limiting factor for the maximum configurable number of backup tunnels to protect a given interface is memory limitations. However, the more tunnels and backup tunnels in the network, the more design and operational complexity is added to the network. Therefore, as a rule of thumb, it is always recommended to not overengineer the network with an extremely high number of MPLS-TE tunnels and backup tunnels (the same concept discussed earlier when having many redundant links will outweigh the benefits at higher layers, such as the network routing layer). Nonetheless, in some scenarios, having multiple TE back tunnels can be beneficial, such as the following:

- **Additional redundant paths:** It may be feasible to consider additional backup tunnels to protect the LSPs of critical traffic with a higher level of redundancy requirements, to cater for a double-failure scenario.

- **Link capacity purposes:** Sometimes the protected interface has a high bandwidth, whereas the alternate paths each individually cannot offer the same level of link capacity. Therefore, distributing the traffic over multiple backup tunnels can protect this high-capacity link over multiple paths (backup LSPs) to maintain adequate bandwidth protection following a failure event.

For instance, if the protected interface is a high-capacity link and no single backup path exists with an equal capacity, multiple backup tunnels can protect that one high-capacity link. The LSPs using this link will fail over to different backup tunnels, allowing all the LSPs to have adequate bandwidth protection during failure (rerouting). In other words, if bandwidth protection is not desired, the router spreads LSPs across all available backup tunnels (load balancing across backup tunnels).

However, technically, if multiple backup tunnels are deployed, the backup tunnel selection process will consider the following selection criteria when choosing the preferred one:

- Tunnel that terminates at the NNHOP over a tunnel that terminates at NHOP.

- Tunnel deployed with subpool reservation will be preferred over a tunnel deployed with any pool (same concept applies to tunnels deployed with global pool reservation, global pool over the one with any).

- Tunnel deployed with the lowest sufficient subpool reservation will be preferred over a tunnel deployed with a higher bandwidth subpool reservation.

- Tunnel deployed with limited backup bandwidth over a tunnel deployed with unlimited bandwidth.

For instance, in the scenario illustrated in Figure 9-10, if an LSP at R1 requires a backup tunnel with a 1-Mbp subpool bandwidth, the preferred tunnel will be selected based on the tunnel selection preferences previously listed, as follows:

1. TE2, TE3, and TE4 are first, because they all terminate at NNHOP.
2. TE2 will be eliminated because it protects only LSPs using global-pool bandwidth.
3. TE3 will be preferred over TE4 because it is deployed with a subpool that has the least sufficient backup bandwidth available as compared to TE4.

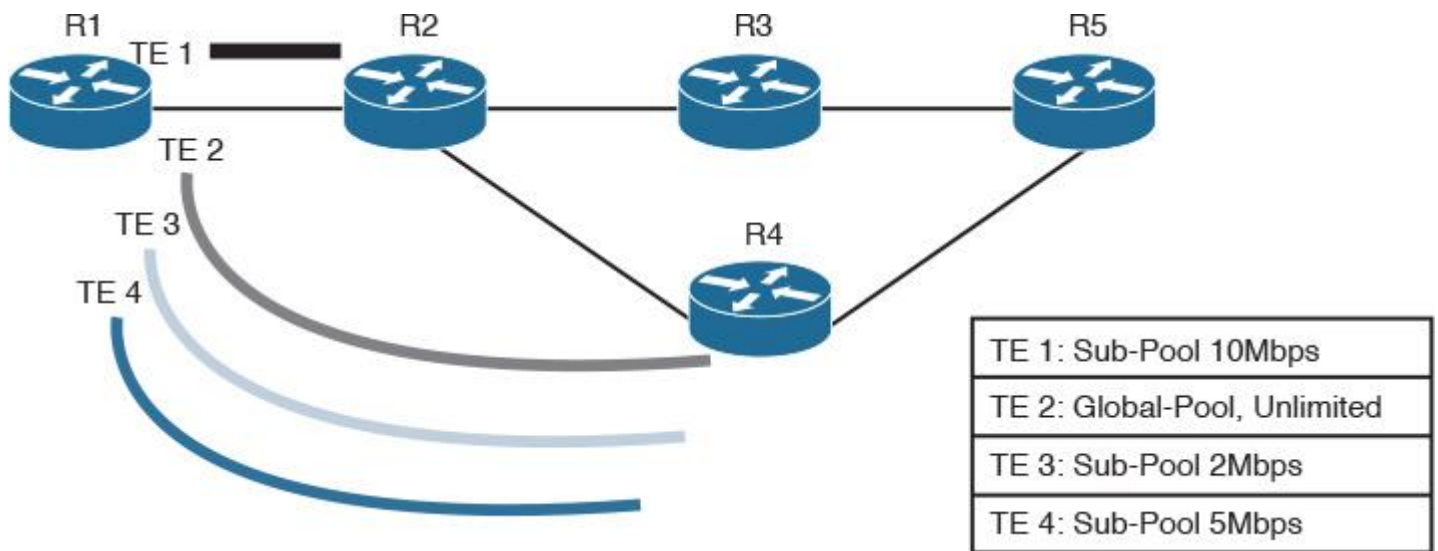


Figure 9-10 MPLS-TE FRR tunnel selection

Therefore, to ensure that the protected traffic flows can be distributed across multiple backup TE tunnels, the attributes of the backup TE tunnels with regard to the selection criteria highlighted above must be the same.

Path Reoptimization and MPLS-TE Tunnel Promotion

One of the most common requirements with MPLS-TE with FRR is that traffic must reuse the original LSP/MPLS-TE tunnel after the failed link or node has recovered. Another common requirement is that MPLS-TE paths should be moved across links/paths that are less congested or optimal when they are available. From a high-level design point of view, both requirements are valid points and offer a value to the business (more optimized and dynamic paths and bandwidth utilization). However, two primary design concerns should be considered:

■ **Network stability:** Moving traffic from one MPLS-TE tunnel to another in a matter of subseconds can impact network stability in some situations, such as when there is a link flap and traffic will keep moving between the primary and backup MPLS-TE tunnels.

■ **RSVP double booking of bandwidth:** In each of the design requirements above, traffic has to be moved from one MPLS-TE tunnel to another. Ideally, traffic transition must be performed in a transparent manner in which no service interruption must occur (unnoticeable by applications). To achieve this MPLS-TE, use a principle called *make before break* (refer to IETF, RFC 3209).

In addition, the conditions of MPLS-TE backup tunnels may change for any given LSP, in which case it will cause reevaluation or what is commonly known as *promotion*. Such scenarios include the following:

■ A new backup tunnel comes up. For instance in Figure 9-11, the MPLS-TE tunnel 1 is protected by TE tunnel 20 (NHOP). When backup TE tunnel 30 (NNHOP) comes up on line, this may cause reevaluation of the current backup tunnel.

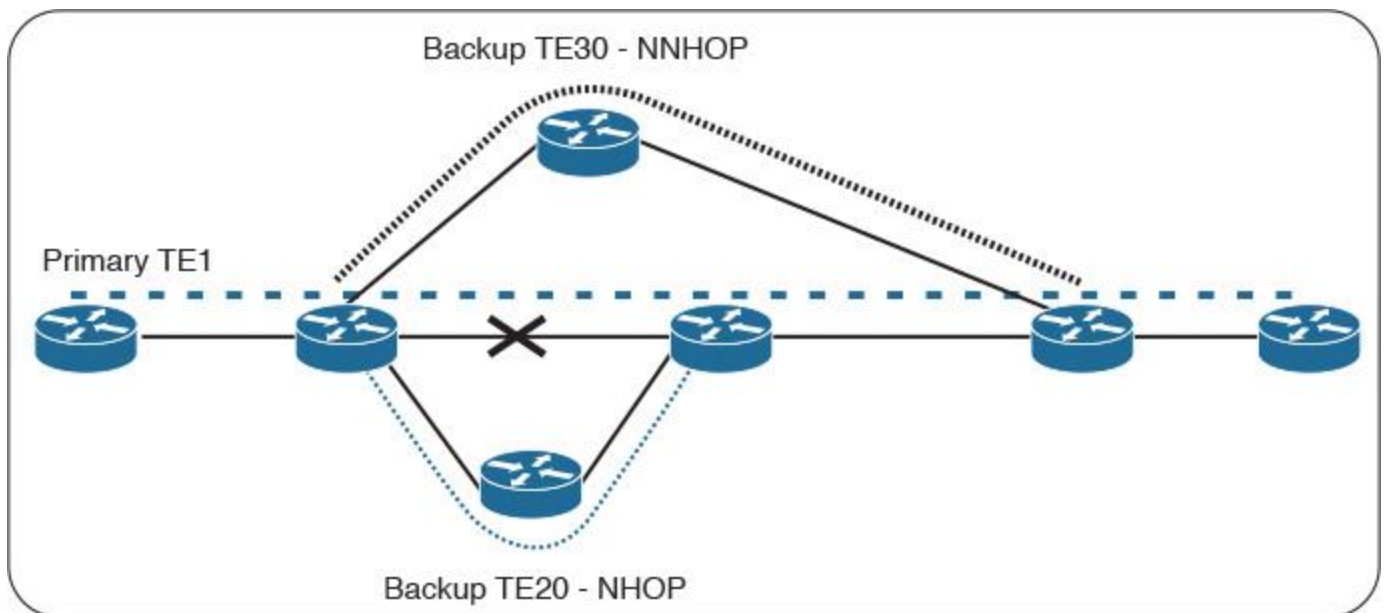


Figure 9-11 *MPLS-TE FRR Tunnel Promotion*

- The actual available bandwidth (or required bandwidth) for the current active backup tunnel may change.
- The current active backup tunnel may go down.

Note

Reevaluation of the current TE backup does not mean the protected LSP must switch over to a new path immediately. In other words, if an LSP is actively using a backup tunnel and a better backup tunnel becomes available, the active LSP is not switched to the better backup tunnel; instead, it might be considered during the next periodic promotion cycle [63].

Path Diversity and Fate Sharing

The main goal of FRR is to provide a precomputed backup path with an extremely fast switchover time from the primary (failed path) to the backup path. However, sometimes the backup path may traverse an SRLG point. This can introduce a risk for a certain type of traffic that must meet a tight SLA with a high degree of end-to-end path reliability and service availability. Therefore, network designers in scenarios like this should cater to a double-failure scenario and make sure that the backup TE tunnel takes a diverse path from the primary path. Network designers should also avoid any SRLG in the path (if possible) to avoid fate sharing during failure scenarios.

Scalability Considerations

One of the main concerns with MPLS-TE and FRR is the scalability limitations when the network grows to an extremely large number of nodes due to the mesh nature of MPLS-TE. Therefore, a combined approach of MPLS-TE FRR and other FRR mechanisms, such as loop-free alternate (LFA), is sometimes used to lessen the impact of having a very large number of primary and backup tunnels across the network.

In addition, the location of the TE headend and tail end and how the backup tunnels handle the backup LSPs are primary influences on the level of scalability of the TE and FRR design (as highlighted earlier in Chapter 5) (in other words, if the TE tunnels are set up between PEs versus across the core only [P] routers and whether one backup tunnel is designed to protect multiple interfaces, where one backup tunnel can protect hundreds or thousands of LSPs [also known as $N:1$]). For example, using MPLS-TE FRR directly between PEs in a service provider network with tens or hundreds of PEs can significantly complicate the design and increase operational cost and control plane complexity.

The aforementioned points are purely design decisions. Practically, it is impossible to generalize and advise that a certain design option is almost always better. Yet by understanding the impact of these points on the overall network and FRR solution scalability, network designers can make a better design decision,

considering the goals that need to be achieved, business priorities, network size, and the supported technologies and capabilities. For example, some networks may have a software code that does not support LFA, and a software upgrade is not an option. In this case, LFA will be eliminated from the design options. Similarly, if a service provider wants to meet certain levels of an SLA for its VIP customers, where no single point of failure must exist along the primary and backup paths, it is more feasible to set up MPLS-TE along with FRR end to end (PE-to-PE), which excludes any SRLG in the path that meets bandwidth and other constrained SPF (CSPF) requirements. As highlighted in Chapter 5, the planning of the MPLS-TE is crucial to generate a viable and reliable design, including FRR considerations.

Bandwidth Protection Considerations

Bandwidth protection with regard to MPLS-TE FRR back tunnels is one of the main design considerations that must be accounted for during the planning phase of the solution. From a design point of view, an MPLS-TE FRR design strategy that includes bandwidth protection offers a more reliable FRR solution. With this approach, the backup tunnel selection takes into consideration bandwidth availability over any given path. As a result, operators can satisfy strict SLA requirements. For example, delay-sensitive traffic, such as VoIP traffic riding a TE tunnel, can be protected with MPLS-TE FRR, which can avoid placing the backup TE tunnel over congested paths following a link or node failure. Bandwidth protection with MPLS-TE FRR commonly uses the following approaches:

■ **Limited bandwidth:** With this approach, the LSPs must have a predefined bandwidth to be protected with an MPLS-TE backup tunnel. However, the aggregate value of the LSP's bandwidth that needs to use the TE backup tunnel must not exceed the assigned bandwidth value of the TE backup tunnel. This approach is simple to deploy, but it may introduce limitations with regard to bandwidth sharing of MPLS-TE backup tunnels protecting different independent failures/LSPs.

■ **Unlimited bandwidth:** With this approach, the MPLS-TE backup tunnel does not provide any bandwidth protection, and the LSPs over this backup TE tunnel will share the same backup tunnel in a best effort fashion. However, only the MPLS-TE LSPs signaled with zero bandwidth can use the backup tunnels that have unlimited backup bandwidth. This approach offers a cost-effective (economical bandwidth usage) solution that supports oversubscription by sharing bandwidth protection used to cater for multiple independent failures. However, this approach does not offer differentiated treatment for different types of LSPs such as VoIP and data. Therefore, it is commonly used when MPLS-TE is intended to provide basic link protection only.

Moreover, it is important to identify the backup protection strategy used, whether the protection is N:1 (facility) or 1:1. For instance, let's assume that an N:1 MPLS-TE backup strategy is used (without DS-TE). This means a single backup tunnel is protecting multiple LSPs. If there is an LSP carrying VoIP (delay-sensitive traffic) while other LSPs carrying data, in this case, even though an explicit bandwidth is signaled for the backup tunnel, in the case of a failure all the LSPs crossing the link will be using the same backup TE tunnel, in which VoIP traffic will not get any priority over other tunnels for data traffic. This does not mean that VoIP will always suffer with this strategy because the backup path might be overprovisioned. However, it is more of a concern that the backup path may introduce congestion in a failure scenario.

Note

As mentioned earlier, when multiple TE backup tunnels are available with different bandwidth allocation models such as global pool, subpool, or no bandwidth limit, the TE backup tunnel selection will be impacted. Furthermore, the LSPs signaled with zero bandwidth might even be demoted. This "demotion" occurs when there are one or more LSPs (without the bandwidth protection bit set) to be removed from the current assigned TE backup tunnel to provide backup to an LSP that has its bandwidth protection bit set. This scenario happens when there is insufficient backup bandwidth.

Loop-Free Alternate Path (IP FRR)

The goal of IP FRR is to reduce failure reaction time to tens of milliseconds by applying the same concept of EIGRP feasible successor to link-state routing protocols. A precomputed loop-free alternate next hop is used if the currently selected primary next hop fails, as soon as the failure is detected. A network with this feature

usually experiences less traffic loss and less microlooping of packets than a network without IP FRR [Loop-free Alternate IETF draft-ietf-rtgwg-ipfrr-spec-base-12], as illustrated in Figure 9-12.

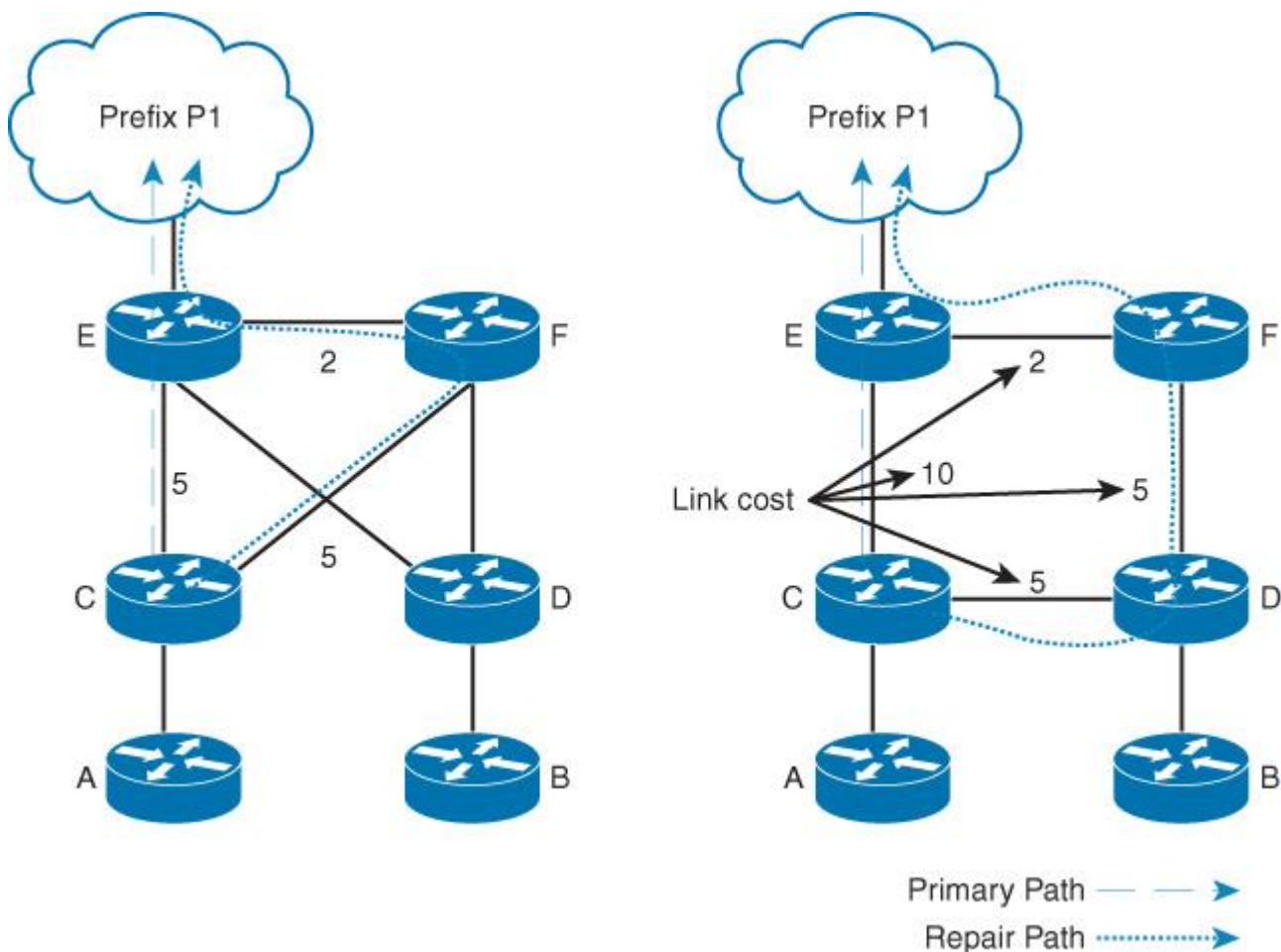


Figure 9-12 LFA Applicability

Note

In Figure 9-12 (square topology), the node D primary path to reach prefix P1 is via node F (lower path cost). Therefore, LFA is applicable in this topology (because forwarding traffic immediately after a C to E link failure will not impose any potential looping).

LFA usually offers per-prefix protection (fast convergence) for simple triangular topologies. Ring topologies, however, require considerations of the other flavor of LFA, known as *remote LFA*. In remote LFA, the headend node uses a tunneling mechanism (typical LDP based) to configure an MPLS path. This alternate path is installed as an LFA in the Cisco Express Forwarding (CEF) table. As a result, the LFA can bring the overconvergence time in the supported topologies to less than 50 ms after the detection of the failure because the protocol does not need to wait for the failure propagation, processing, and SPF calculation. Instead, the node can switch over quickly to the precomputed alternate path within a few milliseconds.

Despite the fact the LFA offers simple and native IP FRR support with link-state routing protocols, the following limitations, in which LFA may not be an optimal solution for every design, must be considered:

- LFA consumes hardware resources, which makes it less than ideal when there are many interconnections between the network nodes, such as with partially meshed and fully meshed topologies, in addition to the added complexity if LFA is in these types of topologies.
- LFA is limited within a single IGP flooding domain.
- Ring and some square topologies require packets tunneling such as MPLS to be enabled for the remote LFA to establish an LDP session, as illustrated in Figure 9-13.

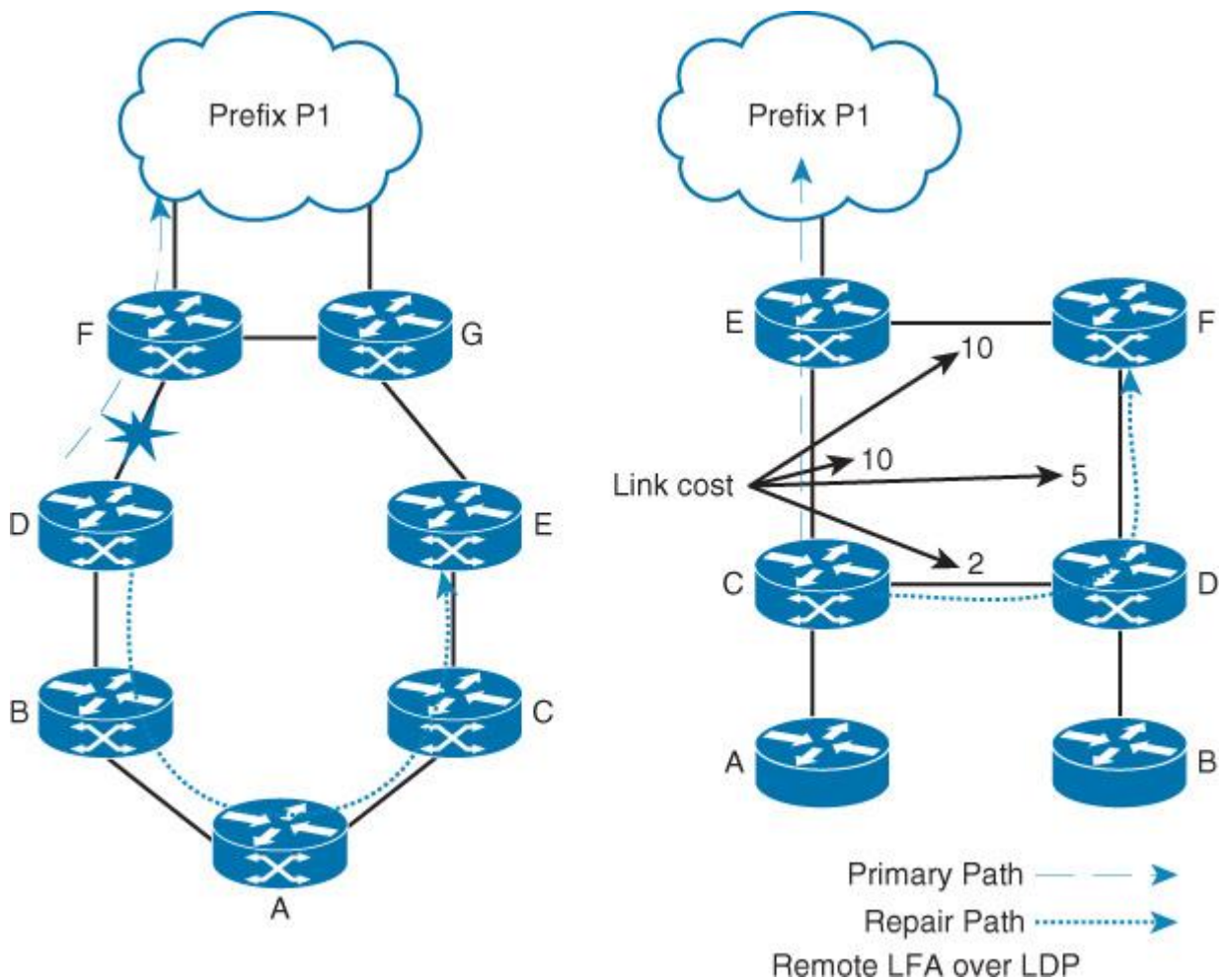


Figure 9-13 Remote LFA

Note

By comparing the square topology in Figure 9-13 to the one illustrated in Figure 9-12, it is obvious that in Figure 9-13 node D's preferred path to reach prefix P1 is via node C. Therefore, remote LFA R-LFA (with LDP encapsulation) is required to avoid any microlooping following C to E link failure.

The decision whether to use LFA needs to be made with input from network designers about the topology, to what extent the solution has to be deterministic, and the required FRR coverage; for example, in an SP environment across the edge networks only (points of presence [POP]), core, or both. Practically, LFA is more applicable to POP design, in particular to the ones designed in triangles. It can offer a fully protected LFA solution against both link and node failures compared to the square topology, where some prefixes may not offer a full LFA without considering the remote LFA concept.

The core of the network, however, might not be as simple and common as with the POPs. For instance, the underlying physical topology can influence the decision of whether LFA applies. Redesigning a POP network topology is almost always easier, cheaper, and less disruptive than redesigning a core network. Therefore, if the core topology is not fully supported by LFA or the network designer aims to achieve a more precise and deterministic path protection mechanism, MPLS-TE can be a better choice here. These issues will drive us to ask the question: LFA versus MPLS-TE: Which one is better? Table 9-3 compares both solutions from different design aspects. Based on the scenario, requirements, and design constraints, network designs can select the most suitable approach.

Design Considerations	MPLS-TE FRR	LFA
Requires separate control protocol in addition to the IGP used	Yes.	No.
Adds control plane complexity	Yes. The level of complexity depends on several factors, such as the number of protected LDPs and nodes.	Yes. The level of control plane complexity depends on the topology (mesh versus triangle) and protected prefixes.
Topology dependent	No.	Yes (some topologies not supported).
MPLS required	Yes.	No. Supports native IP and MPLS, but LDP is required to support R-LFA.
Supports advanced path selection attributes	Yes, such as bandwidth, differentiated services (DiffServ) based, and SRLG exclusion.	Yes (very limited compared to MPLS-TE based).
Offers end-to-end deterministic path protection/selection	Yes, to a high degree (for example, PE to PE).	Yes, to a limited degree (for example, within a POP network).
Requires knowledge for the operations staff	Requires advanced knowledge/expertise in MPLS and MPLS-TE.	Same link-state IGP knowledge/expertise.
Scalability	Depends on several factors discussed in this chapter and Chapter 5, such as the number of LSPs, PE-to-PE versus P-to-P TE tunnel termination, and hardware resources.	Depends on the network size and topology (triangle versus partial mesh), available hardware resources, and number of protected prefixes.
Flexibility	Flexible (topology independent, supports various path/tunnel selection criteria, supports load balancing/sharing, and supports multiple routed domains.	Limited flexibility compared to MPLS-TE FRR (for instance, works within single link-state flooding domains and can be impacted by the underlying physical topology).

Table 9-3 *MPLS-TE FRR Versus LFA*

Like the MPLS-TE planning approaches discussed in Chapter 5, LFA is no different here. Network designers must

- Assess the goal of the solution (for example, only path protection or provide differentiated services with path protection)
- Assess the current environment in terms of
 - Routing protocol in use
 - Physical topology.
 - Native IP or MPLS based
 - Single or multiple routing (flooding) domains
- Level of flexibility (for example, how flexible and easy it is for this topology to be altered or redesigned [based on the situation])

After this foundational information has been identified, network designers can make the call whether to use LFA, MPLS-TE, or a combination of both, considering the design aspect listed in Table 9-3.

For instance, an MPLS VPN service provider with the network architecture depicted in Figure 9-14 requires very low network convergence time (path protection) across the entire network and highly flexible with advanced path protection capabilities across the core without any requirement of service differentiation. Also the suggested solution should cater for ease of deployment requirements of this operator.

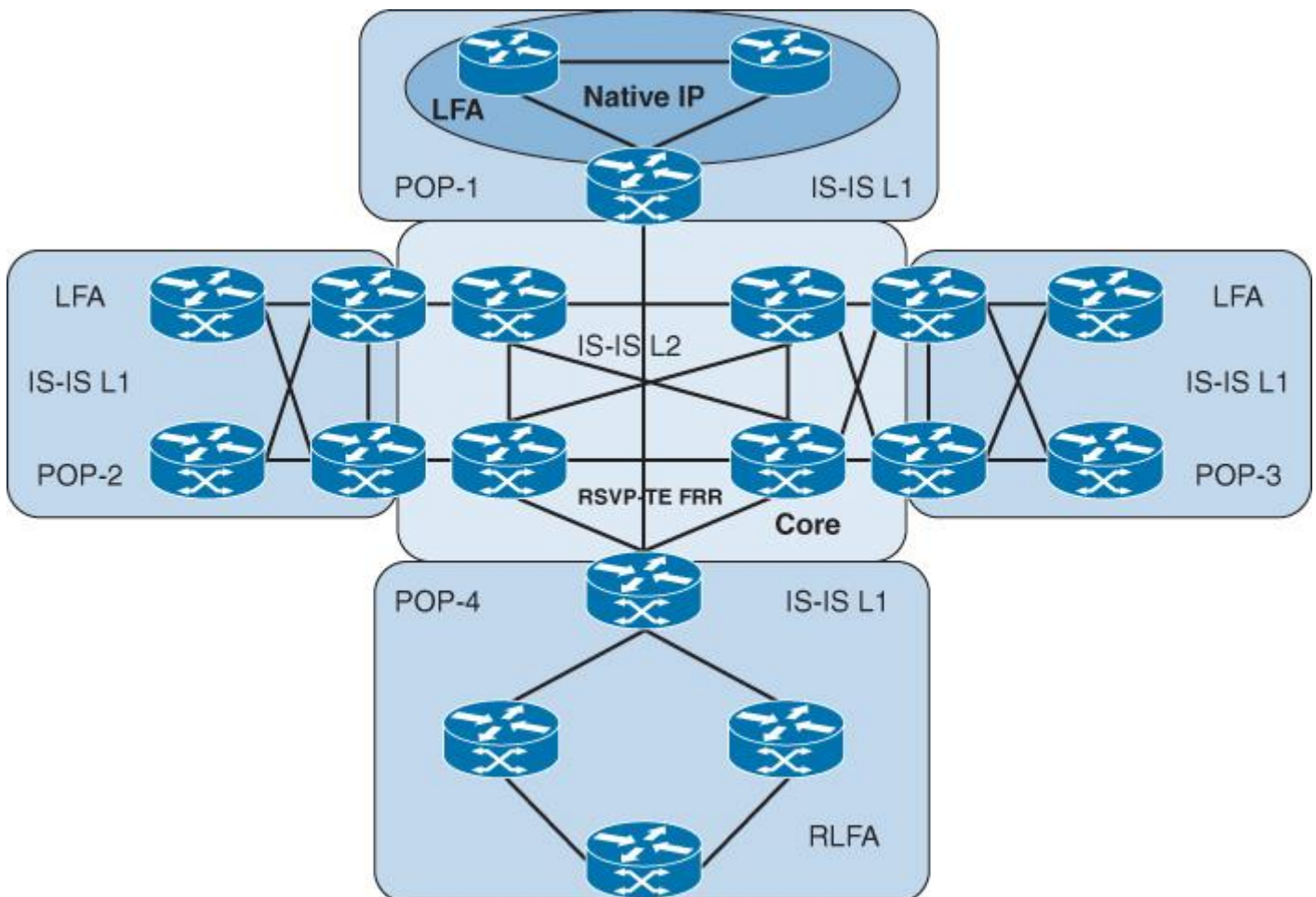


Figure 9-14 FRR Mixed Approach (MPLS-TE FRR and LFA)

Based on the requirements provided and the network architecture, it is feasible to use LFA at the edge of the network (single flooding domain, no additional protocol to design and manage and no physical connectivity change required at the POP or core levels), while one-hop MPLS-TE along with autotunnel backup MPLS-TE FRR can be applied at the core of the network, which provides path protection that offers advanced TE capabilities without being impacted by the underlying physical connectivity (full mesh in this